



Scopism SIAM Body of Knowledge FOUNDATION

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Foreword

As SIAM Programme Director and Head of IT Service Management at Sellafield Ltd, I have experienced first-hand the challenges of integrating multiple service providers in a complex, high-stakes environment. Our mission demands resilient, efficient, and value-driven services that support critical operations every day. Like many organizations, we work with a diverse ecosystem of partners - each with their own specialisms, priorities, and ways of working - and it is our job to ensure that they all come together seamlessly to deliver the consistent quality of service we require.

When I first encountered SIAM, it provided a framework that helped make sense of this complexity. Rather than treating each relationship in isolation, SIAM gave me a way to think holistically about service delivery, focusing not only on contract management but also on collaboration, shared accountability, and the outcomes that truly matter to our customers.

This is why this updated publication of SIAM Foundation is so important. It gives practitioners around the world a clear and structured introduction to SIAM concepts, allowing them to take those first steps with confidence. The guidance, combined with the insight shared by the Scopism SIAM Community, has helped raise the maturity of SIAM adoption across many sectors. The community provides a unique space for practitioners to share experiences, lessons learned, debate new ideas, and find practical solutions to real-world challenges. By connecting professionals across industries and geographies, it keeps SIAM thinking fresh, relevant, and forward-looking - helping organizations avoid common pitfalls and accelerate their progress towards a truly integrated, outcome-focused way of working.

Governments, public bodies, and enterprises globally face similar pressures: to improve resilience, deliver measurable value, and embrace innovation. In many countries, procurement strategies are deliberately designed to open opportunities to a wider range of providers, including small and medium-sized enterprises. This brings diversity and innovation - but also greater complexity. SIAM offers a proven approach to orchestrating this supplier landscape, ensuring that innovation can flourish while maintaining the high standards of quality, security, and performance required.

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Looking ahead, the global shift towards cloud platforms, digital ecosystems, and AI-driven automation will only make service integration more critical. SIAM, with its focus on data, collaboration, and outcomes, provides the tools to navigate this evolving landscape.

This updated publication of SIAM Foundation provides clarity, practical direction, and a shared language for those embarking on or refining their SIAM journey. I commend it to practitioners everywhere and trust that it will serve as a valuable companion as you develop, mature, and ultimately transform the way you deliver services.

Tony Williams

SIAM Programme Director & Head of IT Service Management

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1. Introduction to Service Integration and Management (SIAM)

1.1. What is SIAM?

Service integration and management (SIAM) is a management methodology that is used to create value in an ecosystem where multiple service providers are involved in the delivery of services.

Whether these service providers are internal, external or a mixture of both, challenges can arise when objectives, contracts and targets are misaligned across the end-to-end value chain.

SIAM introduces the concept of the 'service integrator', a capability established to ensure the customer organization gains maximum value from its service providers. The service integrator is a logical entity that is held accountable for the end-to-end delivery of services.

Acting as a vital link between the customer organization and its service providers, the service integrator provides governance, management, integration, assurance and coordination across the ecosystem, enabling consistent, effective, and outcome focused service delivery.

Methodology or framework

There are different definitions of what constitutes a 'methodology' versus a 'framework' (or a 'practice', 'approach', 'model, etc.). The definition used for a methodology is *a set of methods used in a particular area of study or activity*¹.

It's important to understand that SIAM is not a prescriptive or one-size-fits-all methodology. Instead, it offers a comprehensive framework of guidelines, best practices, concepts, and models that can be applied flexibly to suit the needs of a multi-sourced ecosystem.

¹ Source: Cambridge (online) dictionary

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SIAM governance operates at three levels in the ecosystem:

- Strategic
- Tactical
- Operational.

SIAM supports cross-functional, cross-process, and cross-provider integration. It creates an ecosystem where all parties:

- Know their role, responsibilities and context in the ecosystem
- Are empowered to deliver
- Are held accountable for the outcomes they are required to deliver.

Terminology

SIAM is the generally accepted acronym for service integration and management.

Other acronyms that are in use are:

- MSI (Multi Sourcing Integration)
- SMI (Service Management Integration)
- SI (Service Integration)
- SMAI (Service Management and Integration)
- SI&M (Service Integration & Management).

SIAM can be applied to different sizes and types of organization, and to different industry sectors.

SIAM is designed to work across all types of service providers. It can be applied in ecosystems that include only external providers, only internal teams, or a combination of the two. The effectiveness of SIAM and the value it delivers will increase as the number of service providers and the complexity of interactions between them increase.

Some organizational cultures are more able to adapt to SIAM than others. Effective SIAM requires control to be balanced with trust, devolution of responsibilities, openness, and collaboration across all parties. A transition to SIAM is likely to require significant changes in attitude, behavior, and culture

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in ecosystems that previously relied on command and control structures for effective service delivery.

Collaboration versus cooperation

While both cooperation and collaboration involve working together, the definition of **collaboration** is *"the situation of two or more people working together to create or achieve the same thing"*, whereas **cooperation** is defined as *"the act of working together with someone or doing what they ask you"*².

Cooperation is seen as working together to achieve your own goal, whereas collaboration is working together for a common goal. It could be argued that in a contractual situation with external service providers, where each party has their own goals, there is always cooperation. Whereas for internal service providers, the customer organization's goals are the same as theirs, which means collaboration.

Within SIAM the emphasis is on **working together** towards the goals of the customer organization.

The SIAM methodology encompasses:

- Practices
- Processes
- Functions
- Roles
- Structural elements.

The customer organization will transition to an operating model that includes SIAM principles and structural elements.

² Source: Cambridge Dictionary

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1.1.1. The SIAM ecosystem

There are three layers in a SIAM ecosystem:

1. Customer organization (including retained capabilities)
2. Service integrator
3. Service provider(s).

Each layer has a defined role as part of effective end-to-end management of services and the delivery of maximum value. Each layer should have sufficient capability and maturity to fulfill its role.

Ecosystem versus environment

An **ecosystem** is defined as “the interacting community within an environment, functioning as a system”³. An **environment**, on the other hand, is “the surroundings or conditions which influence operations”.

The SIAM ecosystem consists of three layers: customer organization (including retained capabilities), service integrator and service provider(s). The environment refers to ‘the outside’, including any external factors such as legislation, regulations or market conditions, that influence the SIAM ecosystem (see also SIAM model in section 1.1.6).

The focus, activities and responsibilities of each layer are different. **Figure 1** provides an illustration of this.

³ Source: Wikipedia

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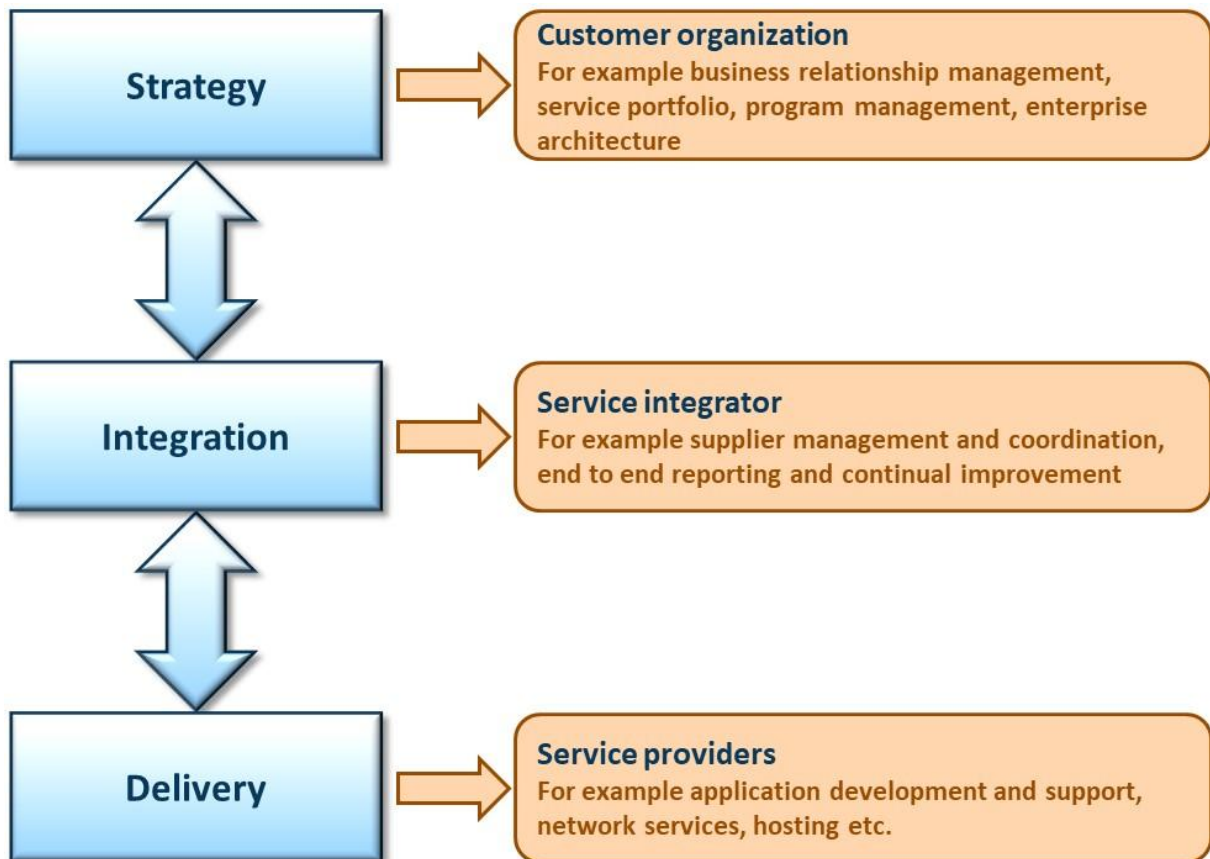


Figure 1: Focus of the SIAM layers

1.1.1.1. Customer organization

The customer organization is the client that is making the transition to SIAM as part of its operating model. It commissions the SIAM ecosystem.

Customer organizations typically contain business units such as human resources, finance, sales and their own internal IT function. They also have their own customers who use their products and services.

Figure 2 shows the layers of the SIAM ecosystem, and the consumers of services from the customer organization.

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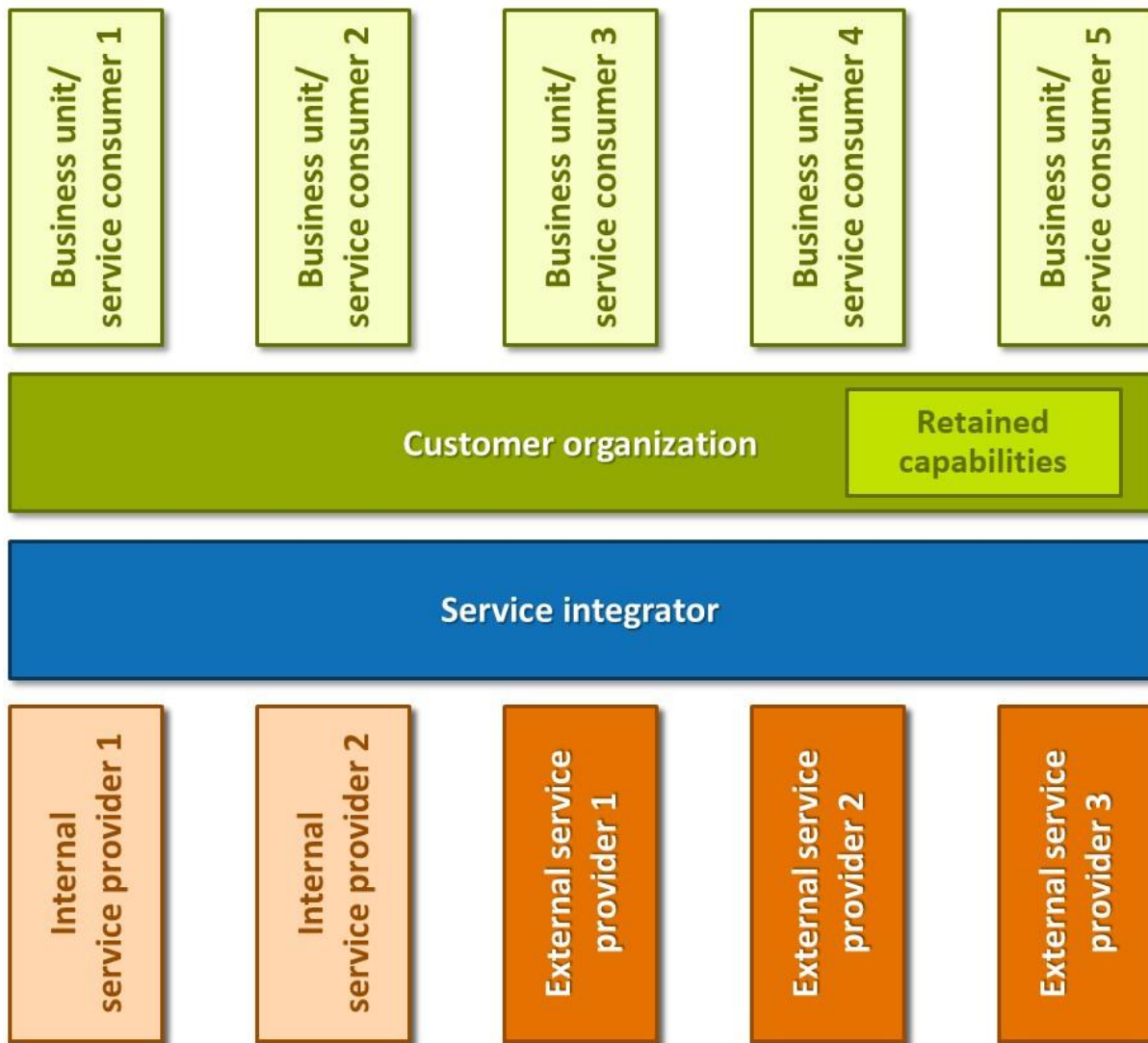


Figure 2: The SIAM layers, including consumers of services from the customer organization

The customer organization will own the contractual relationships with external service providers, and any external service integrator, and will provide corporate governance over teams operating as internal service providers.

1.1.1.2. Retained capabilities

The customer organization will include some retained capabilities. The retained capabilities are the functions that are responsible for strategic, architectural, business engagement and corporate governance activities.

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These business-differentiating functions typically remain under the direct control and ownership of the customer organization. Retained capabilities also include any accountabilities and responsibilities that must remain with the customer for legislative or regulatory reasons.

Examples of possible retained capabilities are:

- Enterprise architecture
- Policy and standards management
- Procurement
- Contract management
- Demand management
- Financial and commercial management
- Service portfolio management
- Corporate risk management
- Governance of the service integrator; based on achievement of business outcomes.

The service integrator is independent from the retained capabilities, even if it is internally sourced, because service integration is not a retained capability.

Instead, retained capabilities, sometimes referred to as the 'intelligent client function', provide the customer with the in-house skills, knowledge, and authority needed to govern the relationship with the service integrator and provide insight and direction on business needs, priorities, and strategic outcomes for the SIAM ecosystem.

1.1.1.3. *Service integrator*

The service integrator layer of the SIAM ecosystem is where end-to-end service governance, management, integration, assurance and coordination is performed.

The service integrator layer focuses on implementing an effective cross-service provider organization, making sure that all service providers are collaborating to deliver the end-to-end service. It provides operational governance over the service providers and has a direct relationship with the customer organization and the service providers.

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The service integrator layer can be provided by one or more organizations, including the customer organization. If the service integrator layer is provided by more than one organization, it should still be considered as a single logical service integrator.

1.1.1.4. *Service provider*

Within a SIAM ecosystem, there are multiple service providers. Each service provider is responsible for the delivery of one or more services, or service elements, to the customer organization. It is responsible for managing the products and technology used to deliver its contracted or agreed services and operating its own processes.

Types of service

A service is a structured set of activities, capabilities, and resources that delivers value by enabling desired outcomes for a customer. Services support the customer by reducing the customer's need to manage associated operational costs and risks.

From the perspective of the service integrator we could identify:

- Customer-facing, end-to-end services, provided to the customer organization, managed by the service integrator
- Supporting services delivered by service providers underpinning customer-facing services by providing part of the end-to-end service.

In all cases, the purpose of a service is to create value through outcome-focused delivery.

Service providers within a SIAM ecosystem are sometimes referred to as 'towers'. This term implies isolation and a monolithic approach, so the term 'service provider' is used as standard in this document.

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Service providers can be part of the customer organization or external to it:

- An **external** service provider is an organization that is not part of the customer organization. Its performance is typically managed using service level agreements and a contract with the customer organization
- An **internal** service provider is a team or department that is part of the customer organization. Its performance is typically managed using internal agreements and targets.

For instance, if the customer retains their own internal IT capability or development teams, this should be treated as an internal service provider, governed by the service integrator in the same way as other service providers.

Examples of services provided by service providers in a SIAM model include:

- Desktop services/end user computing
- Hosting
- Network/LAN/WAN
- Cloud services
- Application development, support and maintenance
- Managed services.

The list above provides technical service examples as SIAM is often used in IT environments. However, service integration is a generic capability and with the growth of Enterprise Service Management (eSM), SIAM is also being applied in an enterprise environment (eSIAM). Additional examples of enterprise focused services could be HR, facilities management or finance.

Service provider categories

It can be helpful to categorize service providers in a SIAM ecosystem, to help define their importance to the customer organization and the approach to governing and assuring their services.

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There are three common categories of service providers in a SIAM ecosystem:

- **Strategic service provider:** provides critical and/or potentially highly customized service(s) which are aligned with the long-term vision and direction of the customer organization (for instance, a bespoke business application). Their agreement with the customer organization may include elements of shared risk and reward.
- **Tactical service provider:** provides specialized or tailored services which focus on the actions needed to achieve the goals of the organization (for instance, service desk or desktop support).
- **Commodity service provider:** provides standardized or easily substitutable services with minimal customization, typically to multiple customers (for example, cloud hosting or off-the-shelf infrastructure services).

SIAM is applied to all three categories, but the nature of the relationship and the amount of management required will be different.

1.1.2. SIAM practices

Practice: *the actual application or use of an idea, belief, or method, as opposed to theories relating to it.*⁴

SIAM includes specific practices that differentiate it from other management frameworks. These practices support governance, management, integration, assurance, and coordination across the layers.

The four categories of SIAM practices ensure a holistic and balanced approach to designing and operating a SIAM ecosystem. Each category encompasses a wide range of individual activities that collectively support the integration, coordination, and continual improvement of multi-provider ecosystems.

⁴ Source: Oxford English Dictionary © 2017 Oxford University Press

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The four categories are:

- People practices
- Process practices
- Measurement practices
- Technology practices.

Examples of practices are described in section **5 SIAM practices**.

1.1.3. SIAM and processes

Process: a documented, repeatable approach to carrying out a series of tasks or activities.

SIAM itself is not a process; it draws on and uses other management processes.

Most management approaches expect processes to be executed within one organization. In SIAM, these processes may also be executed:

- Across organizations in the same SIAM layer
- Across organizations in different SIAM layers.

Many of the processes used within a SIAM ecosystem are familiar processes like change management and business relationship management.

Within a SIAM model, however, these processes require adaptation to support integration and coordination between the different parties. The processes also need to be allocated to the appropriate layers in the SIAM model. Some processes will span multiple layers. For example: the customer organization and the service integrator can both carry out elements of supplier management; the service integrator and service providers will each have responsibilities in the end-to-end change management process. This allocation may be different for each implementation of SIAM.

In a SIAM ecosystem, the focus is on the end-to-end process. The customer organization will not mandate processes, toolsets or other working practices within each service provider. However, it will expect the service providers to integrate effectively with the overarching SIAM model and the end-to-end processes to support service delivery.

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The processes also require alignment with the SIAM practices (see section **5 SIAM practices**).

1.1.4. SIAM roles

Roles and responsibilities need to be defined, established, monitored and improved within a SIAM ecosystem.

This includes the roles and responsibilities of each:

- Layer
- Organization
- Function
- Structural element.

High-level policies for roles and responsibilities are defined during the Discovery & Strategy stage of the SIAM roadmap (see section **2 SIAM roadmap**). More detail is added during the Plan & Build stage.

Roles and responsibilities are allocated to relevant parties during the Implement stage. They are then monitored during Run & Improve and amended as required.

The roles are further described in section **4 SIAM roles**.

1.1.5. SIAM structural elements

Within SIAM, 'structural elements' are collaborative entities that have specific responsibilities and work across multiple organizations and layers in the SIAM ecosystem.

Functions

Function: *an organizational entity, typically characterized by a special area of knowledge or experience.*⁵

Each organization in the SIAM ecosystem will determine its own organizational structure. This structure will include functions that execute specific processes and practices.

⁵ Source: IT Process Wiki

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Functions represent the key areas of expertise and capability required to operate. They describe what needs to be done and by which organizational group, without prescribing the specific processes or roles. A function may be delivered by a single team, multiple teams, or a combination of the customer organization, service integrator, and service provider resources working together as part of the integrated ecosystem.

In SIAM, the structural elements can be seen as functions, forming an organizational entity. But other functions can exist in an organization, for example, service desk, procurement, or supplier and contract management.

These structural elements link the functions with the practices, processes, and roles of SIAM. They foster cross-party communication and shared responsibilities between the service integrator, service providers, and the customer retained capabilities.

The role of the structural elements includes:

- Governance
- Developing and maintaining policies
- Developing and maintaining data and information standards
- Reviewing and improving end-to-end service performance
- Reviewing and improving capability and maturity
- Identifying, encouraging, and driving continual service improvements and innovation
- Resolving shared issues and conflicts
- Delivering specific projects
- Integrating, aggregating and consolidating data to form an end-to-end view
- Recognizing and rewarding success.

Structural elements include representatives from the service integrator, the service providers, and, where required, the customer.

Using structural elements helps to establish relationships between the different parties. This encourages communication and collaboration, as attendees work together to achieve shared goals.

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The use of structural elements differentiates SIAM from other methodologies and helps to facilitate the desired outcomes from SIAM.

There are three common types of structural element:

1. Boards
2. Forums
3. Working groups.

Examples of the structural elements are described within section **4 SIAM roles**.

1.1.5.1. *Boards*

Boards perform governance in the SIAM ecosystem.

They are formal decision-making bodies, and are accountable for the decisions that they take. Boards will convene regularly, for as long as the SIAM model is in place.

In SIAM, governance activities are carried out by boards acting at strategic, tactical and operational levels. Examples are:

- Strategic: approval of funding, contractual and commercial agreements, and strategy
- Tactical: approval of policies
- Operational: approval of changes to services and processes.

1.1.5.2. *Forums*

Forums allow attendees work together on proactive development, innovations, and improvements.

They will convene regularly, for as long as the SIAM model is in place. Their responsibilities include:

- Developing and sharing common working practices
- Developing data and information standards
- Continual improvement
- Innovation.

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Forums are often aligned to processes (see section **1.1.3 SIAM and processes**). For example: a problem management process forum can be established with problem management staff from each service provider and the service integrator. They can jointly develop a set of key performance indicators for the problem management process.

They can be established wherever the SIAM ecosystem would benefit from a proactive, improvement focused element, such as service owner forums or a procurement forum.

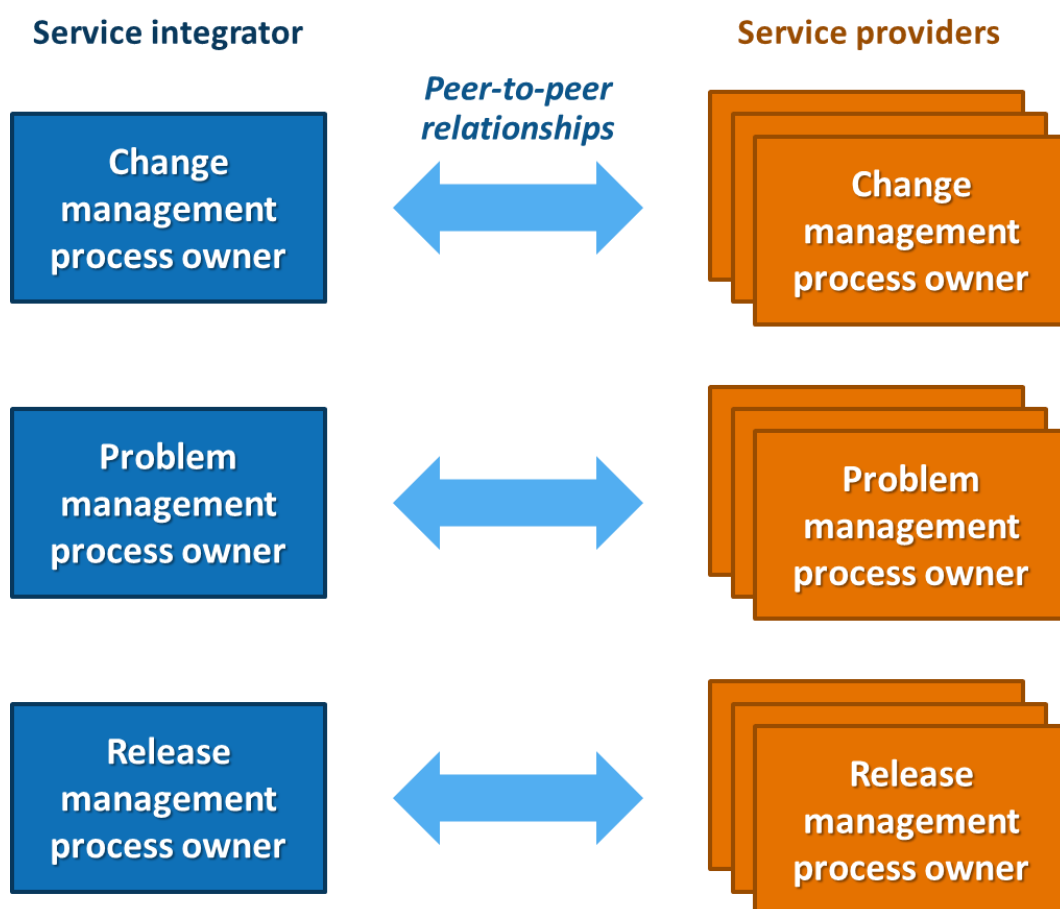


Figure 3: Peer to peer process forums

There is no requirement to create an individual process forum for every process. Where the scope overlaps, a single forum may address multiple processes or be structured around a common theme like service operations.

The value of the forums will diminish if their remits overlap and work is duplicated, or if they are perceived as an unnecessary overhead.

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1.1.5.3. *Working groups*

Working groups are convened to address specific issues or projects. They are typically formed on a reactive, ad-hoc, or fixed-term basis. They can include staff from different organizations and different specialist areas.

For example: an ad-hoc working group could be established with members from several service providers to investigate an intermittent issue with the performance of an integrated service. This could include specialists from capacity management, operations, development, problem management and availability management. Alternatively, a fixed term working group could be established to manage the delivery of an integrated release. The members would be from all layers and from multiple processes and functions.

Forums and working groups often involve the same people, so can be combined into the same meeting if appropriate. In these combined meetings, it is important to ensure that there is a focus on proactive as well as reactive activities.

1.1.6. **SIAM model**

Each organization will develop its own SIAM model, based on the layers in the SIAM ecosystem. The SIAM model that an organization adopts will be influenced by several factors:

- The services that are in scope
- The required outcomes
- The use of proprietary models by externally sourced service integrators.

Because of this, there is no single 'perfect' SIAM model. No model is 'better' than any other, although some may be more suitable to particular implementations than others.

Different organizations adapt models to meet their own needs. All models share common characteristics that are aligned to the methodology described in the SIAM Body of Knowledge.

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The SIAM model as part of an organization's operating model

An organization is a complex system for delivering value. An operating model breaks this system into components, showing how it works. It can help those making changes check that they have thought through all elements and that the whole will work.⁶

The SIAM model fulfills a similar function for the organization's SIAM ecosystem. As such, it will need to be aligned to, and forms a part of, the organization's operating model.

Figure 4 shows a high-level SIAM model, including the relationships between SIAM layers practices, processes, functions, and structural elements.

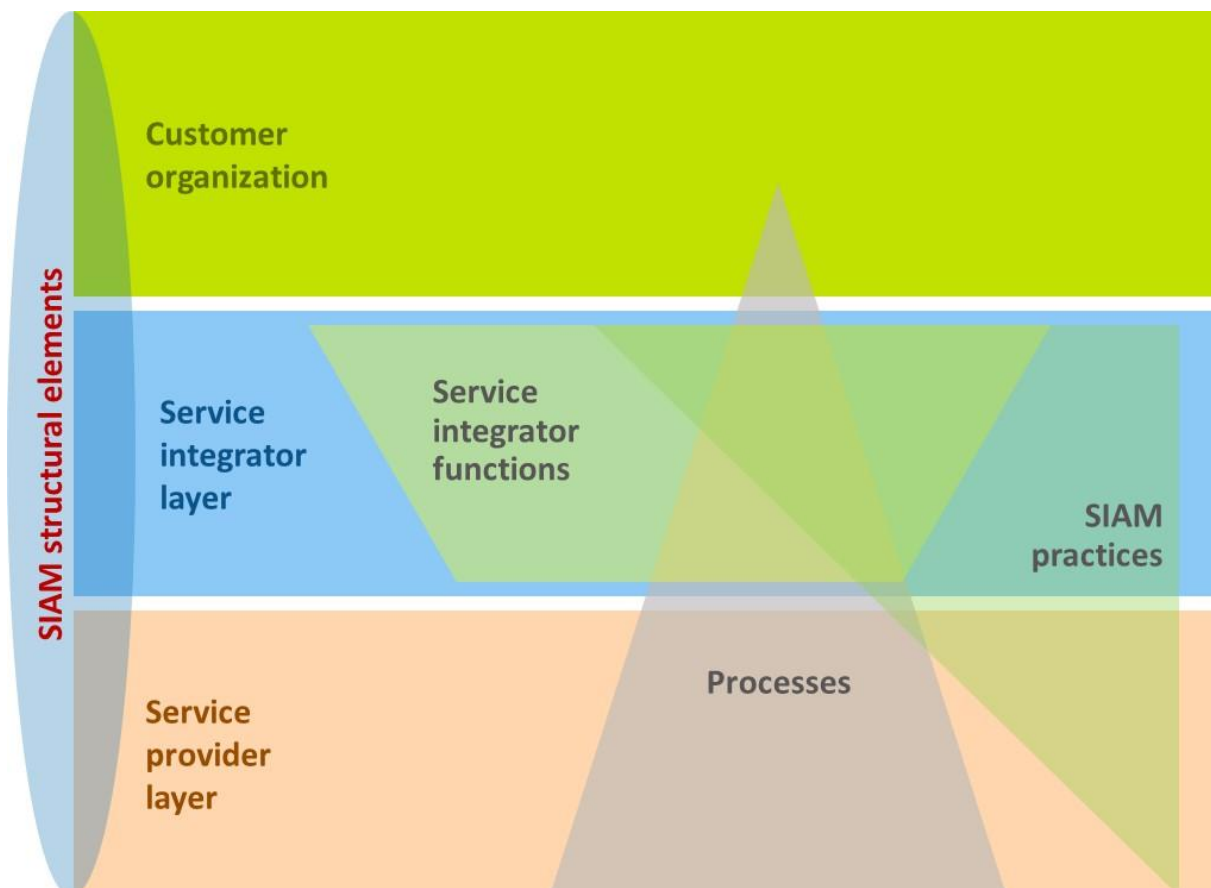


Figure 4: A high-level SIAM model

⁶ Source: Wikipedia

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1.1.7. SIAM contractual and sourcing considerations

Within the SIAM model, the customer owns the contractual relationship with external service providers and any external service integrator.

The service integrator is empowered to act on behalf of the customer, exercising the parts of the contract related to the delivery of the services by the service providers.

The contracts between the service providers and the customer organization must make it clear that the service integrator is the agent of the customer, whether that service integrator is internally sourced or externally sourced.

To provide flexibility to support future business and technology strategies, the contracts should, where possible, be designed to accommodate changes to services and ways of working.

In many existing customer and provider relationships, standard contracts have limited the ability to transition to SIAM. For SIAM to be effective, the customer organization needs to select the right service providers and have suitable contracts in place. Targets within the contracts should encourage service providers to work together, including contributing to service improvements and innovation.

While SIAM models can use shorter, more flexible contracts than traditional outsourcing, this is not always the case. The key is to avoid rigid, highly interdependent arrangements and instead apply the principle of loose coupling. This means structuring contracts so that changes to one service or service provider have minimal impact on others, enabling easier replacement, onboarding, or modification without destabilizing the ecosystem.

In a SIAM ecosystem, the best outcomes are realized when the customer, the service integrator, and the service providers work collaboratively to achieve common goals. Wherever possible, contracts should facilitate this, by being fair to all parties and by applying the terms in a transparent and impartial way.

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Trust-based supplier management is more likely to achieve the necessary collaborative outcomes, rather than simply relying on strict adherence to contractual terms. To establish this trust and fairness, it is necessary to consider the management practices and ways of working needed within the SIAM ecosystem. This is discussed in more detail in SIAM Professional.

1.2. The history of SIAM

1.2.1. SIAM as a concept

Organizations have been using services delivered by multiple service providers for many years. They have recognized the need for service integration across service providers, and used different approaches to try to achieve end-to-end service management.

Historically, models for managing this type of ecosystem were proprietary to very large service providers, developed to meet specific client requirements, and rarely shared outside those providers.

In most cases, these service providers also delivered significant systems integration capabilities, but with no clear separation from service integration. These organizations were typically referred to as Systems Integrators (SI) or IT Outsource (ITO) providers.

1.2.2. The emergence of the term 'SIAM'

The term 'service integration and management', or SIAM, and the concept of SIAM as a management methodology originated in around 2005 from within the UK public sector, which is also the source of other best practice methodologies such as ITIL®.

The methodology was initially designed for the Department of Work and Pensions to obtain better value for money from services delivered by multiple service providers, and more specifically to separate service integration capabilities from systems integration and IT service provision.

This new approach reduced the duplication of activities in the service providers, and introduced the concept of a 'service integrator', which provided the capability for governance and coordination to encourage service providers to work together, to driving down costs and improving service quality.

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SIAM was viewed as a management methodology, not a function or a framework. Within the methodology, a service integrator provided a set of service integration capabilities.

The emerging SIAM methodology facilitated collaboration between service providers, and management of interfaces between them.

The methodology allowed individual service providers to act autonomously and define the specific mechanisms that enabled those activities. These were then audited and assured by the service integrator.

SIAM contributed to a comprehensive operating model that supports end-to-end service integration. It was an evolution of how organizations design and operate integrated service delivery across multiple service providers. It has continued to develop as organizations have moved away from outsourced contracts with a single supplier to an ecosystem with multiple service providers.

SIAM practices have matured as organizations contend with two significant factors:

1. The increasing reliance on technology for business processes and services
2. The increasing complexity of the technical solutions supporting business processes and services

Whether an organization has external service providers, internal service providers, or a mixture of both, SIAM will allow it to reduce silos, improve collaboration, and build on a strong foundation of delivery capability.

Figure 5 shows a simple view of the SIAM model.

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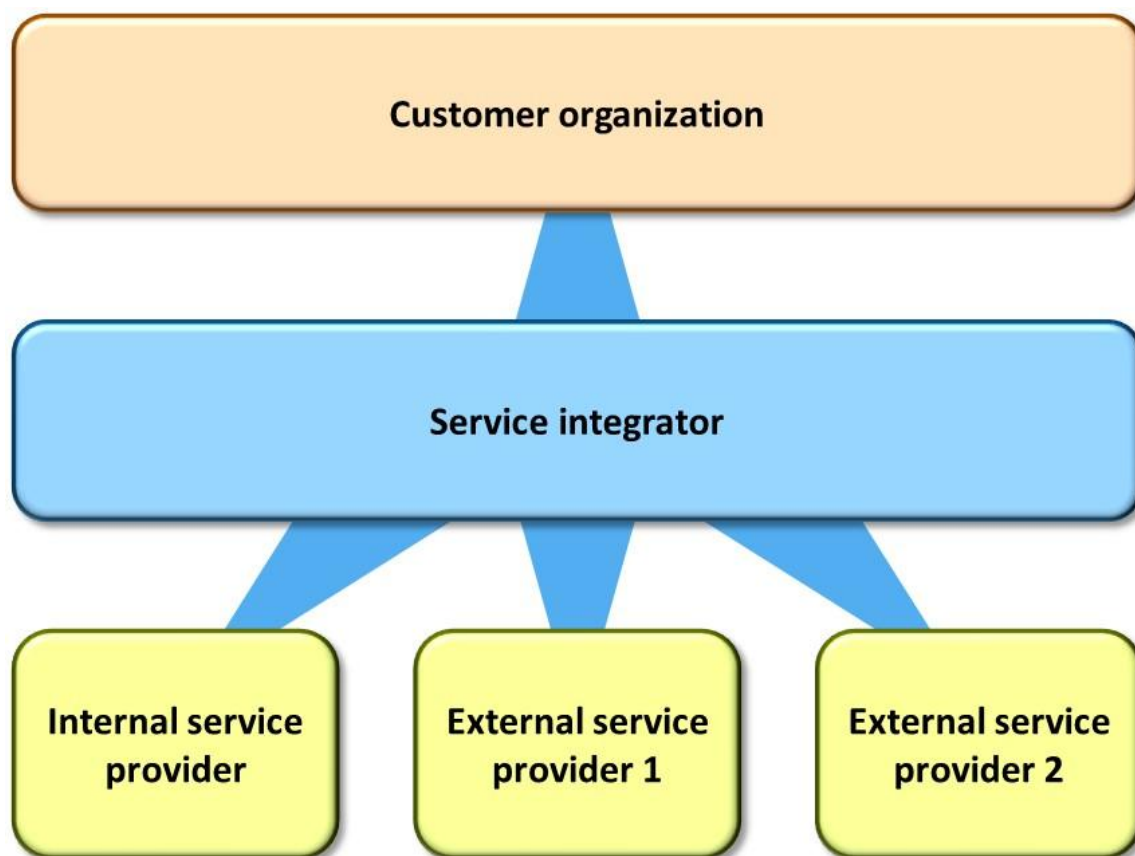


Figure 5: A simple view of a SIAM ecosystem

1.2.3. Adoption of SIAM within the UK Government

In 2010, the UK Government published a new information and communications technology (ICT) strategy. This included moving away from large prime supplier contracts to a more flexible approach using multiple service providers and cloud-based solutions.

This accelerated the development and awareness of SIAM both in the UK public sector and elsewhere. This acceleration led to the publication in 2012 of the UK Government 'Cross Government Strategic SIAM Reference Set', developed from SIAM experience and expertise in the Department of Work and Pensions, Ministry of Justice, NHS Connecting for Health and the Government Procurement Service. The reference set described a wide range of SIAM capabilities and a suggested enterprise model, but encouraged adaptation to suit local requirements.

This was the first widely available description of SIAM. Its publication rapidly increased the awareness, development, and discussion of SIAM worldwide.

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1.2.4. The SIAM Body of Knowledge

In 2016 the SIAM Foundation Body of Knowledge was published. Since then, SIAM adoption has increased globally, driven by strategic factors, including:

- Increased complexity of sourcing models
- A worldwide need to improve value
- A desire to remove reliance on single suppliers
- The need for effective controls
- A desire for the ability and flexibility to use 'best of breed' service providers and services, including the use of commodity cloud-based services.

The SIAM Body of Knowledge has evolved as SIAM has expanded and evolved, from the first iteration, through the SIAM Professional Body of Knowledge published in 2017, via the 2020 refresh, to its current format.

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The Scopism SIAM Body of Knowledge consists of a layered model.

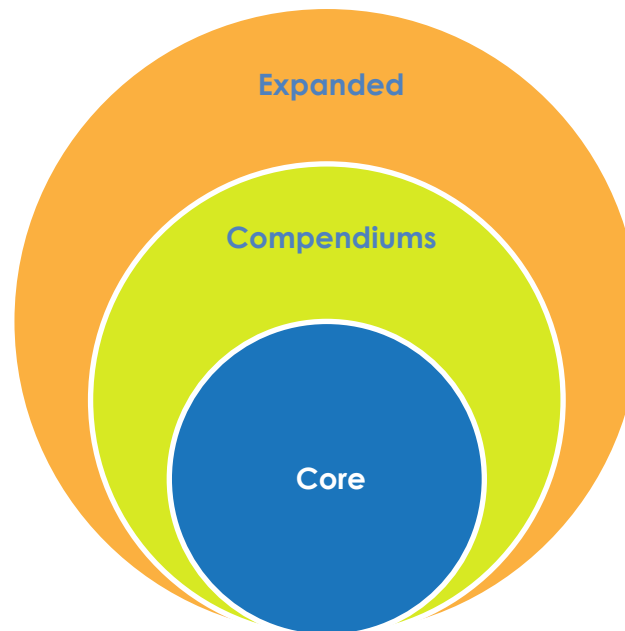


Figure 6: The SIAM Body of Knowledge layered model

- The **core** layer consists of the SIAM Foundation and Professional, explaining SIAM theory and principles.
- Encircling the core is the second layer, which includes, the **Compendiums**. These publications explore specific topics in the context of SIAM, such as this publication.
- The outermost layer comprises community authored content, including whitepapers, case studies, templates and other practical resources.

1.3. The purpose of SIAM

The SIAM management methodology is unique, because it focuses on the specific challenges associated with complex, multi-sourced service delivery models. SIAM enables the flexibility and innovation necessary to support the pace of change demanded by today's fast-moving organizations.

Successful organizations have an operating model that allows them to be flexible and resilient, adapting as market conditions change. An operating model that includes SIAM principles provides this flexibility and resilience by allowing the service ecosystem to be adapted as necessary. Service providers can be added and removed as demand changes, with minimal risk to existing operations and minimal management overhead. A flexible

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operating model provides the foundation for bolder strategic initiatives, as the organization grows in confidence about its ability to deliver new or changed products and services.

The service integrator provides the customer organization with a single point of accountability and control for the integrated delivery of services. This is achieved through the definition and application of controls within a robust governance framework that also provides the necessary coordination between the service providers within the SIAM ecosystem. The service integrator also drives collaboration and improvement across the service providers, acting on behalf of the customer.

The service integrator takes ownership of these activities on behalf of the customer organization, allowing it to focus on the activities necessary for its own business objectives, rather than focusing on service providers and technology.

The service integrator manages the complexities of dealing with multiple service providers, allowing the customer organization to benefit from their specialisms and capabilities without incurring any additional management burden.

The application of the SIAM methodology creates an ecosystem where all parties involved in the delivery of services are clear about their roles and responsibilities and are empowered to deliver within those boundaries.

SIAM also provides visibility and shared understanding of the necessary touchpoints, interfaces, and dependencies between the services and the service providers. This includes defining how information flows, how responsibilities connect across organizational boundaries, and how potential conflicts or overlaps are identified and resolved. This facilitates the coordination of delivery, integration and interoperability, ensuring that these touchpoints and dependencies are clearly highlighted, documented, and proactively managed.

The service integrator provides assurance of the performance of individual service providers and the end-to-end service, ensuring that the expected outcomes are delivered to the customer organization.

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1.4. The scope of SIAM

The scope of SIAM will vary from organization to organization. For the customer organization to derive any benefit from a transition to a SIAM model, the service(s) that are in scope must be defined.

This service definition makes it clear what is being governed, assured, integrated, coordinated and managed by the service integrator.

For each service within the scope of SIAM, these areas need to be defined:

- Service outcomes, value, and objectives
- The service provider(s)
- The service consumer(s)
- The service characteristics, including service levels
- The service boundaries
- Dependencies with other services
- Technical interactions with other services
- Data and information interactions with other services.

A service model should be created that shows the hierarchy of services. This hierarchy must clearly identify:

- Services that are directly consumed by the customer organization
- Underpinning services and dependencies.

Figure 7 shows an example of a service model showing the service hierarchies.

The model shows how business needs in the customer organization are met by services (1-4), and how in turn they are dependent on one or more supporting services (5-7), which may be comprised of service components.

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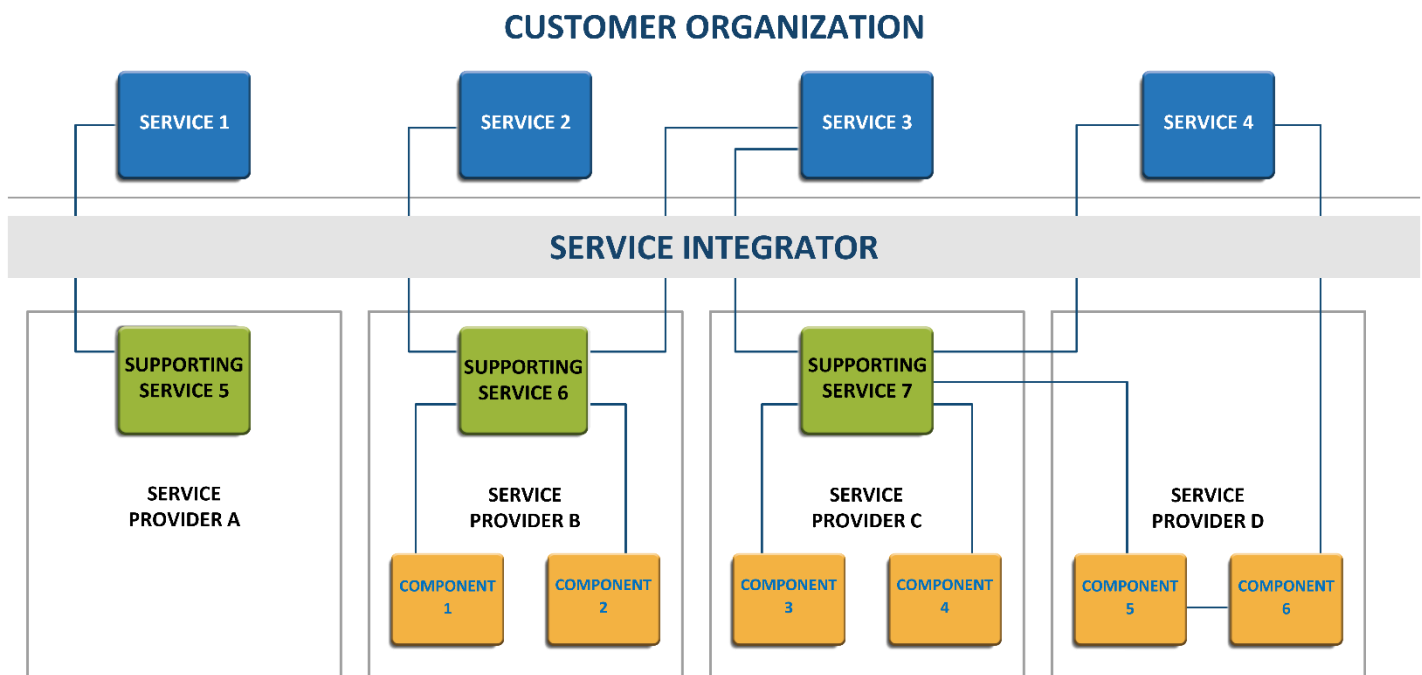


Figure 7: A service model showing the hierarchy of services

1.4.1. Types of service within a SIAM model

SIAM can be applied to both IT services and non-IT services. Different organizations will have different types of services within the scope of their SIAM model. Some models may only include services that were previously provided internally, as part of a strategy to outsource these services to external organizations.

Others may include a wide range of externally provided services but retain, for example, their internal IT department as an internal service provider. The customer organization will determine the scope, based on its strategy and requirements.

Examples of IT services include:

- Office productivity applications
- Customer relationship management systems
- Networks
- Bespoke customer applications.

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Examples of non-IT services that can be within the scope of SIAM are business processes such as sales order management, payroll processing, and consumer help desks.

1.4.2. Commodity services

SIAM can be applied to commodity services, including those provided via the cloud.

Service providers for commodity services typically use the same delivery models for all their customers. It is therefore unlikely that they will adapt their ways of working to align with a customer organization's specific SIAM requirements or accept governance from the service integrator.

If this is recognized and the service integrator can adapt to its approach while delivering customer outcomes, SIAM can still be effective for these services.

1.5. SIAM and the business strategy

Adopting SIAM principles and transitioning to an operating model that includes them is a significant commitment. Top-level management should drive the decision and align this with the organization's broader business goals, operating model, and strategic direction. The expected outcomes must justify the investment required (see section **1.6 Value to the organization – the SIAM business case**).

Key factors to consider include the organization's sourcing strategy, the maturity and flexibility of its current operating model, the product and service roadmap, and the existing internal and external service provider landscape. These considerations are a core part of the Discovery & Strategy roadmap stage (see section **2.1**).

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1.5.1. Why change?

Without effective service integration, many of the benefits anticipated from services delivered by multiple service providers can remain unrealized.

Transforming an organization to a SIAM model ensures that critical artefacts are developed as part of the SIAM roadmap. These will include:

- A clear design for how the overall end-to-end service will operate and integrate
- A defined governance approach
- Definition of accountability for the integrated service
- An end-to-end performance management and reporting framework
- Coordination between service providers
- Integration between the processes of different service providers
- Definition of roles and responsibilities.

Organizations must be clear about why they want to adopt SIAM.

Transitioning to a SIAM model is not an easy task. It will require investment and changes for all involved parties. The changes will affect areas including:

- Attitude, behavior, and culture
- Processes and procedures
- Capabilities
- Organizational structures
- Resources
- Knowledge
- Tools
- Contracts.

Senior level sponsorship and management commitment will be essential. Without management commitment, the transformation to a SIAM model is unlikely to succeed.

There are organizations for whom SIAM is not appropriate. Before any organization embarks on a transition to SIAM, it must fully understand SIAM and the benefits it could derive. This will enable it to make a value-based judgement.

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The organization can gain this understanding in one of three ways, or in combination:

- Educate and train the staff who are leading on SIAM discovery and strategy in the SIAM methodology
- Seek help from outside the organization, either from similar organizations or from organizations experienced in SIAM adoption
- Recruit new staff who have the required understanding and experience.

1.5.2. Drivers for SIAM

In this context, a driver is defined as *"something that creates and fuels activity, or gives force and impetus"*⁷

These drivers are the triggers that create an organization's desire to move to a SIAM model. Understanding the drivers for SIAM will help an organization to gain clarity of purpose, creating a higher chance of success.

The drivers will be used to create a business case for the transition to SIAM. They will also help the organization to maintain focus throughout the SIAM roadmap.

The challenges of delivery in a multi-provider ecosystem

Service providers play a crucial role in helping a customer organization deliver its business outcomes. Poorly delivered services directly affect the customer organization's outcomes, and the services it can offer to its own customers.

This is true whether the services are delivered by one service provider or multiple service providers. However, the challenges of successful delivery are greater when there are multiple service providers, owing to increases in complexity and the interactions that need to take place between service providers.

⁷ Source: Collins English Dictionary – Copyright © HarperCollins Publishers

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Consider these scenarios that illustrate how poor service provision can have wider consequences:

- A hospital has booked in a patient to have an extensive medical scan. The medical machinery has stopped working and the cause of failure is unknown. The patient's appointment must be rescheduled. Will the delay to this appointment have a negative impact on the patient's health?
- A motoring organization cannot dispatch a patrol officer to assist a lone female motorist and her small child on a busy freeway because its command and control systems are unavailable owing to a failed system change. The organization does not know which patrol officers are available or where they are. To how much risk is the woman and her child exposed, and for how long?
- An online retailer is unable to cope with the increase in transactions prior to the holiday season. This makes its retail platform slow down, unnecessarily reject payments and at times show as unavailable. Will customers accept this or buy their holidays elsewhere?
- A hastily implemented, partially tested update to a travel agent's booking system has caused the personal information of its customers (including credit card details) to be hacked. The press has found out and is publishing worst-case scenarios of identity theft and potential financial impact for the customers. Will the reputation of this travel agent recover enough to remain a viable business?

There are generic drivers for SIAM that can be tailored for each organization. These can be placed into five driver groups:

1. Service satisfaction
2. Service and sourcing landscape
3. Operational efficiencies
4. External drivers
5. Commercial drivers.

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1.5.2.1. *Service satisfaction drivers*

These are drivers related to the level of satisfaction the customer has with the services that it receives, and the level of satisfaction that is expected.

There are seven drivers related to service satisfaction:

1. Service performance
2. Service provider interactions
3. Clarity of roles and responsibilities
4. Slow pace of change
5. Demonstration of value
6. Lack of collaboration between service providers
7. Delivery silos.

Service performance

Customers expect guaranteed service performance and availability, irrespective of who provides the service.

On some occasions, customers in a multi-provider ecosystem can experience dissatisfaction with the level of service they receive, even though each of the service providers report that they are achieving their individual service level targets.

One example is incident resolution times, where the time taken to pass an incident from one service provider to another is not considered in the service level calculation.

Without effective governance, coordination and collaboration, there will be service performance issues including:

- A lack of transparency for the end-to-end service
- Incomplete understanding of, and inability to report on, end-to-end service performance
- No management of service levels across the end-to-end service
- Service performance that is not aligned to business requirements.

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Service provider interactions

In a multi-provider ecosystem, service users might have to interact separately and differently with each internal and external service provider.

For example, one service provider might only accept contact from users by telephone, another only by email, and another only via an internet portal.

Clarity of roles and responsibilities

Roles, responsibilities and accountabilities can be unclear in an ecosystem that has multiple service providers. They may be distributed across several parties, making it harder to identify who should be doing what.

Some customer organization departments may have the primary relationship with a service provider; for example, the payroll department may work directly with an external payroll service provider. Some service providers may need to have relationships with multiple customer organization departments; for example, a hosting provider may need to coordinate with operations, the engineering department and the application development department. Without effective governance and coordination, a culture can develop where there is no ownership of issues, leading to customer dissatisfaction and loss of perceived value.

For example, a user might frequently experience slow performance in a business service that relies on multiple technical services from different service providers. Each service provider reports that its own service is performing within expected parameters, suggesting that the cause lies with another service provider. Without clear roles, responsibilities, and escalation paths, the issue remains unresolved and the customer experience suffers.

Slow pace of change

Customers expect changes to be made quickly to meet business requirements.

They also expect that new services, new service providers, and new technologies can be introduced rapidly and integrated with existing services to meet demanding timescales.

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Demonstration of value

Customers expect that services will deliver the outcomes they require, at a reasonable level of cost and quality. In many organizations, the service department is not able to demonstrate this value to the customer.

Lack of collaboration between service providers

As the number of parties involved in service delivery increases, so does the need for collaboration.

The requirement is no longer just about a one-way relationship between a service provider and its customer, but a network of relationships between multiple service providers who all need to work together to provide a customer-focused service.

External service providers have their own commercial interests and drivers, which can conflict with the goals of the customer organization and other service providers.

An example of this is where a business service received by the customer organization relies on the integration of several services from different service providers. An individual service provider may only be concerned with the availability of the service elements for which it is responsible.

If a service provider does not consider how its service interacts with other providers' services, it could make changes that stop the integrated service working.

Delivery silos

Delivery silos can exist where there are multiple internal or external service providers. Each service provider focuses only on its own goals and outcomes.

These silos isolate service providers, processes and departments. Their impact includes:

- Duplication of work
- Lack of knowledge sharing
- Increased cost of service provision
- Potential for degraded service performance
- Inability to identify service improvements.

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A culture of blame can arise between the service providers due to the lack of collaboration between silos. When a service is faulty, each silo focuses on proving it is not at fault, rather than working with other silos to correct the fault.

1.5.2.2. Service and sourcing landscape drivers

These are drivers related to the nature, number, and types of services and service providers, and the complexity of the interactions between them.

There are five drivers related to the service and sourcing landscape:

1. External sourcing
2. Multi-sourcing
3. Increase in the number of service providers
4. Shadow IT
5. Inflexible contracts.

External sourcing

Many of the traditional frameworks and practices used to manage services were designed for an environment where most of the services were developed and supported internally. However, the way that many organizations source their services has fundamentally changed.

Rather than the former insourced approach, many organizations have made the strategic decision to source applications and services externally.

External sourcing of services may enable a customer organization to reduce costs by realizing the benefits of competition amongst a wider network of service providers. This sourcing approach can also provide the customer organization with access to best in class capabilities.

The service landscape typically includes both customized and standardized services. Regardless of type, the customer organization expects appropriate integration across all services it consumes.

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Multi-sourcing

Many organizations have made a strategic decision to transition from single-sourcing to multi-sourcing and multiple delivery channels.

This transition often results in a mix of internal and external sourcing. Multi-sourcing can reduce many of the risks and issues associated with being over-reliant on a single service provider. These risks include:

- Slow pace of change and low levels of innovation
- High cost of services when benchmarked against competitors
- Reliance on specific technology platforms
- Inability to take advantage of new service offerings, service providers or technologies that are available elsewhere
- Long-term contractual restrictions
- Lack of control over services
- Lack of service knowledge in the customer organization
- High risk to service continuity during a transition to a new service provider
- Cost of the transition to a new single service provider
- A risk that the service provider may go out of business.

Increase in the number of service providers

The number of service providers in the market is increasing. More and more options are available to customer organizations that are evaluating different sourcing approaches.

Shadow IT

Shadow IT describes IT services and systems commissioned by business departments, without the knowledge of the IT department (sometimes referred to as 'stealth IT').

These services are commissioned to meet a business requirement, but can cause problems when they require connectivity and alignment with the other services consumed by the customer.

Inflexible contracts

Lengthy, inflexible contracts with service providers lock in customers and prevent them accessing technology developments and innovative practices.

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Moving to a SIAM model will typically include shorter, more flexible contracts that allow customer organizations to add and remove service providers, and adapt how they work with existing service providers.

1.5.2.3. Operational efficiencies drivers

These are drivers that relate to improvements and efficiencies for the end-to-end delivery of services, and the potential to create operational efficiencies through standardization and consolidation.

There are four drivers related to operational efficiencies:

1. Disparate service management capabilities
2. Data and information flows
3. Data and information standards
4. Tooling.

Disparate service management capabilities

In an ecosystem with multiple service providers, each will maintain its own service management capability. The customer organization will also need to retain service management capabilities, which interact with the service providers.

This can result in:

- Duplication of resources and activities
- Low utilization in some areas and high utilization in others
- Inconsistent levels of capability and maturity
- No sharing of knowledge
- Inconsistent processes and procedures
- A blame culture between teams.

These can result in increased costs and degraded service performance for the customer organization.

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Data and information flows

In an ecosystem with multiple service providers, data and information will be transferred between parties during end-to-end service delivery.

If the data and information flows are not mapped and understood, the flow can be interrupted, leading to service performance issues and operational inefficiencies.

The 'integration' element of the SIAM methodology manages the service from end-to-end. This requires an understanding of all data and information sources and interactions between all parties.

Mapping data and information flows provides an insight into the boundaries between the different service providers. This knowledge can then be used to create integrated flows of data and information. SIAM is then used to manage and coordinate these flows. This enables end-to-end delivery of the required level of service to the customer organization.

Data and information standards

If data and information standards are not consistent across all service providers, then extra effort will be required when data and information are exchanged between service providers and with the customer organization. A common data dictionary, introduced as part of an integrated approach to service management, could include:

- Incident severity, categorization and recording
- Service levels and service reporting
- Requests for change
- Capacity and availability recording
- Management report formats
- Knowledge artefacts.

Tooling

Service providers will have their own toolsets to support their internal processes. When there is a requirement to exchange data and information with other providers and the customer organization, lack of integration between toolsets can create problems.

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Without a design for interoperability, these exchanges can be inefficient, leading to:

- Re-entry of data and information by the receiving party (the 'swivel chair approach')
- A requirement to translate data and information
- Inadvertent alteration of data and information
- Loss of data and information
- Time delays in the exchange between the parties, resulting in a poor service experience.

The swivel chair approach

The 'swivel chair approach' is a colloquial term for manually entering data into one system and then entering the same data into another system. The term is derived from the practice of the user turning from one system to another using a swivel chair.

1.5.2.4. External drivers

These are drivers that are imposed from outside the organization and thus beyond their control (for instance, changes to legislation, geopolitical events, or economic factors). The organization must respond to these drivers in some way.

There are two drivers related to external factors:

1. Corporate governance
2. External policy.

Corporate governance

Many customers have corporate governance requirements that demand clarity over the responsibilities of service providers and the controls that are applied to them. An example is the General Data Protection Regulation (GDPR⁸) which came into effect in 2018 to protect personal data within the European Union (EU) and imposes obligations on organizations anywhere in the world to ensure they handle personal data of EU residents in a compliant manner.

⁸ Source: <https://gdpr.eu/what-is-gdpr>

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Effective corporate governance fulfills externally driven compliance requirements and requires a definition of roles, responsibilities, accountabilities and interactions between all parties and systems at a more granular level than in the past.

External policy

For some organizations, the use of SIAM is mandated through a policy created outside the customer organization.

Policy drivers apply to:

- Public sector organizations affected by government or state policies
- Public sector service providers affected by government or state policies
- Private sector organizations that are part of a larger group that has adopted SIAM as part of its strategy.

1.5.2.5. *Commercial drivers*

These drivers apply to organizations who want to offer commercial services related to SIAM.

There are two drivers related to commercial factors:

1. Service providers
2. Service integrators.

Service providers

When a customer organization adopts SIAM, it will need its service providers to align to its SIAM model.

The delivery models of some service providers will not align with SIAM models, because they do not include the requirements for integration with other service providers and a service integrator.

If these service providers want to be able to compete for business in SIAM ecosystems, they must make changes to how they deliver their services.

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Changes may affect:

- Tooling
- Processes and procedures
- Process interfaces
- Data dictionaries and standards
- Service reporting
- Governance approaches
- Data and information standards
- Commercial and contractual standards.

Service integrators

Some organizations want to provide service integration capabilities to customer organizations. They might act as an externally sourced or hybrid service integrator, or they might provide specialist support during one or more stages of the SIAM roadmap:

- Discovery & Strategy
- Plan & Build
- Implement
- Run & Improve.

1.6. Value to the organization – the SIAM business case

Any organization that is considering a transition to SIAM needs to understand the expected benefits. Clarity on these benefits will form the basis for developing the organization's business case for SIAM.

Benefits can be a mixture of tangible (for example: cost savings) and intangible (for example: improved customer service).

The benefits and costs will be different for each organization. They depend on many factors, including:

- Drivers
- Required business outcomes
- Services in scope
- The customer organization's role in the SIAM ecosystem
- Budget
- Organizational culture

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- Appetite for risk
- The legacy contracts in place and their flexibility to accommodate new ways of working.

The costs that will be incurred need to include not just cost of service under a SIAM model, but also the cost of the transition project to achieve the change. There will also be costs associated with developing any capabilities or artefacts, which the organization does not currently have, but that will be required to operate within a SIAM ecosystem.

An organization should consider its own drivers to achieve the necessary clarity for the anticipated business benefits.

There are generic benefits that are likely to be relevant to most organizations making the transition to SIAM.

The benefits can be placed into four groups:

1. Improved service quality
2. Optimized costs and increased value
3. Improved governance and control
4. Improved flexibility and pace.

When defining the expected benefits, organizations should consider how long it will take for them to be delivered. It can be some time after the transition is complete before benefits start to be realized.

A SIAM model leverages experience and input from multiple service providers. It delivers benefits from collaboration between service providers, and from competitive tension between them, and these factors should be incorporated in the business case and rationale for SIAM.

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1.6.1. Improved service quality

Improving service quality often forms part of a SIAM business case. Benefits related to service quality can include:

- A shift in focus from satisfying contractual targets to focus on innovation and satisfying perceived business need
- Consistent achievement of service levels, including end-to-end:
 - Incident and problem resolution times
 - Service availability
 - Service reliability
 - Service experience.
- Improvements in customer satisfaction with the services
- The customer organization can concentrate on delivering its business outcomes, and have confidence in its supporting services
- Improved quality in the delivery of changes, integrated across service providers
- Improved flow of data and end-to-end processes
- Consistency in how end users interact with service providers
- Consistent and understandable management information about the services
- Access to best of breed services and service providers
- Development and sharing of knowledge and best practice
- Continual service improvement.

1.6.2. Optimized costs and improved value

The business case for SIAM must include the costs associated with the transition to a new way of working. The service integrator layer can add additional cost to an organization, whether it is sourced externally or provisioned using internal resources.

However, the increased value associated with a transition to SIAM, and the potential for cost optimization in the service provider layer, may balance out or exceed any overall cost increases, and should be factored into the business case.

If SIAM is correctly designed and implemented, it will provide better service value, with both tangible and intangible benefits.

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Benefits in this group include:

- Cost optimization from:
 - Innovation
 - An understanding of the true cost and value of each service and service provider
 - Competitive tension between service providers
 - Best use of skilled (and often scarce) resources
 - Reduced costs of process execution
 - Identification and removal of duplication of resources and activities
- Improved value for money for individual services
- Consistent performance from all service providers, leading to improved efficiency
- Improved management of resources and capacity
- Faster response to changing business needs
- Faster access to new technologies and services
- Contract optimization and the potential for shorter term, more effective contracts
- Flexibility to accommodate change.

1.6.3. Improved governance and control

SIAM provides an opportunity to apply consistent governance and control over all service providers, both internal and external.

Governance and control benefits include:

- Consistent and visible definition and application of a governance framework
- Consistent assurance of services and service providers
- A single point of ownership, visibility, and control of services
- Clearly defined services, roles, responsibilities, and controls
- Improved management of service provider performance
- The ability to benchmark between service providers
- Contract optimization and standardization related to governance and control
- Improved visibility, understanding and management of service risks.

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1.6.4. Improved flexibility

If correctly designed and implemented, SIAM can provide the flexibility that is necessary to support changing business requirements, balanced with an appropriate level of control.

The benefits in this group include:

- Effective and timely introduction of new and changed services and service providers
- The flexibility to replace poorly performing or uneconomic service providers
- The ability to rapidly accommodate changes to services, technologies, and business requirements
- Increased ability to manage commodity services in a consistent way
- Increased ability to scale service provision.

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2. SIAM roadmap

The SIAM roadmap outlines an example plan for the implementation of SIAM as part of an organization's operating model. It describes the high-level stages and activities required to create and transform to a SIAM model.

Using a roadmap for the implementation has several benefits, including:

- Defining the SIAM requirements
- Providing a planning framework
- Determining the most appropriate SIAM structure and SIAM model
- Guiding the implementation
- Directing ongoing continual improvement.

There are four stages in the SIAM roadmap:

1. Discovery & Strategy
2. Plan & Build
3. Implement
4. Run & Improve.

For each stage, this section provides examples of:

- Objectives
- Triggers
- Inputs
- Activities
- Outputs.

Whilst the activities are presented here in a sequential manner, many are likely to be iterative or may even be undertaken in parallel.

High-level requirements are defined in the first, Discovery & Strategy stage. These are then further developed and designed in the second, Plan & Build stage, before being implemented in the third stage, Implement. The fourth stage, Run & Improve, is where the SIAM model is operated and continually improved.

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Each stage has a checkpoint at the end, which should review areas including:

- The actual outputs from the stage against those intended
- Risks
- Issues
- The plan for the next stage.

This information should be used to validate decisions taken earlier in the roadmap. It might highlight potential issues, requiring a return to an earlier stage for further work.

An example of an iterative roadmap

In the Discovery & Strategy stage, a customer organization might propose an internally sourced service integrator.

In the second stage (Plan & Build), it formulates a plan and designs its SIAM model to support this structure.

However, during the third stage (Implement), it discovers that it does not have resources with the required capabilities.

It returns to the first stage to review its strategy, and changes it to apply the hybrid service integrator structure instead. The Plan & Build stage must then be revisited.

Many organizations use outside assistance during the execution of their SIAM roadmap. This can be helpful during the transition to SIAM, but the customer organization needs to ensure that the model being used by the external organization is suitable for its needs.

If outside help is required, it is a good idea to have a commercial boundary between an organization that is assisting with the Discovery & Strategy and Plan & Build stages, and any external service integrator.

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2.1. Discovery & Strategy

2.1.1. Objectives

The Discovery & Strategy stage initiates the SIAM transformation project, formulates key strategies, and maps the current situation. This enables the customer organization to:

- Determine what it intends to source internally
- Consider any additional skills and resources that may be required
- Determine what it would like to source externally
- Understand the expected benefits.

The objectives for this stage are to:

- Establish the SIAM transition project
- Establish a governance framework
- Define the strategy and outline model for SIAM and the services in scope
- Analyze the current state of the organization, including skills, services, service providers, tools and processes
- Analyze the marketplace for potential service providers and service integrators.

2.1.2. Triggers

There are many reasons for organizations to consider adopting a SIAM model. These drivers are described in section **1.5.2 Drivers for SIAM**.

2.1.3. Inputs

Inputs to this stage include:

- Enterprise, corporate, and IT governance standards
- Current business, procurement and IT strategies
- Business requirements and constraints
- Current organization structure, processes, products and practices
- Existing service provider information, including existing contracts and agreements
- Understanding of market forces and technology trends.

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2.1.4. Activities

The activities in this stage are:

1. Establish the project
2. Define strategic objectives
3. Define governance requirements and the high-level governance framework
4. Define principles and policies for roles and responsibilities
5. Map the existing services and sourcing environment
6. Assess the organization's current maturity and capability
7. Understand the marketplace
8. Define the strategy for SIAM and the outline SIAM model
9. Produce the outline business case.

2.1.4.1. Activity: Establish the project

The SIAM transformation project should be formally established using the organization's selected project management methodology.

This includes:

- Setting up a project management office
- Defining roles and responsibilities for the project
- Setting up project governance
- Agreeing the approach for managing project risks.

The organization will also choose the project delivery approach, for instance, waterfall or Agile.

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2.1.4.2. Activity: Define the strategic objectives

Strategic objectives are the long-term goals of the organization that SIAM is intended to support.

They are related to the drivers for SIAM and the SIAM business case. The objectives defined and agreed in this activity will be used as a basis for items including the:

- SIAM model
- SIAM governance framework
- Sourcing model
- Roles and responsibilities.

2.1.4.3. Activity: Define the governance requirements and high-level governance framework

SIAM requires a specific governance framework that allows the customer organization to exercise and maintain authority over the SIAM ecosystem.

The model should be tailored to the specific SIAM structures, the SIAM model, and the customer organization's overall appetite for risk.

At this stage, the SIAM governance framework will be defined at a high-level. It should include:

- Specific corporate governance requirements that support any external regulations and legal requirements
- Controls to be retained and operated by customer organization
- Definition of governance boards and governance board structures
- Segregation of duties between the customer organization and external organizations
- Risk management approach
- Performance management approach
- Contract management approach
- Dispute management approach.

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2.1.4.4. Activity: Define principles and policies for roles and responsibilities

In this activity, the key principles and policies for roles and responsibilities are created. They will need to take into account the governance requirements and strategic objectives.

Specific roles and responsibilities will not be defined or assigned until more detailed process models and sourcing agreements have been designed in the Plan & Build stage.

Two aspects should be considered here:

1. Segregation of duties if one organization is operating in more than one SIAM layer
2. Boundaries of delegated authority.

2.1.4.5. Activity: Map the existing services and sourcing environment

Before a SIAM model can be designed, the current environment must be understood. This includes:

- Existing services and the service model
- Existing service providers (internal and external)
- Contracts
- Service provider performance
- Relationships with service providers
- Cost of services.

The creation of the service model is a critical activity to support the design of the desired future state. The service model enables the identification of essential business functions, critical service components and dependencies across the ecosystem.

This activity will provide clarity on the current environment. It can also help to highlight issues including:

- Duplications in service offerings
- Misaligned contractual commitments
- Unused operational services
- Uneconomic services
- Service risks that require mitigation.

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Information about service providers can be used to decide whether they are to be retained in the current format, or whether their services should be sourced under new arrangements.

2.1.4.6. *Activity: Assess current capability and maturity*

Capability and maturity

Capability: *The power or ability to do something*⁹

Maturity: relates to the degree of formality and optimization. For example, process maturity can range from ad hoc practices, to formally defined steps, to managed and measured results, to active optimization of the processes.¹⁰

Both capability and maturity need to be assessed to inform the strategy for SIAM.

For example: a customer organization may currently have low maturity in service integration processes, practices, and tools; but have a high capability in these areas. This may influence its choice of preferred SIAM structure, leading it to select an internally sourced service integrator.

A baselining exercise should be carried out to understand the customer organization's current capability and maturity in organization, processes, practices and tools. This will inform the next stage of the roadmap.

This exercise can also identify any issues that require a review of earlier decisions. For example, where there is insufficient capability to run the project management office; or insufficient maturity of the incident management process.

⁹ Source: Oxford English Dictionary

¹⁰ Source: Capability Maturity Model (CMM)

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2.1.4.7. Activity: Understand the marketplace

It is important at this stage to understand the availability and capabilities of potential external service integrators and service providers. This will inform the strategy for SIAM and the SIAM model.

This activity should include a review of available technologies and services against the strategic objectives. For example, legislation or corporate policy could limit the use of technology to store data off shore.

Some service providers will not take part in the SIAM model's boards, forums and working groups. This will impact the workload of the service integrator, and needs to be considered during the selection of the service integrator structure.

2.1.4.8. Activity: Define the strategy for SIAM and the outline SIAM model

This activity will take the information and outputs from previous activities in this stage to define the strategy for SIAM, and an outline SIAM model.

The information and outputs should include:

- Strategy for SIAM:
 - The vision for SIAM
 - Strategic objectives
 - Current maturity and capability
 - Existing services and sourcing environment
 - Marketplace analysis
 - Governance requirements
 - Proposed SIAM structure, including retained capabilities
 - Proposed sourcing approach
 - Justification for proposals.
- Outline SIAM model:
 - Principles and policies
 - Governance framework
 - Outline roles and responsibilities
 - Outline of process models, practices, and structural elements
 - Outline of services
 - Service providers to be retired.

The strategy for SIAM and the chosen SIAM model both need to align with the original business requirements and the business strategy.

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2.1.4.9. Activity: Produce the outline business case

This activity will take the information and outputs from all previous activities in this stage to produce an outline business case for SIAM.

This should include:

- Strategy for SIAM
- Outline SIAM model
- Current state
- Expected financial benefits
- Expected non-financial benefits
- Risks
- Outline costs of the transition to a SIAM model
- Outline costs for operation of a SIAM model
- High-level plan.

The outline business case should be approved in accordance with the customer organization's governance arrangements before the Plan & Build stage begins.

2.1.5. Outputs

The outputs from the Discovery & Strategy stage are:

- An established SIAM transition project
- Strategic objectives
- Governance requirements and high-level SIAM governance framework
- Defined principles and policies for roles and responsibilities
- Map of existing services and sourcing environment
- Current maturity and capability levels
- Market awareness
- Approved outline business case for SIAM
- Strategy for SIAM
- Outline SIAM model.

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2.2. Plan & Build

2.2.1. Objectives

The Plan & Build stage builds on the outputs from the Discovery & Strategy stage to complete the design for SIAM and create the plans for the transformation.

During this stage, all plans and approvals are put in place before the Implement stage begins. The main objectives for this stage are to:

- Complete the design of the SIAM model, including the services that are in scope
- Obtain full approval for the SIAM model
- Appoint the service integrator and service providers
- Commence organizational change management.

2.2.2. Triggers

This stage is triggered on completion of the Discovery & Strategy stage, when the organization confirms its intention to proceed with a SIAM implementation.

2.2.3. Inputs

The inputs to this stage are the outline business case, and the high-level model and frameworks created during the Discovery & Strategy stage:

- Governance requirements and high-level SIAM governance framework
- Defined principles and policies for roles and responsibilities
- Map of existing services and sourcing environment
- Current maturity and capability levels
- Market awareness
- Approved outline business case for SIAM
- Strategy for SIAM
- Outline SIAM model.

In this stage, work will be carried out to further define, refine, and add detail to the outputs from the previous stage. Some organizations may choose to use an agile, iterative approach for this.

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2.2.4. Activities

The activities during this stage are:

1. Design the detailed SIAM model
2. Approve the full business case
3. Commence organizational change management
4. Appoint the service integrator
5. Appoint service providers
6. Plan for service provider and service retirement
7. Review stage and approve implementation.

2.2.4.1. *Activity: Design the detailed SIAM model*

The SIAM model provides the detail for how SIAM will be applied across all parties in the SIAM ecosystem. It contains many elements, including:

- Service model and sourcing approach
- The selected SIAM structure
- Process models
- Governance model
- Detailed roles and responsibilities
- Performance management and reporting framework
- Collaboration model
- Tooling strategy
- Ongoing improvement framework.

Careful design of this model is critical to success. The design activities will not necessarily be sequential. There is more likely to be an iterative cycle, which starts with an initial definition, and becomes successively more detailed as each iteration is agreed.

There must be regular review and feedback across all the design activities. Consideration must also be given to interdependencies between the different design activities.

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Organizations will determine the level of detail they require for their own SIAM model. This will depend on several factors, including:

- Strategic objectives
- Market conditions
- Services and service complexity
- Number of service providers
- Appetite for risk
- Resource and process capability and maturity
- Available tools
- Budget.

2.2.4.1.1. Define service model and sourcing approach

This activity defines the services in scope for the SIAM model, the service models, and how the services will be sourced. Services may be sourced as a single entity, or disaggregated into components that are then sourced from one or more service providers. Creating the service model is a critical activity for an effective transition to SIAM (see section **2.1.4.5 Activity: Map the existing services and sourcing environment**).

These areas must be clearly defined for each service:

- Service outcomes, value, and objectives
- The service provider(s)
- The service consumer(s)
- The service characteristics, including service levels
- The service boundaries
- Dependencies with other services
 - Technical interactions with other services
 - Data and information interactions with other services.

Services or service components may be placed into groups, with groups assigned to specific service providers. The service model shows the hierarchy of the proposed services, and the service provider for each service or service component (see section **1.1.1.4 Service provider**). This service model forms part of the overall SIAM model.

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The contents and sizes of the service groups will vary. Some service providers will be assigned to provide a 'group' containing multiple services or service components (e.g. service desk and end user support), whilst other service providers will provide a single service or service components.

Grouping services and service components has both advantages and disadvantages, as shown in **figure 8**. For this reason, care should be taken when defining the service groups and assigning them to service providers. The number of contact points, interactions, and opportunities for failure, will increase as the number of service groups and service providers increases.

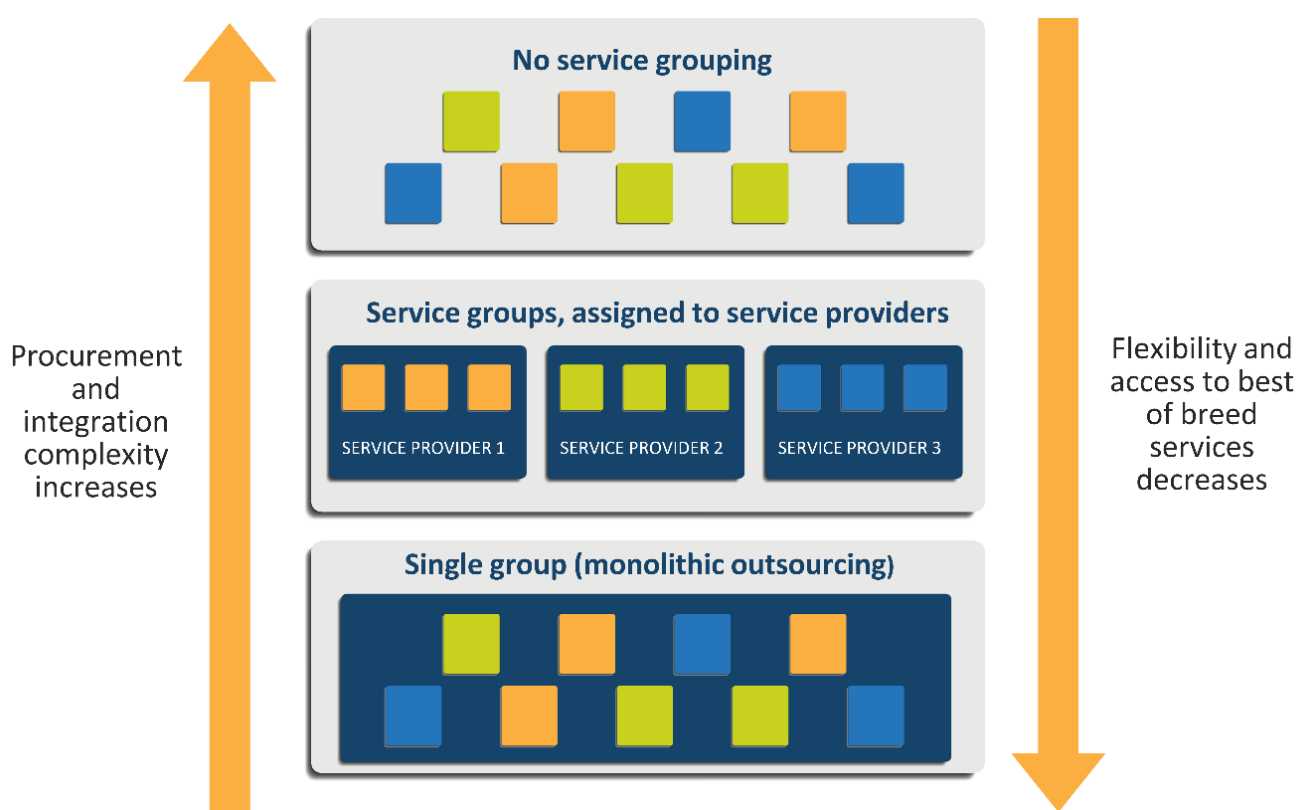


Figure 8: The advantages and disadvantages of service grouping

The service model should also include the expected interactions between the service components, services, groups and service providers. The service model will help to identify omissions, single points of failure, and duplication.

The aim should be to achieve a balance between getting 'best of breed' services, the number of service groups and service providers, and the complexity of the service model and hierarchy. There also needs to be a balance between service complexity and integration complexity. Services

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should be designed to minimize interactions with other service groups, as these interactions drive complexity, risk, and cost.

Service orchestration

Service orchestration refers to the coordination and alignment of people, processes, tools, and providers to deliver a seamless, end-to-end service experience.

Orchestration is achieved by:

- Defining clear interfaces and touchpoints between service providers.
- Establishing and maintaining integrated workflows and data flows.
- Managing effective onboarding and offboarding of service providers and roles to support continuity, knowledge transfer, and rapid stabilization.
- Aligning all providers and capabilities to customer outcomes and priorities.

The ability to source services in groups is one of the benefits of SIAM. Rather than having a single, monolithic contract with one service provider delivering everything, the full range of services can be broken down into the most efficient and best value groupings. Each group is then individually sourced, externally or internally.

Examples of service groups include:

- Service desk
- Desktop support/end user computing
- Networks
- Cloud services
- Managed services.

The design of service groups should try to minimize any dependencies between services. Dependencies create interactions between service providers and potential points of failure, and can increase the workload of the service integrator.

There is no requirement within the SIAM management methodology to separate services that logically stay together. For example, there is no need to divide a Software as a Service (SaaS) offering into 'hosting' and

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'application development and support' if it is more logical to source it as one group. Unnecessary separation can cause issues, such as disputes about who is responsible for performance problems.

There is no limit to the number of different groups within a SIAM model. However, integration complexity will increase as the number of service groups increases.

One of the most important responsibilities of the service integrator is to provide service orchestration across the entire SIAM ecosystem, facilitating collaboration while ensuring consistent adherence to processes and governance. This activity creates the structure, rhythm, and shared direction that enables all parties to work effectively together.

2.2.4.1.2. Select the SIAM structure

The selected SIAM structure determines the sourcing approach for the service integrator. This is a crucial decision that must be taken with care, as any changes to the structure after this point will result in re-work and cost.

All the information gathered so far should be used to select the preferred SIAM structure. If this is different from the proposal created during the Discovery & Strategy stage, it may be necessary to repeat parts of that stage.

See section **3 SIAM structures** for more information on the advantages and disadvantages of each structure.

2.2.4.1.3. Design process models

In a SIAM model, the execution of most processes will involve multiple service providers. Each service provider might carry out individual steps in a different way, but they must align with the overall integrated process model.

End-to-end process models are important SIAM artefacts. Individual processes and work instructions are likely to remain within the domain of the individual service providers.

The process model for each end-to-end process should describe:

- Purpose and outcomes
- High-level activities
- Inputs, outputs, interactions and dependencies with other processes

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- Inputs, outputs, and interactions between the different parties (for example, between the service providers and the service integrator)
- Controls
- Measures and key performance indicators
- Supporting policies and templates.

Techniques such as swim lane models, RACI matrices, and process mapping are commonly used, and are helpful for establishing and communicating process models.

The process models will continue to evolve and improve as further activities are undertaken in this stage, and in the Run & Improve stage. This includes getting input from the selected service providers and service integrator.

Adding granularity

The iterative design and development of the SIAM structure, services and service groups, roles and responsibilities, governance model, process models, performance management and reporting framework, collaboration model, tooling strategy and ongoing improvement framework, all add detail to the SIAM model.

This detailed work and iterative approach is critical to ensure that the SIAM model will work once implemented (and often uncovers working practices not previously considered), and that it aligns with the strategy for SIAM and the customer organization's requirements as they change over time.

2.2.4.1.4. Design governance model

The governance model should be designed using the governance framework and the roles and responsibilities. For each governance body, this model should include:

- Scope
- Accountabilities
- Responsibilities
- Meeting formats
- Meeting frequencies
- Inputs
- Outputs (including reports)
- Hierarchy

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- Terms of reference
- Related policies.

The governance model should also be updated and more detail added as the Plan & Build activities progress. The design of the model is an iterative activity that should be completed before the end of this roadmap stage.

2.2.4.1.5. Design roles and responsibilities

Roles and responsibilities should be designed using the outline SIAM model, process models, the SIAM structure and the governance model.

This should include the detailed design and allocation of roles and responsibilities to:

- Process models
- Practices
- Governance boards
- Forums
- Working groups
- Organizational structures and locations for any retained capabilities.

This work may highlight a need to review earlier designs and decisions.

Roles and responsibilities can be further developed in the Run & Improve stage, but the detail must be confirmed in this stage before any service integrator or service providers can be appointed.

2.2.4.1.6. Design performance management and reporting framework

The performance management and reporting framework for SIAM addresses measuring and reporting on a range of items including:

- Key performance indicators
- Performance of processes and process models
- Achievement of service level targets
- System and service performance
- Contractual and non-contractual obligations
- Collaboration
- Customer satisfaction.

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Measurements should be taken for each service provider and its services, but also across the end-to-end SIAM ecosystem.

Designing an appropriate performance management and reporting framework for a SIAM ecosystem can be challenging. It is usually straightforward to measure the performance of an individual service provider; the challenge is in measuring end-to-end performance as experienced by the users, particularly when there may be limited consistency in how each of the providers measure and report.

The framework should also include the standards for:

- Data classification
- Reporting formats and frequency.

2.2.4.1.7. Design collaboration model

SIAM can only be effective when service providers, the service integrator and the customer can communicate and collaborate with each other.

Section **6 SIAM cultural considerations** has some examples of how to encourage collaboration in SIAM ecosystems.

2.2.4.1.8. Define tooling strategy

A consistent and comprehensive tooling strategy is important within a SIAM ecosystem. The tooling strategy is influenced by:

- The selected SIAM structure
- The SIAM model
- Existing customer organization toolsets and architecture
- Service provider and service integrator toolsets
- Types of service provider
- Budget.

The tooling strategy should focus on enabling effective collaboration, supporting the efficient flow of data and information and ensuring process integration:

- Between the service providers
- Between service providers and the service integrator
- Between the service integrator and the customer organization.

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It must address people, processes, security and governance considerations as well as technology, as these are critical to achieving integration and shared outcomes.

Many organizations use more than one toolset in their SIAM ecosystem, selecting a range of 'best of breed' toolsets for:

- Service management processes
- Data analysis
- Reporting and presentation
- Event monitoring
- Audit and evidence logging.

There are four main options for toolsets:

1. A single toolset is used by all parties, mandated by the customer organization
2. The service providers use their own toolsets and integrate them with the service integrator or customer organization's toolset
3. The service providers use their own toolsets and the service integrator integrates them with its own or the customer organization's toolset
4. An integration service is used to incorporate the toolsets of the service providers and the service integrator or customer organization.

The tooling strategy should include:

- Enterprise architecture
- Functional and non-functional requirements
- Integration requirements (technical and logical)
- Data mapping for each SIAM layer
- Data ownership
- Access control
- Measurement and reporting.

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2.2.4.1.9. Design ongoing improvement framework

An improvement framework needs to be developed and maintained in conjunction with all parties within the SIAM model. This will ensure a focus on continual improvement across the SIAM ecosystem.

It should include structured processes for identifying, prioritizing, and implementing improvements, with clear governance, measurable outcomes, and alignment to business objectives. Service providers should have incentives that encourage them to suggest and deliver improvements and innovation.

Improvement vs. innovation

Improvement: *the process of making something better or of getting better*

Innovation: *a new idea or method*¹¹

Improvement makes existing things better, innovation introduces new things.

2.2.4.2. Activity: Approve full business case

At this point, the design should be detailed and complete enough to enable the full costs of the SIAM transition, ongoing operation and the anticipated benefits to be determined.

The outline business case should be reviewed and updated with detailed information to create a full business case.

This should then be approved using the organization's corporate governance and approvals process. The approval allows the start of procurement activities for any external service providers, service integrator, and tools.

¹¹ Source: Cambridge online dictionary (<https://dictionary.cambridge.org>)

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2.2.4.3. *Activity: Commence organizational change management*

A SIAM transformation is a major business change, affecting the customer organization, service integrator and service providers at every level.

Organizational change management (OCM) will be essential if the transformation is to succeed. For this reason, OCM plans need to be made as a part of the Plan & Build stage, and early activities commenced, before any changes are actually implemented.

During any organizational change, it is important to protect the existing service(s) and minimize the impact on the existing organization.

2.2.4.4. *Activity: Appoint the service integrator*

Ideally, the service integrator should be selected and in place before the SIAM model is finalized and before any service providers are selected.

If this can be achieved, the service integrator can be involved with the Plan & Build activities. The benefits of this approach are:

- The service integrator is involved with the design of service groups and can provide input into selection criteria for service providers, using its experience to assist with these activities
- The service integrator is fully aware of the requirements placed on the service providers during selection and appointment.

The selection process and contractual agreement for an external service integrator may take some time. On occasion, the customer might source the service integrator and the service providers simultaneously.

Alternatively, some service providers might already be in place or undergoing transition from legacy contracts before the service integrator role is confirmed.

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2.2.4.5. *Activity: Appoint service providers*

Service providers cannot be selected until this point, as it will not be possible to fully document the requirements until the SIAM model has been fully defined.

The contracts in place in a SIAM model need to support the overall strategy for SIAM. It is important to ensure that they include appropriate targets and risk and reward models. Detailed requirements should be included in any contracts or internal agreements.

Commodity Services

Where commodity service providers (see section **1.4.2**) have been selected, requirements often need to be adjusted to consider that these are less likely to be aligned to the SIAM model requirements.

For example, a commodity service provider for a cloud service is unlikely to take part in boards, forums or working groups, to change its processes, or to integrate its toolsets with others.

The challenge is to balance the customer organization's requirements against what is offered in the marketplace. Forcing service providers to customize their delivery models can result in increased costs and risks.

The procurement process to appoint external service providers can take some time, which needs to be factored into any project plan or timeline.

It is important to verify that the desired service providers can meet the full set of requirements in the SIAM model, particularly for strategic service providers. If there are issues or gaps, this may require a return to earlier lifecycle stages and activities.

In addition to the service providers that are appointed here, it is important to remember that service providers can be added and removed throughout the SIAM roadmap. Some service providers may not be appointed until after a legacy contract has expired, or may be added when a new requirement emerges.

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2.2.4.6. Activity: Plan for service provider and service retirement

Planning also needs to address retiring services and/or service providers, and any resulting transfer of services, responsibilities or obligations to new service providers.

The relationships with any service providers, service dependencies, contract end dates and potential impact of retiring a particular service or service provider must be carefully considered.

Detailed plans should be developed for any decommissioning, discontinuation, and transfer of services. The plans need to include contractual restrictions, legal requirements, and lead times for service termination.

They must also detail how data, information, and knowledge will be transferred from retiring service providers, including:

- What needs to be transferred
- To whom it will be transferred
- When it needs to be transferred
- How to assess if the transfer is successful.

2.2.4.7. Activity: Review stage and approve implementation

The outputs from this stage should be reviewed against decisions taken in the previous Discovery & Strategy stage, to identify if there are any issues or necessary changes. The roadmap will then progress on to the Implement stage if approval is given.

2.2.5. Outputs

The outputs from the Plan & Build stage are:

- Full design of the SIAM model including:
 - Services, service groups, and service providers (the 'service model')
 - The selected SIAM structure
 - Process models
 - Practices
 - Structural elements
 - Roles and responsibilities

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- Governance model
- Performance management and reporting framework
- Collaboration model
- Tooling strategy
- Ongoing improvement framework
- Approved business case
- Planning and early organizational change management activities
- Service integrator appointed
- Service providers appointed
- Plan for service provider and service retirement.

There may be several iterations during this stage before the outputs are complete and the roadmap progresses to the next stage, Implement. The outputs from Plan & Build must be detailed enough to support the implementation activities.

2.3. Implement

2.3.1. Objectives

The objective of this stage is to manage the transition from the organization's 'as is' current state to the 'to be' desired future state. At the end of this stage, the SIAM model will be in place and in use.

2.3.2. Triggers

This stage is ideally only triggered when the organization completes all activities of the Discovery & Strategy and Plan & Build stages.

However, the timing for the start of the Implement stage can be influenced by events in the existing environment. For example, implementation could be triggered by:

- The end of an existing service provider's contract
- An existing service provider ceasing to trade
- Organizational structure changes due to corporate restructure or takeover.

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The customer organization may have limited control over the timing of these events. It may need to react to them by completing as many of the Discovery, Strategy, Plan and Build activities as possible. There will be an increased level of risk if the activities from these stages are not fully completed owing to a lack of time.

2.3.3. Inputs

All the outputs from the Discovery & Strategy and Plan & Build stages form inputs for the Implement stage.

2.3.4. Activities

The activities in this stage focus on making the transition to the new SIAM model. They include:

1. Select the implementation approach
2. Transition to the approved SIAM model
3. Ongoing organizational change management.

2.3.4.1. *Activity: Selecting the implementation approach*

There are two possible high-level approaches to implementation:

1. 'Big bang'
2. Phased.

Within each high-level approach, variations are possible. For example, a 'big bang' approach for most of the services and service providers, followed by a phased approach for small divisions. This might be needed because of complexity in including certain geographies in the 'big bang' approach or the impracticality of the transition of certain service provider contracts during that time, requiring them to be done in a later phase.

2.3.4.1.1. 'Big bang' implementation

A 'big bang' implementation approach is one that introduces everything at once, including: the service integrator, the service providers (with any new contracts) and the new ways of working.

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The 'big bang' approach can be high risk, because it affects the entire organization at the same time. The resulting impact on the customer organization's business and the services provided can be very high, unless the risks are planned for and carefully managed.

Most organizations who adopt SIAM are introducing it into an environment with existing providers, contracts and relationships. This can mean that 'big bang' is not possible, as different contracts expire at different times.

However, a 'big bang' approach does provide an opportunity to make a 'clean break' from all legacy issues and behaviors at the same time and avoids the complexities of managing a phased approach.

2.3.4.1.2. Phased implementation

A phased implementation approach makes the transformation to the new SIAM model in smaller, more easily managed transition projects and iterations. This can be achieved in several ways, including:

- One service at a time
- One service provider at a time
- One practice at a time
- One process at a time.

This phased approach to SIAM implementation can lower the level of risk associated with the transition, but can be more complex to manage and will extend the total time for implementation.

During this time, part(s) of the organization will have been transitioned to the SIAM model, but other parts will still be waiting to transition. Specific care needs to be given to define and understand the impact of each transition and to ensure that the delivery of existing services continues with no disruption.

A phased approach may also be used to 'pilot' SIAM in one area of an organization before extending the transition elsewhere.

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2.3.4.2. *Activity: Transition to the approved SIAM model*

The transition activities will be dependent on the selected approach; phased or 'big bang'.

This activity involves:

- Establishing processes and supporting infrastructure
- Commencing transition activity to new service providers and services
- Removing service providers who are not part of the SIAM model
- Verifying the successful execution of the transition steps
- Toolset and process alignment between all parties.

This is not a trivial activity. The number of service providers, services, processes and toolsets will all affect the complexity of the transition. It involves the transition to the full SIAM model, including the implementation of new:

- Service providers
- Services
- Service integrator
- Process models
- Roles and responsibilities
- Tools
- Practices
- Structural elements
- Contracts and agreements
- Governance framework
- Performance management and reporting framework.

A robust methodology should be used for this transition, including:

- Testing (both functional and non-functional)
- Data migration
- Service introduction
- Deployment testing
- Service acceptance
- Post-transition support.

The transition normally requires resources who are specifically dedicated to and focused on it.

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The service providers selected during Plan & Build will need to be transferred into the SIAM ecosystem as part of the Implement stage.

Existing service providers who are taking on a new role in the SIAM ecosystem will need to fully understand their new role, relationships and interfaces. New service providers will need to undergo onboarding into the ecosystem in a managed way.

This activity should be managed by the service integrator on behalf of the customer. It is vital that clear ownership and roles and responsibilities are agreed, including reporting lines, escalation paths and mandates to ensure efficient and effective decision-making.

2.3.4.3. *Activity: Ongoing organizational change management*

Organizational change management preparations started in the Plan & Build stage of the roadmap. Activities begin in this stage and continue into Run & Improve.

Specific activities in the Implement stage include:

- Conducting awareness campaigns throughout the organization
- Communicating with and preparing stakeholders for the change
- Ensuring appropriate training is completed
- Continuing with deployment of the organizational change plans
- Assisting affected stakeholders during the implementation and early life support
- Measurement of the effectiveness of communications and organizational change activities.

It is important to focus on protecting the existing service and minimizing organizational impact during this stage.

2.3.5. **Outputs**

The output from the Implement stage is the new SIAM model that is in place and operating, and supported by appropriate contracts and agreements.

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2.4. Run & Improve

2.4.1. Objectives

The objectives of the Run & Improve stage include:

- Manage the SIAM model
- Manage day-to-day service delivery
- Manage processes, teams and tools
- Manage the continual improvement activities.

Depending on its scope, an improvement could trigger another iteration of the earlier roadmap stages.

2.4.2. Triggers

This stage is triggered when the Implement stage is completed. If the chosen implementation approach is 'phased', Run & Improve will take over elements of delivery in an incremental way, as each phase, service, process or service provider exits the Implement stage.

2.4.3. Inputs

Inputs to this stage will include:

- The SIAM model
- Process models
- Governance framework, including the structural elements
- Performance management and reporting framework
- Collaboration model for providers
- Tooling strategy
- Ongoing improvement framework.

These inputs have been designed during the Discovery & Strategy and Plan & Build stages, and then transferred during the Implement stage.

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2.4.4. Activities

The activities in this stage focus on providing consistent, guaranteed service outcomes to the customer organization, which can be managed, measured and improved. They include:

1. Operate governance structural elements
2. Performance management and improvement
3. Operate management structural elements
4. Audit and compliance
5. Ongoing change management.

In the Run & Improve stage, the SIAM model should no longer be seen as 'new'; it is just how things are done.

2.4.4.1. Activity: Operate governance structural elements

Governance boards provide an important role in the control of the overall SIAM ecosystem.

During the Plan & Build stage, the high-level governance framework was created. In Implement, it was transferred to the live environment. In Run & Improve, governance board members adopt their new roles.

See sections **4 SIAM roles** and **1 Introduction to Service Integration and Management (SIAM)** for more information about governance boards.

2.4.4.2. Activity: Performance management and improvement

The performance of all services and processes should be measured and monitored against key performance indicators and, where appropriate, service level targets. The measurements should be both qualitative and quantitative.

Service providers must be encouraged to collaborate rather than protect their own interests. Reward mechanisms can be used to encourage collaboration and communication.

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Good practices for creating a reward system include:

- Use small rewards often, linked to specific actions
- Give rewards at unexpected times
- Reward the behavior, not just the results
- Reward all stakeholders, not just one layer of the SIAM model
- Reward publicly.

Case study

One customer organization has created a CIO Award for Collaboration.

This is given quarterly to the service provider who has demonstrated excellent behaviors, including collaboration, willingness to help others, and ease of working with them. The scores are collated and shared with all parties.

Crucially, service providers are encouraged to nominate each other, encouraging them to recognize good behavior within the service provider layer.

Measurements are used to create meaningful and understandable reports for the appropriate audiences. They provide visibility of performance issues, and support trend analysis to give early warning of possible failures.

Routine service improvement activities should include review and management of actions arising from the information and review of report relevance.

Within SIAM, reports also need to include feedback for how the service is perceived by users, referred to as qualitative reporting. For more information see section **5 SIAM practices**.

Reports can be used to identify opportunities for improvement and innovation.

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2.4.4.3. *Activity: Operate management structural elements*

Forums and working groups are two of the structural elements that unite the service integrator, service providers and the customer.

They provide an ecosystem to work collaboratively on the operation of a specific process or processes, process outputs, issue or project.

In this stage of the roadmap, these forums and groups should be actively working. The frequency and format of meetings will vary, but it is a good idea to have regular contact between the forum and group members in the early stages of implementation, as they will be instrumental in creating the necessary collaborative culture.

See sections **4 SIAM roles**, **6 SIAM cultural considerations** and **1 Introduction to Service Integration and Management (SIAM)** for more information about forums and working groups.

2.4.4.4. *Activity: Audit and compliance*

In addition to the review of reports that takes place in a SIAM ecosystem, a more formal audit schedule should also be introduced.

This can include process audits, service audits, service provider audits; whatever is most appropriate for each organization and the SIAM ecosystem. Some audits will be mandated by regulations, legislation or corporate governance.

These audits may be performed by an external organization. Audits support ongoing assurance of compliance to the customer organization's legislative and regulatory requirements. They can provide valuable information about whether elements of the model are working as they should and help to embed a culture of improvement.

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2.4.4.5. *Activity: Ongoing change management*

After the SIAM model enters the Run & Improve stage, it will change and evolve as the sourcing landscape and business requirements change and evolve.

Ongoing change management will include the onboarding and offboarding of service providers, scaling the services if customer organization's needs grow or shrink, and potentially changing the SIAM structure.

If major change is required, this can include going back to earlier roadmap stages, for example to revisit Plan & Build or Discovery & Strategy.

2.4.5. **Outputs**

Outputs from the Run & Improve stage fall into two categories:

1. Run outputs: business as usual outputs including reports, service data and process data
2. Improve outputs: information used to evolve and continually improve the SIAM model.

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3. SIAM structures

SIAM models are typically implemented using one of four common structures. These differ primarily in how the service integrator layer is sourced and configured.

The structures are:

1. Internally sourced
2. Externally sourced
3. Lead supplier
4. Hybrid.

The customer organization will select a structure, and may adapt it at a later stage, based on factors including:

- Business requirements
- Internal capabilities (including maturity, resources and skills)
- Complexity of the customer's services
- Customer organization's structure and size
- Legislative and regulatory environment
- Customer organization budget
- Current organizational maturity and capability in service integration
- Appetite for external sourcing/loss of direct control
- Required timescales
- Appetite for risk
- Types and numbers of service providers to be managed.

3.1. Internally sourced service integrator

In this SIAM structure, the customer organization takes the role of service integrator, providing the service integration capability. The service integrator role and the customer organization role still need to be defined and managed separately.

If the customer organization role and the service integrator role become indistinguishable, service providers may interact with the customer organization as if it was part of a traditional outsourced ecosystem, undermining the intent and preventing the realization of the benefits of moving to a SIAM model.

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The service provider roles are performed by external service providers and/or internal service providers.

The internally sourced service integrator is exclusively focused on service integration activities.

Figure 9 shows the internally sourced service integrator structure.

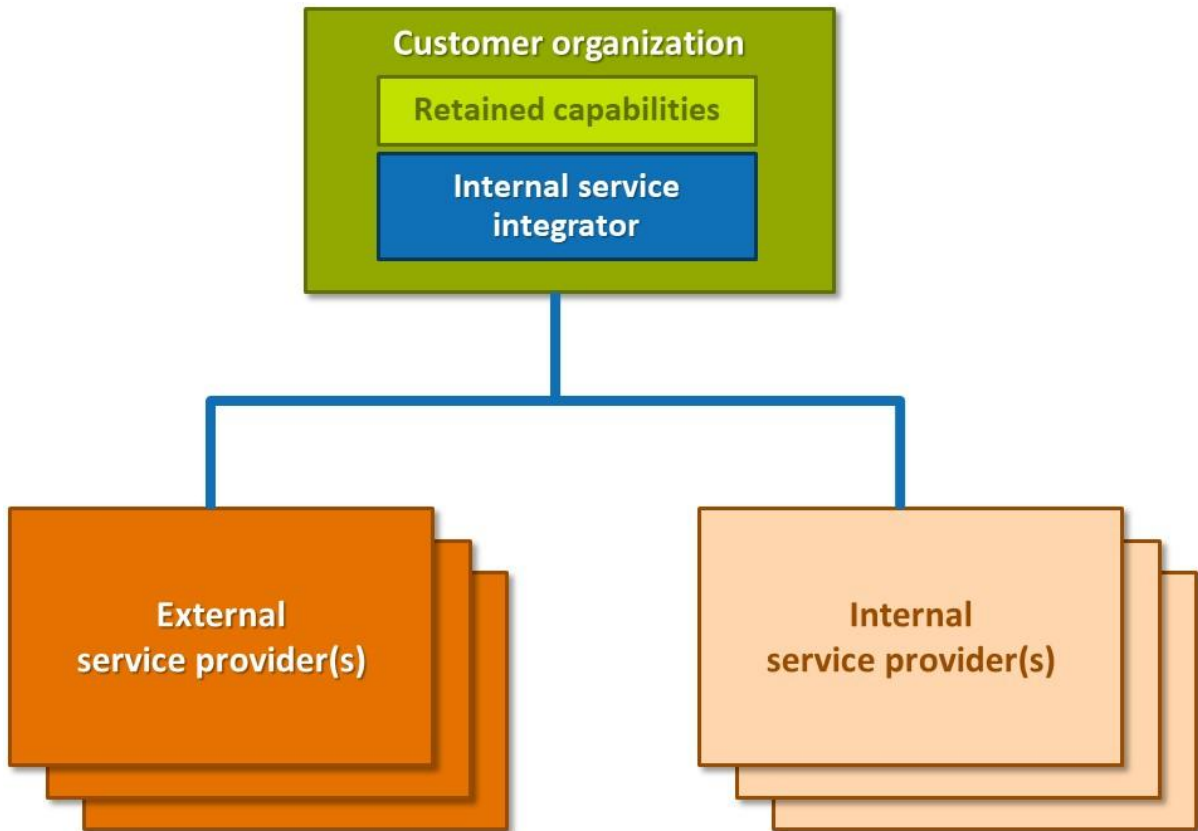


Figure 9: Internally sourced service integrator

3.1.1. When does a customer organization use this structure?

This structure is suitable for organizations where the customer organization already has or intends to develop internal service integration capabilities.

It is typically used where the customer organization wants to retain control and flexibility over the SIAM ecosystem, or where timescales do not facilitate the procurement and establishment of an external service integrator. It is also used by customer organizations that have a business, regulatory or legislative need to retain ownership of the service integration layer.

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As part of this structure, the customer organization may use resource augmentation. This is an approach where individual roles within the service integrator are filled using directly employed internal staff, supplemented by resources provided by an external organization. Even though some of the staff might not be directly employed by the customer organization, this still fulfills the criteria for an internally sourced service integrator as the customer organization has overall ownership and control.

Summary: Internally sourced

Suitable for customer organizations that:

- Have internal service integration capabilities or plan to develop them
- Have business, regulatory or legislative requirements relating to the governance and management of service providers
- Want to retain control and flexibility over the SIAM ecosystem
- Have timescales that do not allow procurement of an external service integrator.

3.1.2. Advantages

The advantages of an internally sourced service integrator include:

- The customer organization has full control over the service integrator role, with no dependency on an external organization, or any of the associated risks and costs
- Valuable skills remain in-house and there is no loss of key resources or key knowledge
- The service integrator shares strategic goals with the customer organization so there is no conflict
- The service integrator can be flexible and accommodate change without a requirement for any contractual amendments
- External service providers will not see the service integrator as a competitor and are thus more likely to collaborate with it

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- The service integrator may be established more quickly because it already understands the customer organization's goals and drivers, and as there is no time required to procure and establish an external service integrator
- The service integrator is part of the same organization that manages service provider contracts so has direct leverage over service providers, their behavior and performance.

3.1.3. Disadvantages

The disadvantages of an internally sourced service integrator include:

- The customer organization must develop and maintain the service integrator capability, resources and skills, and design and implement toolsets, sometimes with no experience of SIAM implementation
- The customer organization may underestimate the number of resources and the expertise required for the service integrator capability
- The service integrator is seen as synonymous with the customer organization; this can make it more challenging for them to mediate between the customer organization and the service providers if there is a conflict
- There is a risk that the customer organization decides to act as the service integrator because they are not fully committed to SIAM, and do not wish to formally establish and outsource the structure. If SIAM is not adopted fully, the benefits will be limited and there will be a further risk that old ways of working continue
- Internal service providers may not accept the authority of the internal service integrator.

3.2. Externally sourced service integrator

In this structure, the customer organization appoints an external organization to take the role and provide the capabilities of the service integrator.

The externally sourced service integrator is exclusively focused on service integration activities and does not take any of the service provider roles. This separation allows for a clear distinction between the service integration layer and service delivery responsibilities, as illustrated in **figure 10**.

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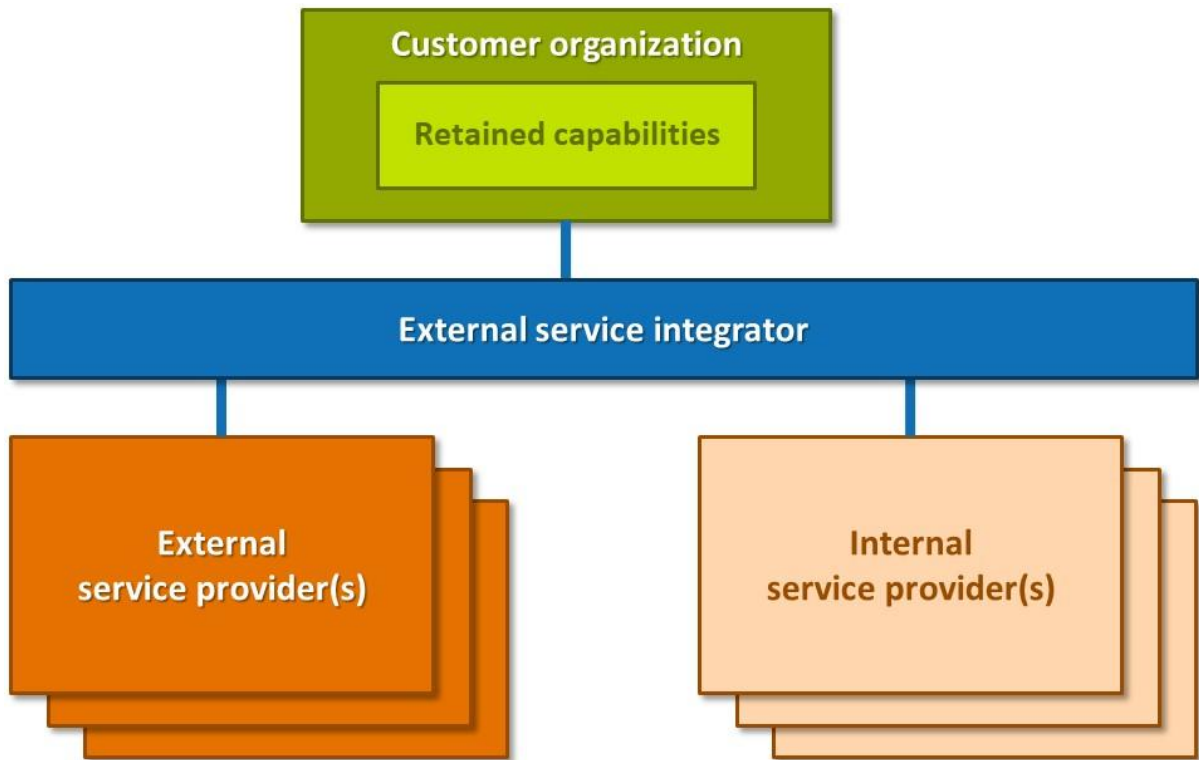


Figure 10: Externally sourced service integrator

3.2.1. When does a customer organization use this structure?

This structure is suitable when the customer organization does not have internal service integration capabilities and does not intend to develop them.

It is also commonly chosen by organizations who do not have the resources available to take on the service integrator role, and do not want to have an increased headcount or the management responsibilities associated with selecting and maintaining service integration resources.

This structure is suitable for customer organizations that are prepared for another organization to take the service integrator role, and that are prepared to place a high degree of trust in their external service integrator, to fully benefit from their expertise.

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Its success depends on the customer organization empowering the service integrator and giving it the responsibilities of coordination and control of service providers, implementing and coordinating processes and managing end-to-end reporting.

For this structure to succeed, the customer organization needs its retained capabilities to provide strong governance over the external service integrator. These retained capabilities must clearly identify the goals and the mandate for the external service integrator, establish expected outcomes and ensure alignment with broader stakeholder objectives.

The customer organization must also enable the service integrator to act on its behalf and should avoid bypassing them by engaging directly in operational relationships with the service providers.

Summary: Externally sourced

Suitable for customer organizations that:

- Are prepared for another organization to take the service integrator role
- Are prepared to have a high degree of trust in an external organization acting as their service integrator
- Do not have service integration capabilities and do not want to develop them
- Do not have service integration resources and do not want to add or manage them.

3.2.2. Advantages

The advantages of an externally sourced service integrator include:

- The opportunity for the customer organization to review multiple service integrators and then select the organization that provides the best fit to their requirements
- The potential for faster benefits realization, as the service integrator's expertise reduces the time to implement the SIAM roadmap; although the time required to select the external service integrator also needs to be considered

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- The potential for improved value, as the service integrator applies its experience to manage the SIAM ecosystem in an efficient and effective way
- Separation of concerns: the service integrator can focus on the end-to-end governance and coordination of the services, processes, metrics and reporting and the customer organization can focus on business outcomes and strategic objectives
- Access to established SIAM models, processes and toolsets, where the service integrator is providing these, leveraging proven capabilities
- Access to innovative practices from the service integrator's experience on other SIAM implementations.

3.2.3. Disadvantages

The disadvantages of an externally sourced service integrator include:

- The high level of dependency on the external service integrator adds a level of risk; including commercial, continuity and security risks
- The potential for higher costs related to the sourcing and management of an external organization
- The potential for resentment from any internal service provider that is part of the customer organization, when being managed by the external service integrator
- The potential for resentment from the external service providers in the SIAM ecosystem, particularly where the service providers and service integrator compete in other markets. This can lead to relationship issues and poor performance
- The external service integrator's models and practices might not be the best fit for the customer organization
- The use of an external service integrator can make it more difficult to change how the service integrator is working, because contractual changes may be required. This means the customer organization may be less agile and may result in higher costs
- There is a risk that the customer organization decides to appoint an external service integrator because they do not fully understand SIAM themselves. This may increase overall costs of delivery and result in poor service because the customer organization has not clearly defined its own objectives

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- The external service integrator must build relationships with the customer organization and with the service providers; the time and effort required to do this is often not accounted for in the initial investment analysis
- The service integrator does not have a contractual relationship with the service providers, so, without empowerment from the customer organization, they can be ineffective.

3.3. Lead supplier as service integrator

In this structure, the role of service integrator is taken by an external organization that is also an external service provider. This can occur when:

- An existing service provider successfully bids to be the service integrator as part of a procurement process
- The existing service integrator successfully bids to be a service provider as part of a procurement process
- One external organization wins two parts of a tender and so becomes the service integrator and a service provider.

The organization that is a service provider and the service integrator is referred to as the lead supplier.

This structure is sometimes referred to as 'guardian' or 'custodian'. It is important to emphasize that the contractual relationship in this structure remains between the customer organization and the service providers. The service integrator does not have a contractual relationship with the service providers.

Prime vendor

The lead supplier structure is different from the 'prime' or 'prime vendor' model, in which the customer organization contracts with a service provider that, in turn, sub-contracts other service providers to deliver services. In this model, the customer's contractual relationship is only with the prime vendor.

Any service provider within a SIAM ecosystem could act as a prime vendor, engaging one or more sub-contracted providers to support its own service delivery. These sub-contracts are not visible within the SIAM model. From a SIAM perspective, formal relationships exist only between the customer organization, the service integrator, and the (prime) service provider. How a

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prime vendor manages its sub-contracts is outside SIAM governance, as long as the agreed service levels and outcomes are met.

Where sub-contractors are used, SIAM policies, corporate standards, and legal or regulatory requirements must still be met and these should be reflected in the prime vendor's contractual agreements.

Figure 11 shows the lead supplier structure.

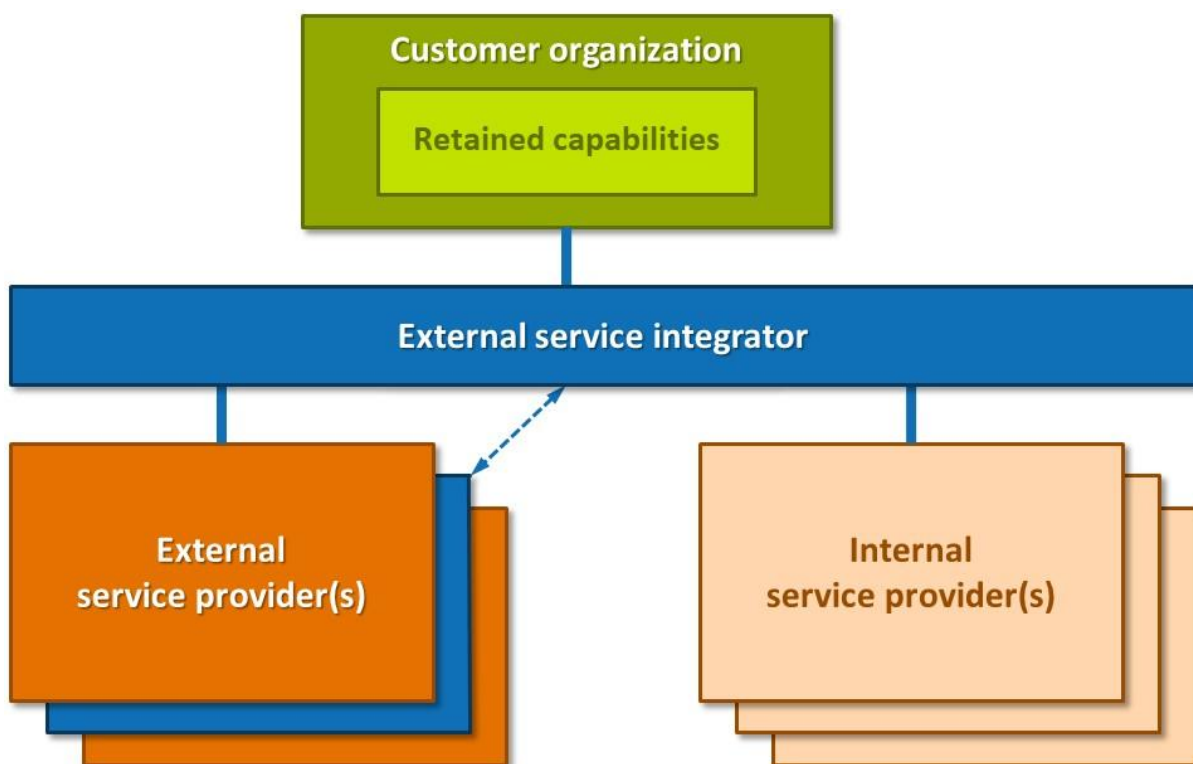


Figure 11: Lead supplier as service integrator

3.3.1. When does a customer organization use this structure?

A customer organization would choose this structure for the same reasons that it would choose an externally sourced service integrator; i.e. it does not have its own service integration capabilities or resources and does not wish to develop and maintain them.

In this structure, when the customer organization evaluates the market for a service integrator, one of its existing service providers may be well-positioned as it already has in-depth knowledge of the customer organization, and a

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level of trust exists. This could result in the service provider being chosen as the service integrator and can facilitate a smoother SIAM transition.

Conversely, the current service integrator may also have expertise in the delivery of one or more of the services (or service elements). If it adds service delivery to its contractual arrangement with the customer organization, it may transition from an external service integrator to a lead supplier as service integrator.

If a single organization is acting as both the service integrator and a service provider, there are management considerations that need to be addressed. These include:

- Making sure there is no unfair advantage for the service integrator or the service provider
- Maintaining the impartiality of the service integrator role
- Ensuring that the customer organization is not being charged twice for the same capabilities.

This requires clear segregation of duties in the lead supplier, to prevent the exchange of confidential or sensitive information between teams or individuals who may have conflicting interests. This is known as 'ethical walls'¹².

The service integrator and the service provider roles should be viewed and managed as two separate entities (as if they were separate organizations). They will each have their own contract or agreement, roles, responsibilities and reporting requirements.

Ideally, different resources will work in the service integrator and service provider entities to reduce the likelihood of any conflict of interest.

¹² Source: https://www.leapxpert.com/glossary_term/ethical-wall

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Summary: Lead supplier

Suitable for customer organizations that:

- Have a trusted service provider that also has service integration capabilities
- Have a trusted service integrator that also has service provider capabilities
- Are prepared for another organization to take the service integrator role
- Do not have service integration capabilities or resources and do not plan to develop them.

3.3.2. Advantages

The advantages of a lead supplier service integrator are mostly the same as those for an externally sourced service integrator.

There are some additional advantages:

- Where the service integrator is currently acting as a service provider, the set-up process can be faster as there is an existing relationship with the customer organization
- From the customer organization's perspective, the service integrator has a vested interest. If the service fails, it will be subject to penalties at the service provider level, so it has an extra incentive to deliver to agreed targets.

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3.3.3. Disadvantages

The disadvantages of a lead supplier service integrator are mostly the same as those for an externally sourced service integrator.

There are some additional disadvantages:

- The organization acting as the service integrator and service provider might not have effective internal governance, leading to knowledge 'leaking' between the two roles. This will create relationship issues between the service integrator and other service providers if this is perceived as an unfair advantage
- The organization acting as the service integrator and service provider might be perceived to be biased, even if this is not the case, which can also lead to the service integrator/service provider relationships suffering
- The organization acting as service integrator and service provider might charge the customer organization twice for the same resources; for example, service desk resources shared between the two roles, or management resources shared between the service provider and service integrator roles
- The service integrator part of the organization could treat its service provider function harshly or unfairly to try and prevent any allegations of bias, which can also create relationship and service management issues.

3.4. Hybrid service integrator

In this structure, the customer collaborates with an external organization (or organizations) to fulfill the role of service integrator and provide the service integrator capability.

The service provider roles are performed by external service providers and/or internal service providers.

The hybrid service integrator structure is shown in **figure 12**.

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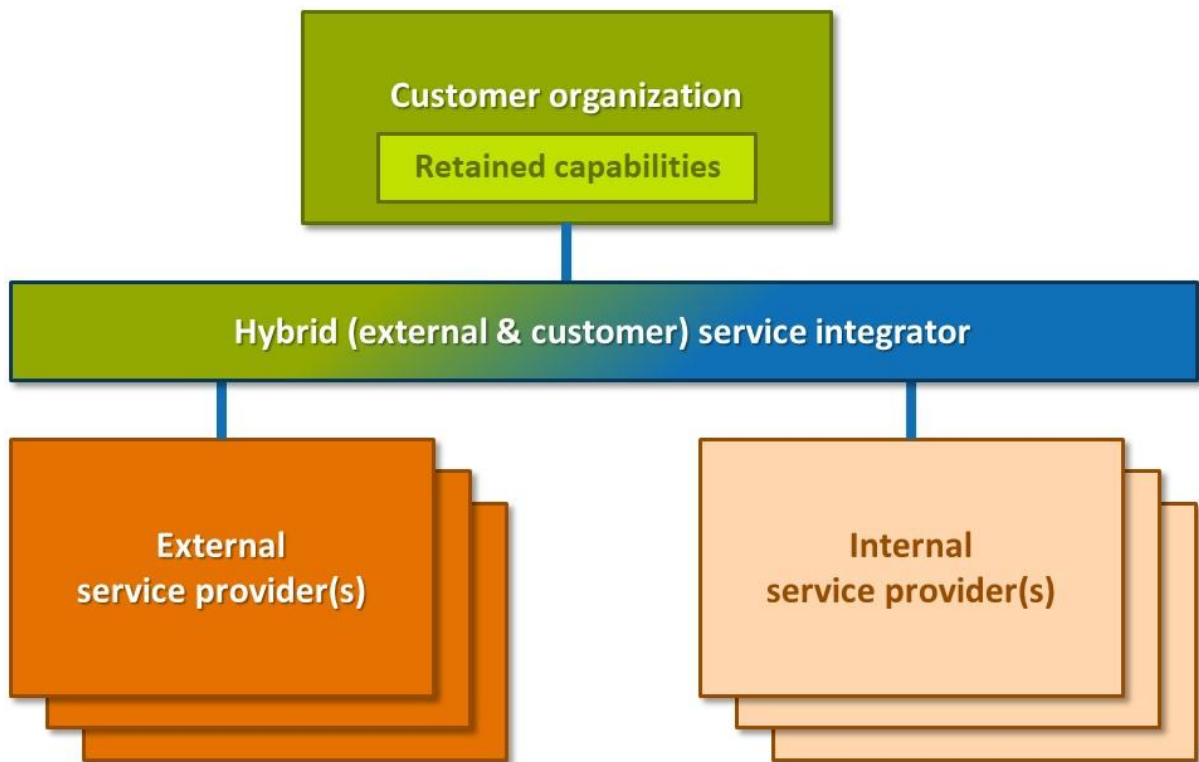


Figure 12: Hybrid service integrator

The hybrid service integrator model can combine elements of the internal, external and lead supplier structures. It can take many forms, including:

- one internal and two external organizations acting as the service integrator
- an internal organization partnered with a lead supplier acting as the service integrator,
- multiple external organizations acting as the service integrator.

The specific structure depends on the customer organization's preferences, prerequisites and constraints. This section focuses on a hybrid structure comprising of one internal and one external organization, with the external party exclusively focused on service integration activities.

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3.4.1. When does a customer organization use this structure?

This structure is suitable for organizations that wish to retain an element of involvement in the service integrator role, but do not have sufficient in-house capabilities or resources.

In the hybrid structure, the service integration capability is created through collaboration between the customer organization and an external service integrator acting as a service integration partner. This can allow the customer organization to learn from an external service integrator that already has expertise in the role.

This structure can be temporary or permanent. If it is temporary, the hybrid approach will end when one of the following has occurred:

- The customer organization has developed sufficient service integration skills and resources in-house, and has transitioned to an internally sourced structure.
- The customer organization has decided that it no longer wants the hybrid structure, and has transitioned to an internal, external, or a lead supplier structure.

In this structure, it is normal to allocate specific service integration roles, functions and structures to either the internal service integration function or the service integration partner. This differentiates this structure from the resource augmentation approach that can be applied to the internally sourced structure.

Summary: Hybrid

Suitable for customers that want:

- To act as a service integrator but do not have sufficient capability or resources
- To learn from an external service integrator
- The flexibility of a temporary or permanent hybrid service integrator.

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3.4.2. Advantages

The advantages of a hybrid service integrator include:

- The customer organization develops skills and resources, and can revert to an internally sourced solution if the service integration partner fails to live up to initial expectations
- Benefits can be realized more quickly, as the service integrator brings expertise and collaborates with the customer organization, reducing the time it takes to transition to a SIAM model
- Access to commercial skills and knowledge; the service integrator can help the customer organization to avoid common mistakes.

3.4.3. Disadvantages

The disadvantages of a hybrid service integrator include:

- The customer organization must develop a service integration capability, and recruit and manage resources
- Without clear design, this structure can lead to duplication of skills, missed activities, confusion about responsibilities and poor definition of where the boundaries of operation lie
- This structure can be confusing for the service providers if a clear governance framework and communication plan have not been implemented
- When the hybrid approach is meant to be temporary, the customer organization may inadvertently build a long-term dependency on the service integration partner
- Customer organizations may adopt the hybrid model because they are reluctant to give up control, not for a valid business reason. This can lead to the benefits of SIAM not being realized.

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4. SIAM roles

This section of the SIAM Foundation looks at roles in a typical SIAM ecosystem. This includes individual roles, as well as the role of each SIAM layer, and the way that roles are grouped into structural elements.

A role is a description of the specific duties, responsibilities and expectations of a particular position within an organization, which must include required skills, knowledge, behaviors, and experience.

A responsibility is defined as: *something that it is your job or duty to deal with*¹⁴.

Accountability is defined as: *the state of being accountable, liable or answerable*¹³.

4.1. Roles and the SIAM roadmap

Within a SIAM ecosystem, roles and their associated responsibilities need to be defined, allocated, monitored and improved.

Principles and policies for roles are defined during the Discovery & Strategy stage of the SIAM roadmap, before detail is added during Plan & Build. Roles are then allocated during the Implement stage and monitored during Run & Improve.

The four main activities related to roles are:

1. Definition of principles and policies
2. Design
3. Allocation
4. Monitoring and improvement.

¹³ Source: Collins Dictionary

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4.1.1. Definition of principles and policies

Definition of the principles and policies for roles is a vital step in the design of the SIAM ecosystem.

During the Discovery & Strategy stage, existing roles and job descriptions are mapped and compared to required responsibilities (for instance, those defined within the governance framework) and the proposed SIAM model.

During the Discovery & Strategy stage, roles are not detailed; they are revisited and more detail is added during the design activities in the Plan & Build stage.

There is no single, ideal mapping of roles for a SIAM ecosystem. Each SIAM model will be different, depending on what the customer organization wishes to retain, and what it is prepared to source externally from the service integrator and/or the service providers.

The customer organization's decision about what to source internally, and what to source externally, will be influenced by several factors:

- The overall objectives for implementing SIAM
- The target SIAM model
- The selected SIAM structure
- The customer organization's strategy and organizational goals, including financial constraints and risk appetite
- The customer organization's capability and skill levels
- What the customer organization regards as a strategic capability that is essential to retain
- Existing service provider relationships and outsourced roles and responsibilities.

4.1.2. Design

During the Plan & Build roadmap stage, detailed roles are designed using the outline SIAM model and outline process models, the SIAM structure and the governance framework. Role profiles are created to clearly define the accountabilities and responsibilities of each role.

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4.1.3. Allocation

During the Implement stage, roles are allocated, using the role profiles.

There are some roles that should always be allocated to specific SIAM layers due to their focus. Examples are:

- The customer organization must retain any roles that are mandated by legislation or regulations
- The service integrator will always be accountable for service governance, management, integration, assurance, and coordination, including end-to-end service management, service provider management, monitoring and reporting
- The service providers will fulfill service delivery roles.

4.1.4. Monitoring and improvement

Once the roles are established, they are continuously monitored and evaluated to determine their effectiveness and to identify any opportunities for improvement. Improvements can be made to roles, their responsibilities, and to the interfaces between them.

Roles will need to be reviewed following any restructuring activities across the customer organization to ensure they remain aligned and effective.

4.2. How is a role different in a SIAM ecosystem?

The definition of roles in a SIAM ecosystem must recognize that they will be applied in a multi-provider environment. Without careful design and management of roles and responsibilities, there is an increased risk that activities could be missed or duplicated as more parties are involved and the ecosystem is more complex than a single-source environment.

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Mapping activities

In a SIAM ecosystem, one process or activity might span the three layers. For example, consider change management:

- Customer layer: initiates changes, has input into change assessment, authorization and scheduling
- Service integrator layer: initiates changes, manages the integrated change management process
- Service provider layer: initiates changes, presents them to the change advisory board, implements changes using internal processes.

There is also an opportunity for multiple roles to be performed by one person. For example, a process manager for a service provider might have:

- A change management role, attending the change advisory board
- A problem management role, attending a problem management working group
- A knowledge management role, providing input into knowledge articles.

The way that roles are allocated will depend on factors including the size and complexity of the SIAM ecosystem and resource availability and capability.

4.2.1. The role of the customer organization

Outside of a SIAM model, it is common for the customer organization to have a direct relationship with its service providers. In the SIAM ecosystem, the customer organization needs to understand that its role is to support and empower the service integrator. If the customer organization continues to work directly with service providers within a SIAM ecosystem, it may inadvertently create a 'shadow IT' culture, which risks the overall success of the SIAM ecosystem.

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4.2.2. The role of retained capabilities

For staff who are part of the retained capabilities, adapting to SIAM means relinquishing direct control of service providers (both internal and external) and stepping back from day-to-day management of service provision. Their role needs to be strategic and proactive, rather than operational and reactive.

The customer retained capabilities role needs to have a strong relationship with the service integrator. Its purpose is to provide direction, and enable service integrator autonomy without creating a dictatorship.

The customer organization owns the contracts with the service providers, but the service integrator is responsible for managing delivery against them. The retained capabilities need to let the service integrator carry out its role without undermining it.

4.2.3. The role of the service integrator

The role of the service integrator involves being the agent of the customer organization, and acting on its behalf. Especially for an external service integrator, this means pursuing the strategic goals of the customer organization, and not solely focusing on its own goals and objectives.

The service integrator also represents the service providers and the end-to-end service to the customer organization, focusing on the value that is delivered.

The service integrator role relies on good relationships. To be effective, it must have good relationships with the customer organization, any retained capabilities, and the service providers.

The service integrator's role is to assure and facilitate service delivery. It needs to be contractually and commercially aware to carry out its role effectively in the SIAM ecosystem. The service integrator needs to focus on service integration and collaboration across multiple service providers.

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4.2.4. The role of service providers

Working collaboratively can be a new approach and a culture change for service providers. External service providers need to adapt to working with potential competitors, and adjust to having a relationship with the service integrator rather than their customer. Internal service providers may have to get used to a new type of performance management.

All service providers may have to change their ways of working and their structure to be effective in the SIAM ecosystem. Their role will require a focus on service objectives, balancing them against their own objectives.

4.3. Role description: Customer organization

Description	The customer organization's role within the SIAM ecosystem is that of the commissioning organization. It also includes the retained capabilities that carry out corporate governance of the SIAM ecosystem.
Typical accountabilities	▪ Strategic direction ▪ Enterprise architecture ▪ Policy and standards management ▪ Procurement ▪ Contract management ▪ Demand management ▪ Financial and commercial management ▪ Service portfolio management ▪ Corporate risk management ▪ Corporate governance, including governance of the service integrator and the SIAM model ▪ Accountability for program and project management.
Typical roles	▪ Head of IT ▪ Head of service/product ▪ Service/product owner(s) ▪ Enterprise architect ▪ Service architect ▪ Chief operations officer (COO) ▪ Chief finance officer (CFO) ▪ Chief information officer (CIO) ▪ Chief information security officer (CISO).

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Typical activities	▪ Defines and assures a core set of policies, standards, procedures and guidelines including architectural, informational, commercial, financial, security and enterprise service architecture ▪ Develops and owns the IT strategy and strategy for SIAM that align with and support the business strategy ▪ Develops and owns enterprise architecture, defines the technology, data and application roadmap, defines the service scope for SIAM ▪ Provides overarching program and commercial management ▪ Assures and governs the service integrator ▪ Manages the service provider relationships at an executive/commercial level ▪ Overall management of risk ▪ Resolves contractual disputes ▪ Owns business relationships and acts as the “intelligent client” function ▪ Defines overall service budget.
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Table 1: Customer organization role description

4.4. Role description: Service integrator

Description	The service integrator layer of the SIAM model is where end-to-end service governance, integration, management, assurance and coordination are performed.
Typical accountabilities	▪ End-to-end service management ▪ End-to-end performance management ▪ End-to-end service reporting ▪ Service governance and assurance ▪ Tracking value for money ▪ Continual service improvement.
Typical roles	▪ Head of service integration ▪ Service delivery manager(s) ▪ Service manager(s) ▪ Process owner(s) ▪ Process manager(s) ▪ Service assurance manager(s) ▪ Performance manager(s) ▪ Security manager(s).

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Typical activities	▪ Responsible for end-to-end service management across the service providers and the interface into the customer organization ▪ Managing service provider relationships at an operational level ▪ Acting as the customer organization's agent and providing a communication path to the service providers ▪ Managing end-to-end performance management of all service providers ▪ Managing performance management of individual service providers against agreed targets ▪ Coordination of the service providers ▪ Assuring service provider performance and service delivery ▪ Governing the service providers, as delegated by the customer organization ▪ Facilitating forums ▪ Managing operational supply and demand for services and capacity ▪ Consolidated service reporting ▪ Providing service communications ▪ Potential responsibility for provision and management of an integrated service management toolset ▪ Managing the performance of service providers against contractual and service targets.
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Table 2: Service integrator role description

4.5. Role description: Service provider

Description	Within a SIAM ecosystem, there are multiple service providers. Each service provider is responsible for the delivery of one or more services, products or service elements, to the customer organization. It is responsible for managing the products and technology used to deliver its contracted or agreed services. The service providers may be internal or external to the customer organization.
Typical accountabilities	▪ Delivering services required by the customer organization to defined and agreed performance targets, standards, policies and architecture ▪ Exhibiting required behaviors for cooperation, collaboration, improvement and innovation ▪ Ensuring cross-service provider service management processes are followed ▪ Working collaboratively with suppliers and the service integrator to resolve issues, incidents and problems, identify improvement opportunities and meet customer outcomes.

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Typical roles	▪ Service manager(s) ▪ Account manager(s) ▪ Process owner(s) ▪ Process manager(s) ▪ Technical staff ▪ Service management staff.
Typical activities	▪ Responsible for the delivery of technology and products to deliver services, at agreed service performance targets and cost ▪ Integrating internal service management processes with the end-to-end service management processes ▪ Adhering to policies, standards and procedures defined by the customer ▪ Adhering to architectural design standards ▪ Working collaboratively with the service integrator and other service providers ▪ Taking part in structural elements, including forums and working groups.

Table 3: Service provider role description

4.6. Governance roles

Governance is a term that is widely used and often misunderstood. In a SIAM ecosystem, governance refers to the evaluation of organizational requirements, the definition and application of policies and standards, and monitoring of the results. These define and assure the required levels of authority, decision-making and accountability.

The governance of IT, including the governance of SIAM in an IT environment, must be integrated into the overall organizational governance to be effective. Without effective support and sponsorship at the organizational level, the implementation and operation of SIAM has an increased risk of being ineffective.

A governance framework includes six key elements: Direction, Capability, Policy, Delegation, Performance, and Accountability¹⁴. These elements are interconnected and collectively ensure comprehensive and effective governance (see **figure 13**).

¹⁴ Source: [ISO/IEC 38500:2024](#)

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Figure 13: ISO/IEC 38500:2024 Framework for the governance of IT

ISO/IEC 38500

ISO/IEC 38500 is the international standard for governance and can be applied in a SIAM model.

Governance according to ISO/IEC 38500 includes direction, oversight, and accountability. This describes the system by which (for example) the current and future use of IT is governed within an organization. This includes ensuring that IT supports and enhances the organization's strategic objectives, manages risk effectively, and complies with relevant laws and regulations.

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ISO/IEC 38500 outlines six principles for good governance:

1. **Responsibility:** individuals and groups within the organization understand and accept their responsibilities.
2. **Strategy:** IT is aligned with the needs of the organization and supports its strategic goals.
3. **Acquisition:** investments are made for valid reasons, based on appropriate and ongoing analysis, with clear and transparent decision-making.
4. **Performance:** IT is fit for purpose in supporting the organization, providing reliable and effective services.
5. **Conformance:** IT complies with all mandatory legislation and regulations. Policies and practices are clearly defined and enforced.
6. **Human Behavior:** IT governance respects human factors, including the needs and behaviors of all people involved in IT.

Refer to the ISO/IEC 38500 standard for further information.

Governance activities are carried out at strategic, tactical and operational levels through governance boards. These boards form structural elements in the SIAM layers.

Boards are decision-making bodies that are responsible for monitoring, evaluating and providing direction, and are accountable for their outcomes.

The boards discussed in this document provide the required level of governance in a SIAM ecosystem. In complex environments with many different service providers, additional boards might be created to address specific areas, for example:

- Information security governance board
- IT service continuity governance board
- Program governance board
- Risk and compliance governance board
- Environmental and sustainability governance board.

The board structure that is put in place in a SIAM model needs to balance the level of overhead created by the board meetings and administration against the governance requirements and the outcomes achieved.

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4.6.1. Strategic governance: Executive boards

Executive boards provide governance and oversight at the most senior level. These boards also play an important role in establishing a SIAM culture, by demonstrating good behaviors at the most senior levels (see section **6 SIAM cultural considerations**).

The attendees for these boards are senior staff with accountability for their organization's role in the SIAM model.

A typical SIAM governance framework will include an executive board attended by all service providers plus representatives from the customer organization and the service integrator. In addition to this, each service provider may have an individual executive board with the customer and possibly the service integrator. This allows a service provider to discuss commercial performance and sensitive issues.

4.6.1.1. Typical attendees

Typical attendees include:

- Customer organization: chief information officer (CIO), chief technology officer (CTO), head of delivery or service delivery director
- Service integrator: operations director, contract and commercial director
- Service providers: operations director, contract and commercial director, account executive, CIO, CTO.

4.6.1.2. Typical frequency

Executive boards are typically held quarterly.

4.6.1.3. Typical agenda

An executive board agenda could include:

- Customer organization strategy: for the next six months, one year and three years
- Service integrator strategy updates, including any possible clashes or synergy, and opportunities for mutual benefit
- Service provider strategy updates, where appropriate, including any possible clashes or synergy, and opportunities for mutual benefit

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- High-level review of last quarter, including service performance, successes and issues
- Contractual performance, including any obligations not being met; these are typically discussed at the individual executive boards, unless there is a common issue across all service providers
- Planning for innovation, considering any new items from the service providers/service integrator
- Any other relevant topics.

4.6.1.4. *Typical inputs*

Executive board inputs could include:

- Quarterly and monthly performance information
- Customer and service satisfaction information
- Customer organization strategy
- Strategic service improvements
- Strategic innovations
- Service integrator and service provider strategies, where relevant
- Service provider technology roadmap.

4.6.1.5. *Typical outputs*

Executive board outputs could include:

- Action and decision logs
- Strategic course corrections or direction changes
- Business change requirements
- Strategic change schedule
- Celebration and communication of success.

4.6.2. **Tactical boards**

A tactical board sits between the strategic and operational boards. It forms part of the preparation for the operational board and can be used to carry out discussions before meeting with the customer organization, for example if a major incident has occurred. It can also be used to identify items for escalation to the strategic board, and acts as a point of escalation for operational boards.

This board is typically not attended by the customer organization.

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4.6.2.1. *Typical attendees*

Tactical board attendees are staff from the service integrator and the service providers. The roles present could include:

- Service delivery managers
- Service managers
- Process owners, as required
- Account managers.

4.6.2.2. *Typical frequency*

Tactical boards are typically monthly.

4.6.2.3. *Typical agenda*

This board is used to discuss service performance and continual improvement, so the agenda will vary depending on any issues that are being experienced.

The service integrator is empowered to interpret the contract on behalf of the customer organization so decisions might be made at this meeting about financial or non-financial remediation, which can then be communicated at the operational board.

This board will take direction from the strategic board and use it to create tactical action plans. It will also review changes escalated from the operational boards.

This board will include coordination, mediation, decision-making, assurance and governance.

4.6.2.4. *Typical inputs*

Tactical board inputs could include:

- Executive board decisions and direction
- Performance data, including customer satisfaction
- Service improvements
- Service provider data.

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4.6.2.5. *Typical outputs*

Tactical board outputs could include:

- Action and decision logs
- Tactical change schedule
- Improvement opportunities
- Information inputs for executive or operational boards.

4.6.3. **Operational boards**

The main operational board convenes to discuss service performance at a lower level than the executive and tactical boards.

It will review service performance and acts as an escalation point for all other operational boards and forums. For example, it may authorize budget or resources to carry out improvement activities identified in a forum that exceed the approval limit of the forum attendees.

Other operational boards will be scheduled as required to support decision-making; the most common example of this is the integrated change advisory board.

Operational boards are not typically attended by the customer organization. There may be exceptions, for example if a major incident has occurred, a customer organization representative can be invited or could request to attend.

4.6.3.1. *Typical attendees*

Operational board attendees could include:

- Service integrator
- Service providers
- User representatives
- Process owners
- Process managers
- Service managers
- Customer retained capabilities, where required.

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4.6.3.2. *Typical frequency*

Operational boards are typically monthly. Additional meetings may be event driven, for example during a major incident.

4.6.3.3. *Typical agenda*

An operational board agenda could include:

- Review of monthly performance reports, including customer satisfaction
- Actions and decisions
- Critical and major incident reviews
- Escalations from other operational boards and forums
- Six-monthly compliance and certification policies and procedures review.

4.6.3.4. *Typical inputs*

Operational board inputs could include:

- Monthly reports
- Process reports; for example, incident reports
- Improvement plans
- Escalations from other operational boards
- Decisions from the tactical and strategic boards.

4.6.3.5. *Typical outputs*

Operational board inputs could include:

- Decision and action logs
- Items for escalation
- Improvement actions.

4.6.4. Example operational board: Integrated change advisory board

The integrated change advisory board is an operational governance board. It meets this definition because it makes decisions and is held accountable for them. It is chaired and managed by the service integrator.

This board reviews all changes within the scope of its authority that could affect the end-to-end service, regardless of which service provider will implement the change. It focuses on changes that affect multiple service providers, associated risks, and unintended impacts to the customer organization.

The board is also responsible for defining change policy. The policy defines the responsibilities for review and approval of different types of change. This includes the definition of standard or self-contained changes that can be approved locally by a service provider.

Where necessary, changes are escalated to the tactical or strategic boards. 'Review' of changes can encompass any action from detailed investigation through to definition of standard changes that gain automatic approval, or approval of systems for automated test and release. The board seeks to facilitate, not prevent change.

The responsibilities of the integrated change advisory board include:

- Ensuring that all service providers and the customer organization are aware of relevant changes
- Confirming that:
 - Changes have been evaluated for risk and unintended impact
 - Remediation plans have been verified
 - Appropriate resources have been allocated and made available to implement the change
 - There are robust communication plans in place
 - Technical and architectural standards have been met
- Collective approval or otherwise of the change
- Creating mechanisms for standard changes and their automatic approval
- Review of completed changes.

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4.6.4.1. *Typical attendees*

Integrated change advisory board attendees could include:

- Service integrator change manager (chair)
- Service provider change managers
- Subject matter experts as required
- Customer organization representation as required.

4.6.4.2. *Typical frequency*

The frequency of the integrated change advisory board varies, relating to the number and scale of changes. Additional emergency meetings can be convened as required.

4.6.4.3. *Typical agenda*

An integrated change advisory board agenda could include:

- New changes to be reviewed
- Update on implemented changes and failed changes
- Improvements to the change management process.

4.6.4.4. *Typical inputs*

Integrated change advisory board inputs could include:

- Change requests and related information
- Change management process performance information.

4.6.4.5. *Typical outputs*

Integrated change advisory board outputs could include:

- Change status updates
- Process improvements.

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4.7. Operational roles

An effective SIAM ecosystem is built on working relationships and cultural alignment between all the SIAM layers.

At an operational level, working groups, boards and forums all help to establish relationships and encourage communication between service providers and the service integrator. These working groups, boards and forums form structural elements of the SIAM ecosystem, spanning the SIAM layers; see section **1 Introduction to Service Integration and Management (SIAM)** for more information.

There are many possible boards, forums and working groups that can be implemented in a SIAM ecosystem, for example:

- Integrated change advisory board
- Problem management forum
- Knowledge management forum
- Continual improvement forum
- Information assurance and security forum
- Service operations forum
- Service owners forum
- Incident management working group (for a specific incident or incidents)
- Release planning working group
- Problem management working group (for a specific problem or problems)
- Innovation working group (for a specific innovation).

The structural elements in place will vary in each SIAM ecosystem. A structural element can be created for any service management process or activity, if it supports improvements in service delivery and outcomes.

Forums can be combined where appropriate – for example, a single ‘process improvement’ forum could be used to assess possible improvements to multiple processes.

Combined forums are of value when processes have similar scope or have dependencies between their activities, for example change, configuration and release management. The number of meetings should always be balanced against the value of the meetings.

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There are generic roles that will attend working groups and forums.

Process owner

The process owner is the accountable role that ensures the end-to-end process is designed, executed and reviewed correctly:

- Accountable for end-to-end process design
- Accountable for process performance.

Both the service integrator and the service providers will have process owners. The service integrator process owner will be accountable for end-to-end process integration across the service providers. For some processes, ownership is within the customer organization, especially for the more strategic processes such as financial and risk management.

A service provider process owner will be accountable for a process within that service provider and for alignment with the end-to-end process. A process owner is a role (rather than a person), so one staff member may act as the process owner for multiple processes.

Process manager

The process manager role supports the process owner and is responsible for the operation of a process or processes. The scope of the role will vary depending on the size and structure of the organization. The scope could include, for example:

- A single process
- Multiple processes
- Processes within a geographical area of the organization
- Processes within a department or business unit
- Processes associated with a particular service
- End-to-end processes across multiple service providers.

In a SIAM ecosystem, the service integrator process managers are responsible for coordinating the end-to-end processes and interacting with the process managers within each service provider.

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Service owner

A service owner is accountable for the end-to-end performance of a service, and the outcomes it provides. The role typically:

- Is accountable for end-to-end service performance
- Defines service strategy
- Forecasts service demand and business requirements
- Is the service budget-holder.

Within a SIAM ecosystem, service owners are part of the customer organization, accountable for the outcomes and value their services delivers. They work with the service integrator as required. External service providers (and the service integrator) may also have service owners within their organization structures, but the focus in SIAM is on the end-to-end service owner.

Service manager

Responsible for service delivery for one or more services in the SIAM model.

This role would typically be carried out by the service integrator, as it has an end-to-end focus. Service provider organizations may have their own service managers, but these roles will focus on the services delivered by that service provider.

Examples of forums and working groups

This section provides some examples of process forums and working groups in a SIAM ecosystem. These examples can be used as the basis for the design of other forums and working groups within a SIAM model.

4.7.1. Knowledge management forum

The knowledge management forum is hosted and managed by the service integrator knowledge management process owner.

It is a regular forum where the performance and effectiveness of knowledge management across the ecosystem is reviewed and assessed.

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4.7.1.1. *Typical attendees*

Knowledge management forum attendees could include:

- Service integrator knowledge management process owner (chair)
- Service provider knowledge management process owners/process managers
- Service integrator service manager as required
- Subject matter experts as required
- Customer organization representation as required.

4.7.1.2. *Typical frequency*

Forums are typically monthly.

4.7.1.3. *Typical responsibilities*

Knowledge management forum responsibilities could include:

- Reviewing accuracy and currency of the knowledge articles in use
- Identifying new knowledge articles that are required based on repeat incidents or requests received by the service desk
- Allowing service providers to collaborate on identifying any incident types that could be resolved at the service desk or via self-help systems rather than by second line teams, improving the end user experience.

4.7.2. **Continual improvement forum**

The continual improvement forum is hosted and managed by the service integrator.

It is a cross-ecosystem forum attended by all service providers and the customer organization. Attendees can present, discuss and agree initiatives for improvement; for example, ways to deliver cost savings or improve customer experience.

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4.7.2.1. *Typical attendees*

Continual improvement forum attendees could include:

- Service integrator continual improvement process owner (chair)
- Service provider continual improvement process owners/process managers
- Service integrator delivery manager/director
- Service owners
- Other process owners as required
- Subject matter experts as required
- Customer organization representation as required.

4.7.2.2. *Typical frequency*

Forums are typically monthly.

4.7.2.3. *Typical responsibilities*

Continual improvement forum responsibilities could include:

- Presenting and reviewing ideas for improvement
- Assessing the potential of initiatives
- Prioritization of initiatives
- Agreeing the responsible party or parties to implement the improvement; this may involve cross-service provider collaboration and implementation
- Approval of any budgetary spend (this may need to be escalated to a governance board)
- Communicating the benefits to the business
- Tracking the progress and ultimate success of the improvements.

4.7.3. **Major incident working group**

The major incident working group may also be referred to as a crisis team, critical incident team or major incident bridge.

It is convened during a major incident, to coordinate the response, facilitate cross-service provider communication and provide regular updates to the customer organization.

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Any lessons learned during a major incident will be discussed in the incident management process forum.

4.7.3.1. *Typical attendees*

Major incident working group attendees could include:

- Service integrator major incident manager (chair)
- Service provider incident management process owners/process managers
- Other process owners as required
- Subject matter and technical experts as required
- Service owners as required
- Customer organization representation as required.

4.7.3.2. *Typical frequency*

A major incident working group will be held when required; typically when a major incident has occurred.

4.7.3.3. *Typical responsibilities*

Major incident working group responsibilities could include:

- Coordinating major incident investigation and resolution
- Coordinating major incident communications
- Encouraging a 'fix first, discuss later' culture.

4.8. **The service desk in a SIAM ecosystem**

The service desk is an example of a function (an organizational entity or team with a specific purpose), which plays an important role in any operating model as the central point of contact for service users. The role of the service desk and how it is sourced will vary from SIAM ecosystem to ecosystem.

The service desk is often seen as a good candidate for external sourcing due to high staff turnover and management overhead, but some companies prefer to maintain it internally or use a hybrid approach.

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The organization providing the service desk will be treated and managed as a service provider in the SIAM ecosystem, whether it is provided by the customer organization, the service integrator or a service provider.

Within a SIAM ecosystem, the service desk acts as a 'single source of truth' and provides important management information about service performance. If the service integrator is not providing the service desk, it must work very closely with it and use the service data it provides.

Some of the potential sourcing options are:

- The customer organization provides the service desk and associated toolset, acting as an internal service provider, and routes incidents to service providers as necessary
- The service integrator provides the service desk and associated toolset
- An external service provider provides the service desk and toolset, but no other services
- An external service provider provides the service desk and toolset in addition to other services; this is often combined with end user computing, applications or hosting
- Different service providers provide their own service desks and toolsets and the service integrator provides a consolidated view; this is only possible where it is clear to the customer organization and end users which service desk to contact for support.

The staff who work on the service desk will require similar skills to those outside of a SIAM ecosystem, but they will also need:

- Supplier management skills
- Commercial awareness.

These skills will allow them to work successfully with different service providers, who may have different contracts, service targets and responsibilities.

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5. SIAM practices

Practices are defined as: *the actual application or use of an idea, belief, or method, as opposed to theories relating to it.*¹⁵

The SIAM practices in this section illustrate how SIAM principles and concepts can be applied. They consider some of the challenges associated with each practice category, and make recommendations that can be used to address these challenges.

Within SIAM there are four categories of practice:

1. People practices
2. Process practices
3. Measurement practices
4. Technology practices.

The four categories of SIAM practice ensure a holistic and balanced approach to designing and operating a SIAM ecosystem. Each category encompasses a wide range of individual activities that collectively support the integration, coordination, and continual improvement of multi-provider ecosystems.

For example, in section **5.1** we consider cross-functional teams. Cross-functional teams are just one example of the challenges in a SIAM ecosystem and the people practices will help to deal with the challenges associated with cross-functional teams. Other challenges and recommendations exist and some are explained in further detail in SIAM Professional.

The SIAM-specific practices in this section should not be confused with other general practices used in IT and business management. Some of these other practices, frameworks and methodologies are described in section **5.5 SIAM and other practices**.

¹⁵ Source: Oxford English Dictionary

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5.1. People practices

Examples of people practices in a SIAM ecosystem are:

- Enabling collaboration
- Defining roles and responsibilities
- Fostering culture
- Supporting capability development
- Managing cross-functional teams.

5.1.1. Managing cross-functional teams

This section will explore people practices using one common SIAM example – the management of cross-functional teams.

A cross-functional team is *a group of people with different functional expertise working toward a common goal*. It may include people from finance, marketing, operations, and human resources departments. Typically, it includes employees from all levels of an organization.¹⁶

The SIAM ecosystem and cross-functional teams

With the SIAM ecosystem, cross-functional teams will have members from different organizations and different SIAM layers. These teams are referred to as structural elements.

There are three common types of structural elements/cross-functional teams:

1. Boards
2. Forums
3. Working groups.

These are described in sections **1 Introduction to Service Integration and Management (SIAM)** and **4 SIAM roles**.

¹⁶ Source: Wikipedia

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In a SIAM ecosystem, examples of cross-functional teams could include:

A major incident working group where the cause of an incident is unclear. The team includes staff from the service integrator and multiple service provider organizations. Team members need to work together towards a shared outcome (incident resolution), whilst meeting service requirements and balancing their own organizational goals.

The integrated change advisory board involving staff from the customer organization, service integrator and multiple service providers. The team members work together to review, prioritize, assess risk and approve or reject changes to an integrated service.

5.1.2. Challenges related to cross-functional teams

Some of the main challenges associated with the management of cross-functional teams are:

1. Conflicting objectives, organizational strategies and working practices
2. Reluctance to share knowledge
3. Lack of automation.

5.1.2.1. *Conflicting objectives, organizational strategies and working practices*

The cross-functional teams in a SIAM ecosystem contain staff from multiple service providers, the service integrator and, in some cases, from the customer organization as well. This can create challenges when staff must balance their own organizational objectives with cross-functional team objectives.

For example, during a major incident, a service provider's organizational goals might be to demonstrate that it is not responsible for causing the incident, and to minimize the resources allocated to resolving it.

However, the end-to-end service targets could be focused on resolving the incident and then assessing what caused it later. This requires the service provider to adopt a 'fix first, discuss later' approach, which may conflict with its individual organizational goals.

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Differences between organizational strategies and working practices can also have an impact on the performance of a cross-functional team.

For example, technical organizations might prioritize resolving incidents above customer communication. In a SIAM ecosystem, they might have to prioritize customer communication over service restoration.

5.1.2.2. *Reluctance to share*

The service provider and service integrator staff working in a SIAM ecosystem need to share information, and collaborate at a people, process and technology level.

In an effective SIAM ecosystem, they may have targets relating to service improvement as well as service delivery. To innovate effectively and improve service delivery, service providers and the service integrator need to work together. Some organizations may be reluctant to do this because they view it as sharing their intellectual property with a competitor.

5.1.2.3. *Lack of automation*

Lack of automation and ineffective toolsets can also be a challenge for cross-functional teams. Where more than one toolset is in use, poor integration between tools is also a challenge.

The issues here can include:

- Inability to measure end-to-end team performance
- Inability to easily share information between teams
- Duplicated work caused by manually entering data into multiple toolsets (the 'swivel chair' approach)
- Reduced likelihood of identifying patterns or opportunities for improvement
- Reduced workflow automation, leading to workflow interruptions, delays, and an inability to monitor workflow.

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5.1.3. Recommendations for managing cross-functional teams

To support effective management of cross-functional teams, the service integrator and the customer need to consider:

1. Roles and responsibilities
2. Clear goals and objectives
3. Knowledge, data and information
4. Communication
5. Toolset integration.

5.1.3.1. Roles and responsibilities

Defining clear principles and policies for roles and associated responsibilities as part of the Discovery & Strategy stage of the SIAM roadmap will lay the foundation for better cross-functional working.

This supports communication within cross-functional teams because all the parties involved have a clear understanding of who the stakeholders are. RACI matrices are a useful tool for mapping roles and responsibilities in cross-functional teams.

RACI matrix

RACI matrices are used to manage resources and roles for the delivery of an activity or task. They can be used to identify all participants in the delivery of a process or function.

Resources can be drawn from different functional areas and organizations, so a RACI matrix is used to track who is doing what, identifying interfaces and engagement with other roles. It provides a clear mapping of roles across the different teams in the SIAM ecosystem.

RACI stands for Responsible, Accountable, Consulted and Informed.

Only one role can be **Accountable** for a task. The role that is accountable for the task has the overall authority - but might not carry out individual pieces of work themselves.

Any number of roles can be **Responsible** as part of the RACI model. These are the workers who will get the actual tasks done, and they will report to the Accountable resource about their progress.

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Sometimes roles are **Consulted** to get a task done. This might be a person within the organization who holds specific knowledge, or it could be a document store or even an internet search engine. These resources need to be tracked to ensure they are available when required.

Other roles need to be **Informed**. These resources are stakeholders who need to track and understand exactly how the task is proceeding, or they may need an output from the activity. Customer organization sponsors, for example, will typically be informed about progress as part of a project.

To build a RACI matrix, these steps need to be followed:

- Identify activities
- Identify roles
- Assign RACI codes
- Identify gaps or overlaps that need resolving
- Distribute the chart for feedback
- Deploy to all relevant parties
- Monitor the roles
- Apply improvements or changes based on feedback and experience.

5.1.3.2. *Clear goals and objectives*

As well as a clear understanding of roles and responsibilities, parties in a SIAM ecosystem need clear goals and objectives.

- The customer organization will define the strategic objectives for the services
- The objectives will be translated into contracts, targets, and service agreements
- The service integrator will work with service providers to:
 - Develop process goals and objectives that drive process execution
 - Develop operational level agreements or targets that align with the contracts, targets, and service agreements.

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Whilst it is important that each service provider has measurable service targets to work towards, they need to be part of an end-to-end performance management and reporting framework. This will, in turn, provide evidence of demonstrable achievement of service objectives, business benefits or value.

If there is no clear definition and communication of value, or end-to-end metrics, service providers may focus only on their own performance and not see the end-to-end view.

In some cases, it might be acceptable for a service provider to miss a target in one area, because it means meeting a target in a different area. The service integrator can help the service providers to prioritize when there is a conflict between individual targets and end-to-end service targets.

5.1.3.3. Knowledge, data and information

Cross-functional teams need access to shared knowledge, data and information.

When these are not shared or readily available:

- Team members will waste time re-discovering or recreating them
- Service issues and customer contacts might be managed in inconsistent ways
- Work will not be carried out in the most efficient way
- Different parties may have different 'versions of the truth'.

The service integrator needs to create a knowledge management strategy and policy to govern how knowledge is gathered, processed, presented, managed and removed.

The service integrator will also make sure that all service providers have access to the knowledge they need as part of a shared knowledge repository. All service providers should contribute to this repository for the benefit of all other parties.

Checks need to be in place to make sure knowledge is updated, relevant, and being used. See also **4.7.1 Knowledge management forum**.

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5.1.3.4. Communication

The service integrator and the service providers need to communicate regularly and work to build relationships and trust. The RACI matrix developed as part of the roles and responsibilities definition is useful to define the 'who', 'what', 'when', 'where', 'how' and 'why' of communication.

A communication plan is key to ensure that:

- All stakeholders and their communication requirements are identified
- There is an appropriate level of regular communication for all stakeholders, for example meetings and levels of reporting
- Communication takes place at the right level for each layer of the SIAM ecosystem
- Communication is consistent across service providers
- Effective communication channels are selected to support timeliness, relationship building, ease of execution and access.

Use of the various structural elements in the SIAM ecosystem (including boards, forums, and working groups) will help to build relationships and encourage better cross-functional working.

Virtual teams

In a SIAM ecosystem, team members are likely to be in different geographical locations. These are referred to as 'virtual teams'.

The resources in the teams might also have multiple customers to work with; for example, a service provider's technical support staff might be involved in more than one SIAM engagement.

The service integrator needs to carefully consider how to manage communication within these teams. Even more care is required if teams are virtual as well as cross-functional.

Virtual teams need to build relationships between team members. This can be challenging if there is no regular face-to-face contact between them. It is recommended to have at least one face-to-face event where team members can get to know each other, to foster trust and create good working relationships.

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Tools can be used to support communication in virtual teams. Examples are messaging, video conferencing, collaboration, social media, and chat tools.

5.1.3.5. Toolset integration

For cross-functional teams, integration between toolsets will save time and resources, and reduce the possibility of errors. It can also support workflow automation.

Integrating toolsets will reduce the need to re-enter and translate data. There is less chance of information errors leading to friction between teams.

5.2. Process practices

Within the context of this document, a process is a *documented, repeatable approach to carrying out a series of tasks or activities*.

Examples of process practices in a SIAM ecosystem are:

- Ensuring end-to-end flow
- Resolving process ownership conflicts
- Integrating processes across service providers.

5.2.1. Integrating processes across service providers

This section will explore process practices using one common SIAM example – integrating processes.

SIAM ecosystems and integrated processes

In a SIAM ecosystem, processes must operate effectively and efficiently across multiple parties. This includes service providers, the service integrator, and sometimes, the customer organization.

For example, during change management, the service integrator is accountable for changes to integrated services, across all the service providers involved with the change.

Change management includes change recording, assessment, prioritization, planning, approval, and post-implementation reviews.

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The service providers, the service integrator, and potentially the customer organization will all be involved. This requires a change management process that is integrated across all parties.

5.2.2. Challenges related to integrating processes

Challenges associated with integrating processes across service providers include:

1. Service providers do not integrate their processes or share process details
2. Gaps between process activities
3. Time-consuming and manual reporting
4. Poor relationships between service providers/blame culture.

5.2.2.1. *Service providers do not integrate their processes or share process details*

Within a SIAM ecosystem, data and information must flow between all parties. This does not mean that all parties must use the same process. Instead, each service provider and the service integrator must work together to ensure their processes are aligned to deliver the required outcome.

This requires processes from the service providers, the service integrator, and the customer to be aligned and integrated (see section **2.2.4.1.3 Design process models**). Some of the service providers in a SIAM ecosystem may be unwilling, or unable, to make the adaptations necessary to support this integration.

This may be acceptable if outcomes and performance meet the pre-defined targets. However, unless this is considered in the design of the integrated processes it can result in:

- Adversely affected outcomes
- Failure to meet end-to-end service levels
- Inefficiencies in the execution of the integrated processes
- Unforeseen additional overheads in the service integrator
- Miscommunication.

Change integration across a cloud-based email service

The service provider will publish planned changes and service outages on its website. It will not directly inform the service integrator, seek approval for changes, or attend any change management boards.

Instead, the service integrator must regularly check the service provider's website. The service integrator informs the other service providers and the customer organization of any changes that will affect them through the change schedule.

5.2.2.2. Gaps between process activities

Process integration fails when there is a gap or a break in the process flow.

This could be a simple action; for example, an incident being assigned to a queue for a service provider, and not being picked up, resulting in increased downtime for the customer organization or end user. Gaps are often identified when process performance targets are failed, for example, when incident resolution times are missed.

These gaps need to be identified and addressed during Plan & Build and on an ongoing basis. The development and agreement of process flows and RACI matrices (see section **5.1.3 Recommendations for managing cross-functional teams**) will help to identify and avoid such gaps.

Gaps should also be identified during the service integrator assurance activities.

5.2.2.3. Time-consuming and manual reporting

Where different providers use different processes, it is likely that they will also use different toolsets. The use of different toolsets can affect the ability, effectiveness, and efficiency of the monitoring and reporting on performance of end-to-end processes.

Unless this is recognized and managed during the Plan & Build stage, monitoring and reporting for end-to-end processes can be time consuming and laborious. Design activities must recognize this, to ensure that the value of the information produced justifies the effort required to collect and process it.

5.2.2.4. Poor relationships between service providers/Blame culture

The success of an integrated process depends on all parties contributing to its design, execution, and improvement. Service providers are less likely to contribute if their relationships with other service providers and with the service integrator are poor.

Service providers need to adopt a 'fix first, discuss later' mentality to resolve issues. This needs to be supported by a 'no blame' culture so that service providers are ready to be open about their faults, rather than trying to hide them.

The 'no blame' culture needs to start with the customer organization and then be continually reinforced by the service integrator to create a collaborative ecosystem. This will assist with building the necessary good relationships.

5.2.3. Recommendations for integrating processes

Recommendations for integrating processes across service providers include:

1. Focus on process outcomes
2. Continual process improvement
3. Establishing process forums.

In addition to these practices, the RACI matrices mentioned in section **5.1.3.1** will also be helpful for identifying the role and responsibility of each stakeholder for each process activity.

5.2.3.1. Focus on process outcomes

The service integrator needs to be clear about the outcome that a process is expected to deliver. This can then be communicated to the service providers so that they all understand their role and responsibilities within the process.

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It is better to start with the outcome and then work back, rather than to start with lower level steps and activities in the hope that they can be brought together into a process. For each process that involves multiple parties, these items should be documented and understood:

- Inputs
- Outputs
- Outcomes
- Interactions
- Dependencies
- Controls
- Data and information standards
- Process steps
- Process flow.

RACI matrices can help to document these, and are a commonly used and widely understood technique.

It is important to recognize and reward positive outcomes when processes are performing well.

5.2.3.2. *Continual process improvement*

All processes should be subject to review and improvement measures. This continual improvement can be managed on multiple levels:

- Within each area responsible for the provision and fulfillment of the process
- At the process level, for example via the process forums or the process owner.

These levels should also feed into an overall process improvement program run by the service integrator. This is particularly relevant when an improvement is dependent on resources external to the process or is likely to have a significantly beneficial impact.

Each process will have a process owner who will be accountable for continual improvement across the end-to-end process, and the service integrator has ultimate accountability for process improvement.

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Process improvements should be assessed, justified and approved using an agreed mechanism, often in the process forum. Once improvements are implemented, their benefits should be tracked to confirm a change has been delivered. This can be more challenging in a SIAM ecosystem than for a process that exists within a single organization.

5.2.3.3. *Establishing process forums*

Process forums are a type of structural element within a SIAM model. They are used to bring together process owners and practitioners from the service providers and the service integrator. Their purpose is to work together on the design and improvement of how their (process-)activities support end-to-end delivery.

This includes:

- Defining data and information standards
- Identifying and managing process improvements
- Developing and sharing good practice
- Sharing information
- Assessing and improving capability and maturity.

Process forums are invaluable for building relationships and trust between all parties. They can be established for a single process, a group of related processes, or a practice in the SIAM ecosystem.

5.3. **Measurement practices**

Examples of measurement practices in a SIAM ecosystem are:

- Defining common measures
- Aligning reporting practices
- Driving transparency
- Enabling improvement based on data
- Reporting on end-to-end services.

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5.3.1. Reporting on end-to-end services

This section will explore measurement practices using one common SIAM example – reporting on end-to-end services.

End-to-end service measurement refers to the ability to monitor an actual service, not just its individual technical components or providers. Effective measurement practices support the performance management and reporting framework.

SIAM ecosystems and end-to-end service measurement

In a SIAM ecosystem, examples of end-to-end measurement could include:

The percentage of service downtime related to failed changes: based on the amount of downtime (minutes, hours, days) and the percentage of this caused by changes

Responsiveness of the service against defined targets: based on measuring the actual experience of the service, not just individual elements, such as network speed or application responsiveness

End-to-end measurement is more complex in a SIAM ecosystem because more than one service provider is involved with service delivery. The end-to-end view is aggregated by the service integrator using data from all service providers.

5.3.2. Challenges related to reporting on end-to-end services

Challenges associated with measurement of end-to-end services include:

1. Lack of strategic requirements
2. Reluctance to share information
3. Inability to map service architecture or end-to-end workflow
4. Not measuring the correct amount of data and information.

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5.3.2.1. Lack of strategic requirements

An effective performance management and reporting framework can only be built once it is clear what needs to be measured.

If the overall strategic requirements for the services are unknown, it will be difficult to create a meaningful set of end-to-end measurements and reports.

5.3.2.2. Reluctance to share information

Poor relationships or competitive tension between service providers can lead to an unwillingness to share information. Service providers might also be reluctant to share information if they feel it will be used to punish them, rather than as a source of learning and improvement.

In some situations, the customer organization withholds information from the service integrator. If the service integrator is an external organization, for example, the customer organization might not want to share some information that it considers to be confidential.

5.3.2.3. Inability to map end-to-end service architecture

Many organizations struggle to map an end-to-end service, and understand what is in scope for measurement and what is not. With the addition of multiple service providers and a distributed architecture, this can be even more challenging.

As part of the SIAM roadmap, the service model needs to be mapped, including and how the services are grouped for sourcing (see sections **2.1.4.5** and **2.2.4.1.1**). As part of onboarding, the service integrator needs to work with each service provider to confirm what needs to be delivered as part of the end-to-end service.

5.3.2.4. Not measuring the correct amount of data and information

Some organizations do not collect enough data, and some collect too much.

If an organization does not collect enough, there is a risk that important information will be missed. If they collect too much, there is a risk that there is too much data to analyze, which can also lead to important information being missed.

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The same is true for how much information is included in reports. Small amounts of information may seem easier to understand but may hide important information. Large amounts may be difficult to understand and can complicate the ability to present an accurate picture.

The challenge is to identify the optimum amount of information to collect and report on. A useful technique is to report at a summary level but have the detailed reports available to support any requirement for more in-depth information.

5.3.3. Recommendations for reporting

Recommendations for enabling and reporting on end-to-end services include:

1. Create a performance management and reporting framework
2. Make reports visual
3. Use qualitative and quantitative measures
4. Apply agile thinking.

5.3.3.1. *Create a performance management and reporting framework*

A performance management and reporting framework provides a way to structure data and information from service measurement and link them to the customer organization's strategic requirements.

The performance management and reporting framework will be created during the Plan & Build roadmap stage.

Performance management and reporting frameworks can be structured in a variety of ways, depending on the available toolset, the strategic requirements and the service contracts.

Possible framework structures include:

- By SIAM ecosystem layers:
 - Service provider metrics
 - Service integrator metrics
 - Customer organization metrics.

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- By type:
 - People metrics
 - Process metrics
 - Service provider metrics
 - Technology metrics.
- By hierarchy, allowing for information to be expanded or shown in more detail when needed:
 - Strategic metrics
 - Tactical metrics
 - Operational metrics.

5.3.3.2. *Make reports visual*

Information is most effective when it is visual and easy to understand. Using service dashboards and scorecards will increase the impact of reporting.

A picture can be easier to understand than a long report, but care must be taken to clearly identify each visual and what it indicates.

5.3.3.3. *Use qualitative and quantitative measures*

Quantitative measures are numerical, and factual; for example, the number of incidents that were resolved in the agreed timescales, or a reduction in the number of target breaches.

Qualitative measures are usually descriptive and often in non-numerical form; for example, customer satisfaction surveys.

Whilst it is relatively easy to measure and report on quantitative measures, they do not often reflect the quality of the service accurately. One of the drivers for SIAM is the 'watermelon' effect, where service providers report they are meeting all their targets, but the customer is still not happy. Using a mix of qualitative and quantitative measures will help to provide a balanced view. Care should be taken to ensure that these remain aligned to strategic requirements and service objectives.

The watermelon effect

The watermelon effect occurs when a report is 'green on the outside, red on the inside'.

The service provider(s) meet individual targets, but the end-to-end service is not meeting the customer's requirements. This does not deliver a good outcome for the customer, and should also be a concern for the service provider.

It may be good for the service provider to be meeting its targets, but if the customer is not happy, it will not support a good long-term relationship.

In this situation, the target is not aligned to business requirements.

5.3.3.4. Apply agile thinking

The application of agile techniques can help to determine the optimum amount of information in reports. Start by reporting on a minimum set of viable metrics. These can provide the minimum amount of information to assess performance, with no unnecessary additions or duplication.

These reports can then be used as the basis for discussion and learning, with more measurements being added if required.

It is often beneficial to start small and then develop the performance management and reporting framework. This approach should initially use less resource than trying to measure every single element of the ecosystem.

5.4. Technology practices

Examples of technology practices in a SIAM ecosystem are:

- Designing integration strategies
- Managing tooling interoperability
- Automating interfaces
- Creating a tooling strategy.

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5.4.1. Creating a tooling strategy

This section will explore technology practices using one common SIAM example – creating a tooling strategy.

A tooling strategy outlines the requirements for a toolset or toolsets to support the SIAM ecosystem. It will include functional and non-functional requirements, the processes that need to be supported, standards for interfacing to the toolset(s) and a roadmap for future development.

Typically, organizations will focus on the IT service management tool, which will support processes including incident, problem, change, configuration, release management and request fulfillment. However, there are other areas where a tooling strategy will provide considerable benefit, such as:

- Discovery, monitoring and event management tools
- Classification and correlation tools
- Analysis and reporting tools
- Remote administration, diagnosis, deployment, and other infrastructure and software management tools
- Work planning and prioritization tools
- Workflow management, communication and collaboration tools
- Knowledge management tools
- Survey tools.

SIAM ecosystems and the tooling strategy

An optimized tooling strategy will make it easier for the service providers in a SIAM ecosystem to work together. It can also:

- Help the service integrator to get a 'real time' view of end-to-end service performance
- Improve the efficiency of workflow
- Support data integration, which is critical in establishing aggregated service views from data provided by multiple service providers

There are several possible approaches for tooling. These are listed in section **2 SIAM roadmap**. The aim is to have integration between all toolsets.

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Integration is difficult to achieve, relying on sophisticated data mapping between service providers and the service integrator. Toolset integration requirements need to be documented and assessed in the context of the broader technology architecture.

In some circumstances, it may be acceptable to use less sophisticated and more manual methods (often referred to as 'loose coupling' of data exchange). For time-critical activities like major incident management, there may be little alternative other than to build integration between toolsets (referred to as 'tight coupling' of data exchange).

The integrated toolset acts as a single version of the truth for all the parties in the SIAM ecosystem, simplifying data transfer, reporting and accuracy.

5.4.2. Challenges related to creating a tooling strategy

Challenges associated with creating a tooling strategy include:

1. Ineffective legacy tools
2. Defining the toolset scope
3. Non-integrated toolsets
4. Lack of architecture.

5.4.2.1. *Ineffective legacy tools*

The customer organization may require the service integrator and/or service providers to use legacy toolsets that it already has in place. This can lead to several challenges:

- The toolset may not support all the processes in the SIAM ecosystem
- It may not support the use of integrated processes
- It will contain legacy data, which may be challenging to adapt to the new ecosystem
- It may be difficult to interface with the service providers and service integrators toolsets
- If the service integrator is external, they may not have any expertise in the toolset.

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5.4.2.2. *Defining the toolset scope*

A SIAM ecosystem can include many processes, some of which are outside the 'standard' set of IT service management processes.

The tooling strategy should encompass all the processes in the SIAM model, and recognize that the ideal solution may be a hybrid of various tools, to support the functional requirements of each process and the broader SIAM ecosystem.

The toolset also needs to support end-to-end process control, not just operational execution. More tool vendors are now creating functionality that supports a SIAM ecosystem.

5.4.2.3. *Non-integrated toolsets*

Where the tooling strategy requires that all parties use the same toolsets, some potential or existing service providers may be unwilling to do this, particularly if it is not mandated in their contract.

If the tooling strategy is that service providers integrate their own tools with the service integrator's toolset, some may be unwilling or unable to configure the integration. For example, providers of commodity services may have little flexibility in their offerings.

The tooling strategy needs careful consideration at the Discovery & Strategy and Plan & Build stages of the SIAM Roadmap, as it has influence on and dependencies within the SIAM structures and the overall SIAM model. The strategy must also consider the data and information standards.

Once they have been agreed, requirements from the tooling strategy should be included in any contracts with service providers as well as any external service integrator. This helps to reduce the risk of service providers failing to adopt the tooling strategy, and minimizes inefficiencies in cross-provider processes, reporting, and gaps between service providers. Where service providers do not integrate their toolset, it is often the service integrator who must undertake additional activities to overcome these gaps.

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5.4.2.4. *Lack of architecture*

The absence of an enterprise architecture and technical architecture for the SIAM ecosystem and services will create challenges related to the selection of toolsets, and the definition of interfaces between toolsets.

The architecture documents need to address:

- The need for data sovereignty/visibility requirements to be addressed through role based access controls. For example, service providers may not be able to view each other's targets or performance
- The need for robust data integration capabilities. Some organizations choose to build an 'enterprise service bus' or messaging engine into their technology architectures to cater for this requirement
- The need for all data update activities to be auditable and traceable
- The need for all parties in the SIAM ecosystem to be familiar with the tooling strategy and the specific tools to be deployed, not only so they can develop any integrations as necessary, but also to ensure that their staff are trained adequately in their use.

The toolset architecture must support the tooling strategy.

The service providers of the toolsets themselves must be treated as a service provider within the SIAM ecosystem, because the effective operation of the SIAM model is dependent on their services.

5.4.3. **Recommendations related to creating a tooling strategy**

Recommendations associated with creating a tooling strategy include:

1. Technology strategy and roadmap
2. Industry standard integration methods
3. Ownership of data and toolsets
4. Ease of adding and removing service providers
5. Adopting a common data dictionary.

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5.4.3.1. *Technology strategy and roadmap*

The customer organization needs to outline its technology strategy and roadmap, to help the service integrator and the service providers understand how the SIAM toolset will integrate and evolve.

The customer organization also needs to share any functional and technical requirements, for example if the toolset must meet certain security specifications.

5.4.3.2. *Industry standard integration methods*

Using industry standard integration methods will make it easier for service providers to share information between their own tools and an integrated SIAM toolset. This will simplify interface creation and should reduce the need for expensive development and customization.

The integration approach adopted should not just cater for data transmission, but also for error handling in the event of issues occurring.

Given the potential criticality of the integration, consideration should also be given to service continuity requirements. Both the production and any back up or continuity environments should be tested to ensure that they meet the functional and non-functional requirements required by the customer organization.

5.4.3.3. *Ownership of data and toolsets*

When the service integrator role is being taken by an external organization, the tooling strategy needs to clarify who owns the toolset, and the data within it.

If the external service integrator owns the toolset (for example), the customer organization needs to ensure it still has data access if the commercial relationship ends; or define how the data will be migrated at such time.

In addition, the toolset must be placed under change control, particularly if data integration exists. If changes to toolsets are made by any party, this can have unexpected effects on the integrity of the supporting data integrations described above, if data fields or values change.

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5.4.3.4. *Ease of adding and removing service providers*

One of the benefits of a SIAM ecosystem is the ability to add and remove service providers easily.

The tooling strategy needs to support this. When a new service provider is added, it needs to be easy for that organization to adopt the toolset, including set-up of local toolset interfaces and training its staff.

When a service provider is removed, it must be simple to remove its access to the toolset and ensure that data is stored or (re)moved as required.

5.4.3.5. *Adopting a common data dictionary*

The toolset should be used to enforce a common data dictionary. This will deliver several benefits, for example giving consistency and a common understanding of incident priority and severity classifications.

There will be confusion if one service provider's 'priority 1' incident is another service provider's 'severity 3'. The common data dictionary allows the service providers to 'translate' their own terminology into one that is common across the SIAM ecosystem. This activity should be undertaken for all data fields in the toolset.

The need for a common data dictionary must be part of the tooling strategy, as the selected toolsets must be able to support its use. The data dictionary must be in place before the SIAM model is operational, as it supports the exchange of data and information across the SIAM ecosystem.

5.5. **SIAM and other practices**

There are many other best or enabling practices, sometimes also referred to as management frameworks, methodologies, approaches, models or standards.

These other practices can complement and support the implementation, operation and improvement in a SIAM ecosystem. Most practices share a common goal, of delivering high quality products and services that lead to customer satisfaction. Each practice provides its own benefits, techniques, and applications.

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Some examples of such practices include:

- Service management practices (including ITIL® and VeriSM™)
- Standards (including ISO® standards)
- Governance practices (like COBIT®)
- Project management practices (like PRINCE2® or PMBOK®)
- Agile, Lean and DevOps
- ADKAR®: for organizational change
- TOGAF®, IT4IT™, and other architectural practices
- CMMI®: for services or process assessments
- SFIA®: the global IT skills and competency framework for the digital world.

Organizations need to assess what exists, or is already in use, and how it can be used within the SIAM model. Two of the most common are existing service management and ISO® certifications

5.5.1. Service management practices

SIAM can be considered a service management practice, but there are many others available (including ITIL® and VeriSM™). Service management practices define the capabilities that support the implementation and management of quality services that meet the needs of the business.

Service management is predominantly performed by the service providers, delivering their services through an appropriate mix of people, processes and information technology. Of course, different service providers may use different service management practices, and indeed the service integrator (or customer organization retained capabilities) may utilize some of these practices too.

In most cases, transitions to SIAM will take place in an environment that already uses some service management practices or elements. SIAM does not necessarily replace these; instead, it builds on service management elements and extends them across the ecosystem where they are relevant to the SIAM model. This might include service management processes, or concepts, practices and techniques.

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SIAM recognizes the need to allow service providers to utilize their own approaches and practices. Using other practices can help these services providers individually, and even the service integrator and customer retained capabilities; but within the SIAM model, these practices need to be adapted to work effectively in a multi-provider ecosystem.

5.5.2. ISO® standards

The International Organization for Standardization (ISO®) is an independent, non-governmental organization, made up of the standards organizations of its member countries. It is the world's largest developer of voluntary international standards.

The standards help businesses increase productivity while minimizing errors and waste. The standards also serve to safeguard consumers of products and services, ensuring that certified products conform to the minimum standards set.

ISO/IEC 20000 is the international standard for IT service management. It was developed in 2005, originally to reflect best practice guidance contained within the ITIL® framework, although it supports other IT service management frameworks and approaches equally. ISO/IEC 20000, Part 14 – Service Integration and Management, is a more recent update which provides guidance on incorporating Service Integration and Management (SIAM) into an existing or developing service management system (SMS).

Whilst standards provide guidance, some are normative or prescriptive. This means they provide independently auditable criteria an organization must adhere to in order for the organization to be certified against the standard.

There are many standards and it will depend on the organization, its service providers and external requirements, to determine which standards need to be integrated into the SIAM model.

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6. SIAM cultural considerations

The SIAM ecosystem and the relationships between the customer organization, service integrator and service providers create a unique environment. From sourcing and contractual negotiations through to governance and operational management, there are specific SIAM considerations.

The cultural aspects of SIAM are one such consideration. An effective SIAM ecosystem is underpinned by effective relationships and appropriate behaviors. The ecosystem culture needs to encourage and reinforce these relationships and behaviors.

Service providers that compete in other areas of a market may find themselves working together to meet overall customer objectives in a SIAM ecosystem. Some service providers might be internal departments of the customer organization, working together with external service providers.

There are specific challenges when an external organization is fulfilling the service integrator role, because they are governing service providers who may also be their competitors.

The cultural considerations examined in this section are:

1. Cultural change
2. Collaboration and cooperation
3. Cross-service provider organization.

6.1. Cultural change

6.1.1. What does this mean in a SIAM ecosystem?

An organization that moves from either an insourced or a traditional outsourced environment to an ecosystem based on SIAM will undergo a large program of change and transformation. If the cultural aspects of the change are not managed effectively, it can create disruption in the customer organization.

Adopting a new SIAM structure can include internal role changes in the customer organization and staff being transferred from the customer organization to the service providers or the service integrator (or vice versa).

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This can have a significant impact on staff at a personal level; they will be concerned about their role, their career and their skillsets. Psychological safety should be actively fostered during this period to reduce fear, build trust, and support successful transition.

Moving to an ecosystem that includes multiple service providers will require the customer organization to build SIAM expertise and capabilities, and an understanding of the ecosystem, the technical landscape and the future roadmap and strategy. This expertise and knowledge might already exist in the organization, but in many transitions to SIAM it does not. Staff will need commercial, contractual and supplier management skills, in addition to more traditional service management skills.

Cultural change will also come from a change of management style. The customer organization needs to manage service provider performance at an executive, not an operational level. Its role is to step in and resolve contractual issues when required and to provide corporate governance. This is a shift away from managing activities to managing outcomes; in other words, managing the 'what', not the 'how'. This requires reinforcing a shared purpose across all service providers to maintain focus on desired outcomes, not just contractual compliance.

The customer organization needs to empower its service integrator to manage the service providers at the operational level. These changes in relationship dynamics and responsibilities will lead to, and depend on, changes in culture.

For the service providers, culture change is driven by the need to work collaboratively. All the service providers need to work with the service integrator and with other service providers towards the shared goal. A culture of transparency, trust, and belonging must be cultivated to enable collaborative success across a diverse ecosystem.

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6.1.2. Why is it important?

No organizational change can succeed without cultural change. If the culture and organizational behaviors stay the same, new processes and ways of working will not be adopted and expected benefits will not be delivered.

Effective management of cultural change will provide the basis for a successful SIAM transformation program, and will help the customer to retain skilled and motivated staff in key roles. It also builds resilience within the ecosystem to adapt to future changes beyond the initial transformation.

6.1.3. What challenges will be faced?

Some of the challenges related to cultural change are:

- Staff who are moving to a new organization can experience concern at a professional and at a personal, emotional level. Professionally, there will be a level of uncertainty over their role and their skills and, emotionally, they will be concerned about the possible impact on their life and career. This can lead to staff leaving, absenteeism and loyalty issues.
- Organizations can suffer from change fatigue if too much is happening at once, leading to a higher chance of changes failing and new behaviors not being adopted.
- People continue using old processes or move back to old ways of working. It is important for every stakeholder in the SIAM ecosystem to reinforce behaviors; for example, at the service provider level, staff need to be encouraged to contact the service integrator, and not the customer. Leaders must model and reward the desired behaviors to embed new ways of working.
- The customer organization's own business outcomes may be affected negatively if the changes are disruptive and have an impact on service delivery.

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6.1.4. How can they be resolved?

These cultural issues can be addressed in several ways, including:

- Having a clear definition of the SIAM model and all associated roles and responsibilities at organization, team and individual levels
- From the customer organization perspective:
 - Implementing a good business change or organizational change management process, reinforced with a strong communication plan to prevent misinformation and rumors spreading
 - Applying program management to the SIAM roadmap, tracking progress and identifying where course corrections are needed to help increase confidence in likely success
 - Considering the use of external consultancy to provide guidance, advice, and an objective view
 - Understanding what retained capabilities are needed and putting plans in place to keep the skilled people in their roles. Retention efforts should also recognize and support the emotional and psychological wellbeing of staff during transition.
- From the customer organization and service integrator perspective, implementing a strong overarching governance structure, supported by processes, which work in practice and not just in theory
- From the service integrator and service provider perspective, aligning their own communication plans with the overall communication plan and measuring the effectiveness of communication
- From a service provider perspective, understanding the organizations it will be working with, how they want to work together, and committing to the level of collaboration required in a SIAM ecosystem. Service providers should actively invest in building trusted relationships across organizational boundaries to enhance collaboration.

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6.1.5. Cultural change and the SIAM structures

Externally sourced	A key challenge for this structure relates to staff who are moving from the customer organization to another organization as part of the transition to SIAM; the professional and personal impact will need to be managed.
Internally sourced	To deliver effective cultural change and a successful transition to SIAM, the customer organization will need skilled people. These may not exist within the organization and could be difficult to recruit.
Lead supplier	As with the externally sourced service integrator structure, a key challenge for this structure relates to staff who are moving from the customer organization to another organization as part of the transition to SIAM; the professional and personal impact will need to be managed.
Hybrid	A key challenge for this structure is that confusion about roles and responsibilities can make it difficult for staff from the customer organization to change their behavior; this applies if the interfaces between the customer organization and external organization at the service integrator level are not clearly defined. Staff need to be clear on their role and the role of the service integrator's employees.

Table 4: Cultural challenges in the SIAM structures

6.2. Collaboration and cooperation

6.2.1. What does this mean in a SIAM ecosystem?

In many cases, a transition to SIAM means that service providers that are used to competing must work together to deliver customer organization outcomes. The relationship moves from competitive to collaborative and this often requires a change in mindset.

In an outsourced environment with no service integration element, service providers may pursue their own objectives. Silos and blame culture are commonplace. Within a SIAM ecosystem, the focus is on relationships, particularly cross-provider relationships, governance controls and pursuit of common goals, rather than achievement of specific individual organizational service levels and objectives.

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In a SIAM ecosystem, service providers need to set aside competitive considerations and adapt to a new way of working. The customer organization and the service integrator must also be clear on their role and the boundaries of their responsibilities. These organizations are likely to be operating in new and unfamiliar ways. Shared purpose and transparency become critical enablers of ecosystem success.

Cultural considerations for effective collaboration and cooperation include:

- Fix first, discuss later: when there is an issue affecting service, the service providers need to work together rather than assign blame or pass issues around
- Service providers must acknowledge that the service integrator is the voice of the customer, and has the autonomy to direct and make decisions and govern without being undermined
- From the customer organization's perspective, it must empower the service integrator to manage the service providers, and not interfere or duplicate effort
- Creating an ecosystem that is focused on business outcomes and customer value, rather than individual service provider's contracts and agreements. Recognition and reward mechanisms should reinforce collaborative behaviors.

6.2.2. Why is it important?

In a SIAM ecosystem, the service integrator does not usually have a contractual relationship with the service providers, but it does need to be able to manage and govern their behavior on behalf of the customer organization.

If the parties in the ecosystem are not prepared to collaborate, the service integrator will not be able to control service delivery effectively.

For example, it will be very challenging for the service integrator to manage a major incident end-to-end and within service targets if the service providers fail to share information or accept responsibility for investigation.

6.2.3. What challenges will be faced?

Some of the challenges related to collaboration and cooperation include:

- From the service integrator's perspective, the challenge of service providers bypassing it and going straight to the customer organization. The customer organization needs to support the service integrator by reinforcing correct communication paths, and the service integrator needs to build relationships and promote correct ways of working
- From the service provider's perspective:
 - 'Fix first, discuss later' being misused so that it incurs additional costs. This can happen if issues are identified and not corrected by the customer organization or the service integrator, so that the service provider must deal with them repeatedly
 - Being reluctant to collaborate and share with the other service providers
- A lack of trust can be a challenge, as trust is a critical success factor for collaboration and cooperation. Trust between service providers (some of which may be internal, and some external), trust between the service providers and the service integrator, and trust between the service integrator and the customer organization must be built and maintained. Trust should be treated as a measurable, strategic objective, not just an aspiration.
- In a SIAM ecosystem that includes internal and external service providers, the internal service providers are part of the same organization as the customer organization. They may be reluctant to collaborate with the service integrator and with the external service providers; they may also have less mature delivery capabilities and so are less able to cooperate
- In a SIAM ecosystem that includes internal and external service providers, the internal service providers may lack the contractual obligations that compel external providers to collaborate.

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6.2.4. How can they be resolved?

These cultural issues can be addressed in several ways, including:

- For all parties:
 - Creating a 'code of conduct' or 'rules of the club' agreement, with input from all parties in the SIAM ecosystem. These govern behaviors on a day-to-day basis; for example: how staff will behave in meetings, they will maintain professional and courteous behavior always and will attend forums and make effective contributions. (see section **6.2.4.1**)
 - Signing collaboration agreements that are part of each contract, or agreed between parties after the contract is signed, to add more detail about how they will work together (see section **6.2.4.2**)
- Between the service integrator and each of the service providers, use operational level agreements (OLAs) to break down service targets and agreements into more detail, helping them to understand their role and their interfaces with other parts of the ecosystem and when collaboration and cooperation is required (see section **6.2.4.3**).

Operational level agreement (OLA)

An operational-level agreement (OLA) defines *"interdependent relationships in support of a service level agreement (SLA). The agreement describes the responsibilities of each support group toward other support groups, including the process and timeframe for delivery of their services. The objective of the OLA is to present a clear, concise and measurable description of the internal support relationships."*¹⁷

In a SIAM model, OLAs establishes more formal relationships where there are no contracts, such as between the service providers and the service integrator, or in some instances, between service providers.

¹⁷ Source: Wikipedia

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6.2.4.1. *Example code of conduct*

A code of conduct (or 'rules of the club') document is not a contractual agreement. It provides high-level guidance describing how the parties in the SIAM ecosystem will work together. All the parties can then hold one another accountable, for example, by addressing inappropriate behavior in meetings.

This document is typically informal and concise, often no longer than a single page. Where required, it may include:

- Title page
- Document control: author, date, status, version, change log etc.
- Contents
- Introduction and document purpose
- Parties to the document
- Validity
- Approval/signatories.

The suggested key content is:

Partnership aims

- What are the expected business outcomes?
- What is the SIAM ecosystem meant to deliver?

For example:

- Better value for money
- Greater efficiency and cost-effectiveness
- Greater flexibility to respond to evolving business requirements.

Partnership ethos

- What values do the parties need to uphold?

For example:

- Maintain professionalism
- Work together as one team
- Share knowledge and ideas
- Embrace change
- Put the customer first
- Be courteous/respectful to others.

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6.2.4.2. *Example collaboration agreement*

An effective collaboration agreement will help to create a culture based on working together to deliver shared outcomes, without continual reference back to contracts.

Collaboration agreements should be carefully constructed to clearly define how service providers are expected to collaborate with each other, and with the service integrator.

Sufficient detail should be included to minimize ambiguity, and to reduce the risk of future disputes, particularly where a service provider may not have been aware of specific collaboration expectations such as, the need to take an active role in forums.

Consideration must be given to the remedial approach that will be used if one of the parties does not align with the collaboration agreement. This can form part of the contract with the customer organization but, to be truly effective, the parties should embrace the collaboration agreement as part of the SIAM culture, and not see it as a contractual requirement.

A typical collaboration agreement will contain the following sections:

- Title page
- Document control: author, date, status, version, change log etc.
- Contents
- Introduction
- Document purpose
- Parties to the document
- Validity period
- Termination
- Required behaviors: for example, avoid unnecessary duplication of effort, do not hinder or withhold information from other service providers
- Mechanisms to support collaboration: for example, commitment to support forums to review, improve and innovate processes and service delivery; commitment to triage and work together to resolve issues/challenges when requested by the service integrator; commitment to be involved in reviews and assurance activities
- Where relevant, toolsets to support collaboration

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- Expected areas of collaboration: for example, review of proposed changes, incident investigation, taking part in working groups, innovation
- Dependencies between parties
- Any non-financial/non-contractual remedies: for example, where one or more service providers agree with the service integrator and the customer to carry out actions to address an issue rather than trigger a contractual target and accept a financial penalty
- Change control
- Dispute resolution and escalation points.

6.2.4.3. *Example operational level agreement*

Operational level agreements (OLAs) between the service integrator and service providers break down service targets into more detail. They include guidelines and common ways of working that support effective integration and delivery across the ecosystem. Whilst the contents of OLAs are not themselves contractual obligations (all contracts are held by the customer organization), they should be formal agreements that are documented and controlled.

OLAs support end-to-end service delivery. For example, end-to-end incident management might include a four-hour resolution time for priority one incidents. In the OLA, the service provider might agree to a target of 30 minutes to either accept an incident or pass the information to a different service provider.

Within a SIAM ecosystem, it is important to define each service and its associated targets fully. OLAs support that definition and provide control and visibility. OLAs are prepared by the service integrator with input from the service providers. The service provider referenced in the OLA must have agreed the contents. The OLA supports the customer organization's overall objectives, but the customer organization may not be interested in the document detail.

OLAs can also be set up between two or more service providers, to agree or add some detail about how they work together.

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An operational level agreement should include content such as:

- Title page
- Document control: author, date, status, version, change log etc.
- Contents
- Introduction
- Document purpose
- Parties to the document
- Validity
- Approval/signatories
- Rules for OLA termination
- Rules for OLA governance and escalation criteria
- Review schedule
- OLA change management
- Service description
- Scope of OLA: in- and out-of-scope activities
- Dependent services
- OLA details:
 - Name of service (for example, service desk, capacity management)
 - Service description
 - Service hours
 - Service provider
 - Service consumers
 - Service outcomes
 - Contact points and roles
 - Agreed activities of all parties (for example, party A will send an incident record to party B, party B will confirm receipt)
 - Service targets
 - Measurement: availability, performance targets etc.
 - RACI matrix
- Service boundaries
- Quality assurance and service reporting
- Service reviews
- Glossary.

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6.2.5. Collaboration and cooperation and the SIAM structures

Externally sourced	- Internal service providers may be unwilling to collaborate and cooperate with the external service integrator - If the customer organization lacks SIAM maturity, the service integrator may struggle to establish the cultural alignment needed across the ecosystem. Without strong sponsorship and cultural support from the customer, collaboration and cooperation can be undermined.
Internally sourced	- External service providers may be more willing to collaborate and cooperate as it will perceive the service integrator as the customer organization - If the customer organization lacks SIAM maturity, the service integrator may struggle to establish the cultural alignment needed across the ecosystem. Without strong sponsorship and cultural support from the customer, collaboration and cooperation can be undermined - If the internal service integrator is perceived (or seen) to treat internal service providers differently this can lead to reduced collaboration and cooperation by the external service providers - Internal service providers may be unwilling to collaborate and cooperate with the internal service integrator.
Lead supplier	- Internal service providers may be unwilling to collaborate and cooperate with the external service integrator - If the lead supplier is seen to favor its own service provider in its service integrator role, this can lead to reduced collaboration and cooperation from the other service providers - If the customer organization lacks SIAM maturity, the service integrator may struggle to establish the cultural alignment needed across the ecosystem. Without strong sponsorship and cultural support from the customer, collaboration and cooperation can be undermined.
Hybrid	- The roles and responsibilities of the customer organization acting as service integrator and the third-party service integrator need to be clear. It is challenging for service providers to collaborate if they do not understand the structure and boundaries of responsibilities - Internal service providers may be unwilling to collaborate and cooperate with the hybrid service integrator because it includes external elements - If the customer organization lacks SIAM maturity, the service integrator may struggle to establish the cultural alignment needed across the ecosystem. Without strong sponsorship and cultural support from the customer, collaboration and cooperation can be undermined.

Table 5: Collaboration challenges in the SIAM structures

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6.3. Cross-service provider organization

This section addresses the cultural aspects of cross-service provider organization only. For more detail about managing cross-functional teams and managing conflict see section **5.1.1 Managing cross-functional teams**.

Cross-service provider organization describes the cultural considerations associated with managing a service across multiple service providers.

6.3.1. What does this mean in a SIAM ecosystem?

The SIAM ecosystem can include an internal, hybrid, external, or lead supplier service integrator, plus several internal or external service providers.

Each service provider will have its own strategies, objectives and ways of working. The customer organization does not always have the ability (or desire) to mandate that all service providers follow a common set of processes or use the same toolset. They do, however, require all service providers to be able to interface with and integrate into the end-to-end service management processes. A well-defined process model provides the bridge that enables this integration while respecting each service provider's individual approaches.

From a cultural perspective, a cross-service provider organization requires service providers to have appropriate behaviors and attitudes to support the customer organization and help them achieve its goals, rather than focusing on individual goals. This includes openness to co-create solutions and share knowledge across organizational boundaries.

6.3.2. Why is it important?

Successful cross-service provider organization supports delivery of the end-to-end service. It starts with the customer organization which needs to articulate a clear vision of what success looks like to all the service providers in the SIAM ecosystem.

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The vision needs to be cascaded through all layers and across the ecosystem. This will then enable consistent:

- Strategies
- Objectives
- Processes; this does not preclude service providers using their own processes and procedures, but assures that the overall end-to-end process is integrated, can be managed, and is driving the correct outcomes (see section **2.2.4.1.3 Design process models**).

6.3.3. What challenges will be faced?

Some of the cultural challenges related to cross-service provider organization include:

- The customer retained organization may step back into its old role and get involved with delivery, rather than focusing on corporate governance and its own business objectives. This creates confusion, duplication and does not allow the service providers, the service integrator, or the customer to work effectively
- Service providers focus on their own targets at the expense of end-to-end service targets
- Service providers do not subscribe to the culture of collaboration and do not share innovations and potential improvements with other parts of the SIAM ecosystem
- The service integrator does not treat service providers equally, leading to resentment and disengagement
- If service providers have poor interfaces with end-to-end service management processes and tools, the role of the service integrator becomes more challenging; for example, monitoring, reporting, measurement will all be less effective.

6.3.4. How can they be resolved?

These cultural issues can be addressed in several ways, including:

- For all parties:
 - Establishing consistent contractual targets/service level agreements and common performance measures/key performance indicators across the supply chain, so that all service providers feel they are equal and not disadvantaged
 - Having performance measures that encourage partnering and shared innovation with other parties
 - Cross-service provider processes must be based on a common language that all parties can understand (see section **5.4.3.5 Adopting a common data dictionary**)
 - Facilitating forums for cross-provider retrospectives, allowing collective identification of improvements and fostering shared ownership of outcomes
- From the customer organization perspective, empowering the service integrator to have ownership, responsibility and accountability
- From the customer organization, service integrator, and service provider perspective, celebrate success, praise excellent service performance, delivery and innovation, to emphasize and reward the desired behaviors
- Establish 'champion' forums with representation from the service integrator and all service providers to discuss and improve the effectiveness of end-to-end processes, tools, interfaces and integration.

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6.3.5. Cross-service provider organization and the SIAM structures

Externally sourced	External service providers may be reluctant to share information about how they work with an external service integrator, if they view them as a competitor.
Internally sourced	Internal service providers may be unwilling to work with and integrate with external service providers.
Lead supplier	The external organization that is acting as the service integrator and a service provider could be seen as a 'favorite' of the customer organization. If other service providers do not feel they are being treated fairly, they are less likely to work together.
Hybrid	Effective cross-service provider organization requires clear direction from the customer organization and the service integrator. If the roles within the hybrid service integrator are not clearly defined, meetings and structures to support cross-service provider organization may not be put in place.

Table 6: Collaboration challenges in the SIAM structures

7. Challenges and risks

Adopting a SIAM model requires an organizational transformation. The changes that are involved will affect people, processes, technology and the interfaces between them.

As with any organizational change, there will be challenges to face. These challenges can have a significant impact on the transition to a SIAM model and will require concerted effort to overcome.

Each challenge has associated risks, which need to be recorded, evaluated, managed and mitigated (where appropriate) using a risk management methodology. The impact of the challenge and associated risks to an organization will influence the amount of time and resources that will be used to address them.

The challenges and risks described here should be considered by any organization planning to adopt SIAM. They may not all be relevant, but can provide useful input for SIAM planning.

7.1. Challenge: Building the business case

Organizations must be clear about their business case for SIAM. This should include the expected benefits and costs.

It is not always possible to have a complete picture during the Discovery & Strategy stage of the roadmap, as some of the detail may not be known until the Plan & Build stage. However, it is usual to create an outline business case before starting a SIAM implementation that will be developed throughout the SIAM roadmap into a full business case.

The business case should include the drivers that apply to the organization, drawn from the five SIAM driver groups (documented in section **1.5.2 Drivers for SIAM**):

1. Service satisfaction
2. Service and sourcing landscape
3. Operational efficiencies
4. External drivers
5. Commercial drivers.

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The business case also needs to articulate the benefits that the customer organization expects to achieve by adopting a SIAM model. These could include, for example:

- Mitigating the risk of procuring services from a single provider by leveraging best of breed services from several providers
- Improving its capability to add and remove service providers
- Improving the quality of service
- Increasing the value derived from services.

These benefits will only be achieved through clear objectives, robust planning, and effective management.

7.1.1. Which parties will this challenge affect?

This challenge mainly applies to the customer organization that is creating the business case. It can also apply to an organization that intends to be an external service integrator, as it will need to be able to justify the cost of its services in relation to the benefits as stated in the business case.

7.1.2. Which roadmap stage will this challenge affect?

This challenge starts early in the SIAM roadmap, during Discovery & Strategy, and continues all the way through.

At the end of the Discovery & Strategy stage, executive backing is required to approve the outline business case for SIAM, and to allocate resources to the next stage.

At the end of the Plan & Build stage, executive backing is required to authorize any procurements and allocate resources to the remaining stages. The business case will also be used during the Implement and Run & Improve stages to verify that the anticipated benefits are being realized.

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7.1.3. Associated risks

Without a strong business case, there are several risks, including:

- The customer organization's executives do not approve the transition to SIAM
- The customer organization's executives approve the transition to SIAM, but do not allocate enough resources or provide sufficient support
- The customer organization starts the transformation process without a clear picture of the benefits it expects to achieve; this will make it difficult to verify if the transition to SIAM has been successful
- The success of the program cannot be measured because anticipated benefits have not been clearly defined
- The costs of the transition to SIAM are understated, so there may not be enough budget available to complete the transition.

7.1.4. Potential mitigation

These risks can be mitigated by:

- Allocating skilled resources to build the business case
- Executing all the activities in the Discovery & Strategy and Plan & Build stages
- Linking the strategy for SIAM to the customer organization's high-level strategy and objectives
- Identifying and addressing each of the appropriate drivers for the SIAM transformation
- Adding as much detail as possible and refining the business case as the roadmap proceeds
- Identifying existing contracts that are inefficient, to ensure these are not replicated in new contracts within the SIAM ecosystem
- Identifying contracts that deliver good value and encourage a good cultural fit, to replicate these within the SIAM ecosystem
- Using industry data/benchmarks where available to show the benefits of SIAM in other organizations
- Including the proposed SIAM structure and SIAM model
- Documenting the expected benefits.

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7.2. Challenge: Level of control and ownership

During the Discovery & Strategy stage of the roadmap, the customer organization needs to consider how it will balance the level of control it wants to have over service provision, processes, tools and data against the benefits it will obtain from delegating them to a service integrator. This decision is then confirmed in the Plan & Build stage.

This balance also depends on the level of trust in the ecosystem. Trust between all parties is essential in SIAM ecosystems. A lack of trust can result in duplication of roles and activities, such as the customer organization continually checking what the service integrator has done. The customer organization might be unable to let go of activities it used to perform.

A lack of trust may also result in micro-management, for example, the service integrator reviewing every aspect of every service provider action.

Such issues will increase cost, leading to savings and efficiencies not being realized. It can also cause confusion, inconsistency, poor relationships, and a lack of collaboration.

7.2.1. Which parties will this challenge affect?

This challenge mainly applies to the customer organization while it decides on its preferred SIAM structure and model, and sets policies related to roles, responsibilities, data and tooling.

If this challenge is not resolved, it can make the definition of the SIAM model, and the role of the service integrator and the service providers, more challenging because responsibilities and accountabilities are unclear.

Micro-management or a lack of trust embedded within the model can affect the customer organization, the service integrator and the service providers.

7.2.2. Which roadmap stage will this challenge affect?

The level of control and ownership needs to be decided at a high-level during Discovery & Strategy, and more detail added during Plan & Build.

This challenge will usually be identified after SIAM has become business as usual, in the Implement and Run & Improve roadmap stages.

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7.2.3. Associated risks

If the level of control and ownership is not clearly defined, associated risks include:

- If the customer organization is not prepared to relinquish ownership of service activities and processes, it may not be possible to realize the anticipated benefits from SIAM, as the service integrator may be unable to perform its role
- The SIAM organization is larger than needed as process and service activities are duplicated by the customer organization and the service integrator
- The customer organization wastes time and resources micro-managing and checking the work of the service integrator
- The service integrator wastes time and resources providing extra, unnecessary customer reports
- The service integrator wastes time and resources micro-managing and checking the work of the service providers
- The service providers continue to interact directly with the customer organization, because they see the customer does not value the service integrator (or they do not trust the service integrator)
- If the customer organization relinquishes all control and accountability, the service integrator may lack the strategic direction to allow it to carry out its role.

7.2.4. Potential mitigation

These risks can be mitigated by:

- Defining a clear vision and selecting an appropriate SIAM structure and SIAM model during the Discovery & Strategy, and Plan & Build stages of the roadmap
- Ensuring that the customer organization understands the difference between governance and management, so it is clear what activities it needs to monitor, and what it needs to do. This will form part of the governance framework
- Careful design of the SIAM model in the Plan & Build stage, particularly roles, responsibilities, and the governance framework
- Implementing clear policies for data, tooling and processes
- Defining ownership of processes, tools, data, information and knowledge

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- Adopting a phased approach to the implementation of SIAM to allow the customer organization to develop trust in the service integrator and test the level of control (not too much, not too little ... just right)
- Having regular communication and a culture of improvement to identify and discuss micro-management and duplication of effort
- Establishing effective structural elements to support relationships, communication and build trust.

7.3. Challenge: Commercial challenges

Commercial challenges relate to how the SIAM model is established and the structure that is chosen. The customer, the service integrator and the service providers all need to feel they are getting what they expect and are being treated fairly.

If a customer organization does not have mature SIAM capabilities, the commercial agreements it puts in place may not be appropriate.

For existing service providers, there may be challenges associated with legacy contracts. The two main challenges are:

1. Not fit for purpose: some legacy contracts might still be in place after the SIAM implementation. The contractual requirements in the legacy contract with the service provider are unlikely to align with the new SIAM model and may not support collaboration, shared risk, or end-to-end service commitments required in a SIAM ecosystem.
2. Expiry: continuity of service may be compromised if legacy service provider contracts expire before the implementation of new service provider contracts.

It is important to recognize that service providers who are not going to be part of the future operating model may be challenging to deal with. Clear transition planning and communication strategies are critical to manage these exits and minimize operational disruption.

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7.3.1. Which parties will this challenge affect?

All the parties in the SIAM ecosystem can be affected by this challenge:

- The customer organization needs to feel they are getting value for money
- Externally sourced service integrators and all service providers need to be profitable and not incur penalties they see as unfair
- The service integrator and the customer organization need to have an appropriate commercial framework to govern and incentivize the service providers.

7.3.2. Which roadmap stage will this challenge affect?

This challenge will span the entire roadmap. Legacy contracts that may not be fit for purpose should be identified during the Discovery & Strategy roadmap stage. Commercial decisions will also be made during Discovery & Strategy, and then detail will be added and contracts defined during Plan & Build.

The effects will need to be monitored during the Implement and Run & Improve stages, and remedial activities identified where necessary.

Contract expiry will affect the Plan & Build stage, and can also affect the subsequent Implement and the Run & Improve stages.

7.3.3. Associated risks

Commercial risks include:

- Unrealistic targets and service levels for service providers may result in their withdrawing from the ecosystem
- Lack of clearly defined boundaries between service providers make it challenging to allocate responsibility for service failures
- The service integrator is managing the service providers from a SIAM perspective but does not have any direct contracts with them; unless the right level of empowerment is in place, the service integrator may not be effective
- The service providers impose their own contracts that have targets that do not align with end-to-end service requirements (for example, when the service provider is a very large organization, it may have a standard set of service levels, see section **5.2.2.1**)

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- Misalignment between commercial terms and SIAM governance processes, leading to delays in decision-making or inability to enforce corrective actions
- Increased service integrator workload, when unexpired legacy contracts need to be integrated into the SIAM model
- A gap in service if a legacy contract expires before a new SIAM-based contract is in place
- Additional customer costs for extensions to existing contracts during the transition to SIAM, or for early release from an existing contract.

7.3.4. Potential mitigation

Commercial risks can be mitigated by:

- Getting the right skills and experience involved during the procurement process, including contractual negotiations
- Defining service boundaries and service interactions (see section **2.2.4.1.1**)
- Including in service provider contracts that the service integrator is the managing agent of the customer organization, with devolved authority for managing delivery against contracts
- Making sure targets and service levels flow down and are apportioned across service providers
- Building flexibility into contracts to allow for continual service improvement, changes to scope, and onboarding/offboarding of providers without material disruption
- Ensuring that penalties and service credits can be calculated correctly
- Having clear and unambiguous contracts
- Scheduling regular reviews to assess if contracts are performing as they should.

Risks associated with contracts considered to be not fit for purpose or expiring too early can be mitigated by:

- Understanding to which contracts this relates and creating a timetable to project how long the risk will exist for
- Developing a roadmap for the transition to SIAM that aligns with existing expiry dates as far as possible
- Sharing the new SIAM vision with the existing service providers

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- Renegotiating/amending contracts where possible; it is possible that requirements, SLAs, measurements or end dates could be changed
- Investigating the cost of termination.

7.4. Challenge: Security

Security is a generic challenge for any modern organization. This is particularly true in a SIAM ecosystem, which requires sharing of data and information about services across multiple service providers. A "security by design" approach should be adopted from the outset, not added as an afterthought, with security embedded in every layer, through roles, responsibilities, communication and reporting.

The customer organization needs to be clear about what data and information exists in the ecosystem, where it is, and how it will be managed and secured.

7.4.1. Which parties will this challenge affect?

This challenge affects the customer organization, the service integrator and the service providers; each party has a responsibility for the overall security of the service. However, ultimate accountability for security breaches and regulatory compliance will usually rest with the customer organization.

7.4.2. Which roadmap stage will this challenge affect?

If security related roles and activities are not clearly defined during Plan & Build, the impact will be felt in later roadmap stages.

For example, a security incident during 'Run' might take longer to discover because no one party is responsible for detecting it. The response could also be slow because the service providers do not have clarity on individual responsibilities.

7.4.3. Associated risks

Potential risks associated with security include:

- A lack of understanding of the customer organization's legislative and regulatory responsibilities, and a lack of education for the service integrator and service providers to make them aware of these
- A lack of understanding of the criticality of information, and no agreed approach to managing information
- Inability to map data flows and the end-to-end service, to identify what is in scope for security
- Security roles and responsibilities not mapped and allocated
- Process inadequacies, including a lack of access management for service providers to ensure they can only access what is necessary
- Ineffective data segregation, particularly in relation to a service provider's commercially sensitive data that should not be visible to other service providers
- If roles are not clear, security tasks might be duplicated, leading to wasted effort, or not managed, leading to service unavailability and security breaches
- Misalignment between provider-specific security policies and the overall SIAM security framework.

7.4.4. Potential mitigation

These risks can be mitigated by:

- Having a clear security strategy and supporting policies that are cascaded to all service providers via the service integrator
- Using other practices and frameworks, that focus on governance, service mapping, and information flow modelling, to help identify information assets and data flows
- Designing and implementing end-to-end security management
- Implementing effective processes such as access management
- Identifying and completing security activities when adding and removing service providers; service providers being added need enough access to be effective, and service providers that are being removed need to have access terminated
- Establishing clear reporting lines and escalation paths for security incidents within the SIAM model
- Creating a schedule for audits and testing

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- Encouraging a culture of openness so service providers are confident to share information about a breach
- Establishing a process forum for security
- Embedding shared responsibilities for detection, response, and recovery across providers, with joint playbooks and simulation exercises.

7.5. Challenge: Cultural fit and behaviors

Different service providers will have different corporate cultures, which all need to work within the SIAM ecosystem.

Service providers need to work together to meet customer organization outcomes, often with organizations with whom they are in competition in the broader marketplace.

Existing service providers might not be willing to change to adapt to the SIAM model. When a major organizational change happens, it is easy and tempting for staff to revert to old ways of working with which they are more familiar.

This might mean that the intended value of the SIAM implementation is not realized, because the implementation does not become business as usual or an accepted way of working. An effective SIAM ecosystem relies on much more than just contractual agreements. It also relies on good relationships between the customer organization, the service integrator and the service providers.

New service providers also need to be a good cultural fit and display the required behaviors (see also section **6 SIAM cultural considerations**).

7.5.1. Which parties will this challenge affect?

This challenge affects the customer organization, the service integrator and the service providers; each party has a role to play in cultural change and should adopt required behaviors to make the SIAM model successful.

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7.5.2. Which roadmap stage will this challenge affect?

This challenge needs to be addressed initially in Plan & Build. The issues will increase during the Implement stage, and may worsen during Run & Improve.

In the Implement stage, SIAM is often the new way of working and all parties need to work together to implement the SIAM model. In Run & Improve, it needs to become business as usual. Behaviors need to be reviewed continually, and revisited if key staff change or a new service provider is added.

7.5.3. Associated risks

Potential risks associated with cultural fit include:

- Service provider staff circumventing the service integrator to talk directly to the customer, and vice versa
- One or more service providers not engaging fully
- Frustration for all parties if a service provider says one thing and does another
- Service providers not working well together
- Service providers not interacting with end-to-end processes and procedures
- The service integrator being unable to fulfill its role because the service providers are not working together effectively
- The service integrator is perceived as being biased
- The customer organization or service integrator acts in a dictatorial way and does not have good relationships with the service providers
- The customer organization and the service integrator do not present a united front
- Cultural issues can lead to SIAM benefits not being delivered to the customer organization
- Loss of trust across the ecosystem, leading to operational inefficiencies and finger-pointing during service failures.

7.5.4. Potential mitigation

These risks can be mitigated by:

- Being aware of the risk of cultural mismatch, and planning to identify it and intervene where required
- Assessing cultural fit during procurement and selecting service providers who will be a good cultural fit

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- Regular behavior reviews and audits
- Rewarding good behaviors
- Encouraging a culture of collaboration
- Using collaboration agreements (see section **6.2.4.2 Example collaboration agreement**)
- Demonstrating correct behavior at the customer and service integrator level, presenting a united front to service providers
- Continual reinforcement of correct behavior at all levels
- Ongoing training and awareness for staff
- Establishing SIAM structural elements (boards, forums and working groups) to build relationships and reinforce culture
- Providing regular communication to build relationships, based on a communication plan that identifies relevant stakeholder groups and describes a communication strategy for each of them
- The customer and service integrator being realistic about what can be achieved by working with service providers, rather than punishing them
- Fostering a one team mindset through integrated objectives, shared KPIs, and collective celebration of successes.

7.6. Challenge: Measuring success

To show that SIAM is delivering value, it must be measured. Developing an end-to-end performance management and reporting framework that spans multiple service providers can be a significant challenge.

7.6.1. Which parties will this challenge affect?

This challenge will affect the customer organization if it is unable to validate whether SIAM is delivering value and services are performing and the service integrator that has the task of building the end-to-end reports.

7.6.2. Which roadmap stage will this challenge affect?

This challenge will usually happen during Run & Improve, when the customer organization tries to measure the effectiveness of SIAM in the business as usual environment.

The measures should be defined during Plan & Build, linked to the original drivers for SIAM identified in Discovery & Strategy. Measurements will need to evolve when improvement activities take place.

7.6.3. Associated risks

Potential risks associated with measuring success include:

- Measures not aligned with the anticipated benefits from the business case
- Not measuring and reporting on the right things
- Measuring too much, which wastes resources and can obscure important information
- Not measuring enough to identify the required information
- Being unable to measure services from end-to-end
- Difficulty attributing performance issues when accountability crosses multiple service providers.

7.6.4. Potential mitigation

These risks can be mitigated by:

- Creating an effective performance management and reporting framework
- Clearly defining who needs to measure what, when, how and why
- Regularly reviewing reports to confirm they are still fit for purpose
- Using a mix of qualitative and quantitative measures
- Building customer-centric outcomes into all reporting, beyond operational service-level metrics
- Establishing measurement hierarchies to connect tactical service metrics to strategic business goals.

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Appendix A: Glossary of terms

This glossary provides a list of terms, and their descriptions, used in this document.

Term	Description
Accountability	The state of being accountable, liable or answerable
Aggregation	Bringing together components and elements to create a group or service (also referred to as Service aggregation)
Agile	A set of values and principles under which requirements and solutions evolve through the collaborative effort of self-organizing cross-functional teams
Artificial Intelligence (AI)	The simulation of human intelligence in machines that are programmed to think and learn
Assets	The resources and capabilities of an organization, used to create value in the form of goods and services
Automation	The practical application of technology to reduce human effort, increase speed of processing and response, and improve accuracy in managing service delivery and service provider interactions
Benchmark	A standard or point of reference against which things may be compared
Best of breed	The best system or provider in its referenced niche or category
Board	A type of structural element that is a formal decision-making body, and is accountable for governance in the SIAM ecosystem
Business As Usual (BAU)	The normal activities of a business that are necessary to maintain its operations
Business case	A document that supports decision-making by outlining a proposed course of action, its potential costs and benefits
Business Relationship Management (BRM)	A business framework that strategically nurtures positive relationships to achieve three capabilities: driving value, evolving culture and building partnerships (source BRM Institute)
Capability	An organization's ability to collectively use its resources to achieve its goals and deliver value.
Cloud services	Services that are provided over the internet, including Software as a Service (SaaS) , Infrastructure as a Service (IaaS) and Platform as a service (PaaS)
Code of conduct	A document that is not a contractual agreement, but provides high-level guidance for how the parties in the SIAM ecosystem will work together (also referred to as 'rules of the club')
Collaboration	The situation of two or more people working together to create or achieve the same thing
Collaboration agreement	An understanding that helps to create a culture based on working together to deliver shared outcomes, without continual reference back to contracts
Commodity service	A service that can easily be replaced; for example, internet hosting is often a commodity service
Common data dictionary	A central repository or tool that provides detailed information about the business or organization's data and defines the standard definitions of data elements, their meanings and allowable values

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Term	Description
Competency	The combination of knowledge, skills, abilities, behaviors, and experience that a person has that contributes to individual performance of a job role
Consumer	Provides the requirements for products and services, receives products and services, gives feedback and participates in the verify/review/improve activities
Contract	A legally binding agreement between two or more parties that establishes their rights and obligations for a specific exchange of goods, services, or promises
Cooperation	The act of working together with someone or doing what they ask you, to achieve your own goal (i.e., win-win)
Cross-functional team	A group of people with different functional expertise working toward a common outcome
Customer (organization)	An organized entity with a particular purpose. In a SIAM context, the customer organization is the end client that commissions the SIAM model, owns contracts with any external service providers, and receives the service/SIAM benefits.
Customer experience (CX)	The sentiment expressed by the customer when acquiring or using the service. 'Customer' is usually interpreted as the part of the client organization that purchases the service for its end users, but may be identical to the user in some contexts (see also User experience).
Dashboard	Provides 'at a glance' views of data, metrics etc. relevant to a particular objective or business process
Data	Raw facts or figures collected from systems, processes, or events (for instance, operational logs, user survey responses, or transaction counts)
DevOps	Represents a change in IT culture, focusing on rapid IT service delivery through the adoption of Agile , Lean practices in the context of a system-oriented approach
Driver	Something that creates and fuels activity, or gives force and impetus
Early life support (ELS)	Last phase of a transition, before handover to operation, during which specific support is provided for the implemented change
Ecosystem	An ecosystem is a network of diverse, interdependent entities that collaborate and compete to create collective value. The SIAM ecosystem includes three layers: customer organization (including retained capabilities), service integrator and service provider(s)
Enabler	Any capability that facilitates or accelerates the achievement of desired outcomes. Enablers do not directly deliver value themselves but support, enhance, or unlock the ability of other components to do so.
Enterprise	A collection of organizations that have common goals
Enterprise architecture	A definition of the structure and operation of an organization, mapping the current state and can be used to support planning for desired future states
Enterprise service bus	A type of 'middleware' that provides services to link more complex architectures
Environment	The surroundings or conditions which influence operations
Escalation	Raising the focus, seriousness or superiority of a subject
Ethical wall	Screening mechanism that prevents a conflict of interest by ensuring people and/or organizations do not share certain information regarding another person or organization

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Term	Description
External service provider	A separate legal entity that provides services and is not part of the customer organization
Externally sourced service integrator	Type of SIAM structure: the customer appoints an external organization to take the role and provide the capabilities of the service integrator
Forum	A SIAM structural element, often aligned to specific processes, whose members work together on proactive development, innovations and improvements
Framework	A system of rules, ideas or beliefs that is used to plan or decide something
Function	An organizational entity, typically characterized by a special area of knowledge or experience
Governance	The rules, policies, processes, legislation and regulations by which businesses are operated, regulated and controlled. These define and ensure the required levels of authority, decision-making and accountability.
Governance board	See Board
Governance framework	Within a SIAM ecosystem allows the customer organization to exercise and maintain authority over the ecosystem. It includes corporate governance requirements, controls to be retained by the customer, governance structural elements, segregation of duties, and risk, performance, contract and dispute management approaches
Governance model	Based on the governance framework and roles and responsibilities, this includes scope, accountabilities, responsibilities, meeting formats and frequencies, inputs, outputs, hierarchy, terms of reference and related policies
Hybrid service integrator	Type of SIAM structure: the customer collaborates with an external organization to take the role of service integrator and provide the service integrator capability
Improvement	The process of making something better or of getting better
Information	Refers to the data that is essential for effective decision-making and operations
Infrastructure as a Service (IaaS)	A type of cloud service that allows customers to access virtualized computing resources
Innovation	A new idea or method
Insourcing	Sourcing from within the organization
Intelligent client function	See Retained capabilities
Interdependency	The dependence of two or more people or things on each other
Internal service provider	A service-providing entity that is part of the customer organization, its performance is typically managed using internal agreements and targets
Internally sourced service integrator	Type of SIAM structure: the customer organization takes the role of service integrator, providing the service integration capability
International Organization for Standardization (ISO®)	An independent, non-governmental international organization that develops and publishes international standards

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Term	Description
ISO/IEC 20000	A service management system (SMS) standard that specifies requirements for the service provider to plan, establish, implement, operate, monitor, review, maintain and improve an SMS
Iterative approach	A method of working that involves making incremental improvements through repeated cycles of planning, execution, review, and refinement
Key Performance Indicator (KPI)	A metric used to measure performance of services, processes and business objectives
Layers	There are three layers in the SIAM ecosystem: customer organization (including retained capabilities), service integrator and service provider(s)
Lead supplier service integrator	Type of SIAM structure: the role of service integrator is taken by an external organization that is also an external service provider
Lean	The core idea of Lean is to maximize customer value while minimizing waste
Loose coupling	A system where components have minimal dependencies on each other, making them more independent and easier to modify or replace without affecting other parts of the system
Management methodology	A management methodology describes methods, rules and principles associated with a discipline
Maturity	The degree of formality and optimization
Measure	A generic term for the act of quantifying something or the result of doing so
Measurement	The recorded value of a metric at a point in time, showing actual performance (e.g., "94% of incidents resolved within SLA in July")
Metric	A defined indicator used to assess performance against a specific objective or standard (e.g., "% of incidents resolved within SLA")
Model	A representation or abstraction of a system, process, or concept that helps in understanding, analyzing, and communicating complex information (see also SIAM model)
Multi Sourcing Integration (MSI)	Alternative term for SIAM
Multi-sourcing	Sourcing of goods or services from more than one service provider
Offboarding	The process for exiting/removing service providers from the SIAM ecosystem in a controlled manner
Onboarding	The process for bringing new service providers into the SIAM ecosystem in a controlled manner
Operating model	Represents and guides how an organization delivers value to its customers by aligning its people, processes, technology, and infrastructure to execute its strategy and business goals (see also Model)
Operational Level Agreement (OLA)	Agreement between parties (for instance, the service integrator and a service provider) in support of a service level agreement (SLA), to break down end-to-end service targets into detail and individual responsibilities
Organization	A group of people and resources who work together in an organized way for a shared purpose
Organizational Change Management (OCM)	The process of managing change within an organization to ensure successful implementation and adoption

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Term	Description
Organizational Structures	Define the key decision-making entities and their relationships within the organization
Outcome	The desired, measurable change in behavior, value, or result for the user or business that is achieved because of an (IT) output.
Output	The direct, tangible results produced by a process, task, or activity. They represent what has been created, completed, or delivered as a result of work performed
Outsourcing	Procuring goods or services from an external organization
Performance	The degree to which a service, process, or service provider delivers against agreed targets, combining both the metric definition and the measurement results
Performance management and reporting framework	A framework for measuring and reporting that can be tailored to different levels and perspectives. In a SIAM context, this includes drilling down from end-to-end performance to show e.g. individual service provider performance information.
Platform as a Service (PaaS)	A type of cloud service that allows customers to use virtual platforms for their application development and management, removing the need for them to build their own infrastructure
Practice	The actual application or use of an idea, belief or method, as opposed to theories relating to it
Prime vendor	A sourcing approach where the service provider sub-contracts to other service providers to deliver the service, and the customer only has a contractual relationship with the prime vendor
Process	A documented, repeatable approach to carrying out a series of tasks or activities
Process forum	See Forum
Process manager	Supports the process owner and is responsible for the operation of a process
Process model	Describe the purpose and outcomes for a process, as well as activities, inputs, outputs, interactions, controls, measures and supporting policies and templates
Process owner	The accountable role that ensures the end-to-end process is designed, executed and reviewed correctly
Program management	The coordinated management of a collection of related projects, each with its own objectives, to achieve broader strategic business goals and organizational benefits that are not possible by managing the projects individually
Project management	A process that provides a repeatable approach of applying specific knowledge, skills, tools, and techniques to achieve the project's goals within defined constraints of scope, time, and budget
Reputation	The overall opinion others have formed of you, based on past experiences in built up overtime
Responsibility	Something that it is your job or duty to deal with
Responsible, Accountable, Consulted, Informed (RACI)	A responsibility assignment matrix that describes the participation by various roles in completing tasks or deliverables for a project or business process.
Retained capabilities	The functions that are responsible for strategic, architectural, business engagement, and corporate governance activities, including those which cannot be delegated or outsourced (also referred to as the intelligent client function)

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Term	Description
Retrospective	A meeting during which the team reflects on what happened and identifies actions for improvement going forward
Risk management	The forecasting and evaluation of uncertainties, together with the identification of procedures to avoid or minimize their impact
Roadmap	A plan or strategy intended to achieve a particular goal. The SIAM roadmap has four stages: Discovery & Strategy, Design & Build, Implement, Run & Improve.
Role	A description of the specific duties, responsibilities and expectations of a particular position within an organization, which must include required skills, knowledge, behaviors, and experience. A role is distinct from a job, as it is possible for one individual to perform multiple roles as part of their job.
Separation of duties/concerns	Defines what each role is authorized to do and when more than one person must (or cannot) be involved in a task (to prevent errors or fraud). For example, a requestor may not approve their own request
Service	A structured set of activities, capabilities, and resources that delivers value by enabling desired outcomes for the customer. Services support the customer by reducing the customer's need to manage associated operational costs and risks
Service aggregation	See Aggregation
Service architect	A practitioner of Service Architecture, the discipline that provides the structural framework and technical specifications for new or changing services
Service boundaries	A definition of what parts make up a service (what is 'inside the boundary'), often used in technical architecture documents
Service consumer	See Consumer
Service credits	(or service level credits) A mechanism by which amounts are deducted from the amounts to be paid under the contract to the service provider if actual performance fails to meet the standards set in the service levels
Service dashboard	See Dashboard
Service definition	A document intended to fully define the service and processes necessary to provide and support the service
Service element	(or component) Part of a service that can be allocated to a specific service provider
Service group	A logical grouping of related services (or components) assigned to a specific service provider to optimize management and sourcing (also referred to as Tower)
Service hierarchy	A structured representation of services, showing their relationships, dependencies, and levels within the SIAM model
Service Integration (SI)	Alternative term for SIAM
Service Integration And Management (SIAM)	A management methodology that can be applied in an environment that includes services sourced from a number of service providers
Service integrator	A single, logical entity accountable for end-to-end service governance, management, integration, assurance and coordination of the end-to-end delivery of services and the business value that the customer receives

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Term	Description
Service integrator layer	The service integrator layer of the SIAM ecosystem is where end-to-end service governance, management, integration, assurance and coordination is performed
Service management	The management practices and capabilities that an organization uses to provide services to consumers
Service Management And Integration (SMAI)	Alternative term for SIAM
Service Management Integration (SMI)	Alternative term for SIAM
Service manager	Responsible for the delivery of one or more services
Service map	A visual representation of how services interact, including dependencies, data flows, and technical connections
Service model	A way of modelling the hierarchy of services, defining service structure, relationships, service providers, interactions, and integration within the SIAM ecosystem.
Service orchestration	The coordination and alignment of people, processes, tools, and providers to deliver a seamless, end-to-end service experience
Service outcomes	A definition of what a service is meant to achieve or deliver
Service owner	A role that is accountable for the end-to-end performance of a service, and the outcomes it provides to the business
Service provider	Responsible for managing the products and technology used to deliver one or more contracted or agreed services, or service elements, to the customer. They can be internal or external to the customer organization. (also referred to as Vendor or Supplier)
Service provider category	Service providers can be categorized as strategic, tactical or commodity
Shadow IT	IT services and systems commissioned by business departments, without the knowledge of the IT department (sometimes referred to as 'stealth IT')
SIAM ecosystem	See Ecosystem
SIAM environment	See Environment
SIAM layers	See Layers
SIAM model	Describes the practices, processes, functions, roles and structural elements within a SIAM ecosystem (see also Model). Organizations develop a SIAM model based on the SIAM ecosystem, but tailored to their operating environment and objectives.
SIAM structures	The four structures describe how the service integrator is sourced: internally, externally, lead supplier or hybrid
Skill	The learned and applied ability to use one's knowledge effectively and readily in execution or individual performance
Skills Framework for the Information Age (SFIA®)	A framework that provides a common language for skills and competencies in the digital and IT workforce
Software as a Service (SaaS)	A cloud service where software is paid for monthly as a subscription rather than purchased as a one-time payment
Sourcing	The procurement approach an organization adopts; for example, sourcing services internally or externally. Adopting SIAM will affect how an organization sources services and the types of contracts it puts in place with service providers.

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Term	Description
Stakeholder	A person or group of people that have a particular interest, or are impacted, by something
Strategy	A plan of action designed to achieve a long-term or overall aim
Structural element	Structural elements are multi-functional teams with a specific objective. In SIAM the structural elements include boards, forums and working groups, and the members would come from different organizations and different SIAM layers.
Subject Matter Expert (SME)	(or domain expert) A person who is an authority in a particular area or topic
Supplier	See Service provider
Swim lanes	A visual element used in process flow diagrams to distinguish job sharing and responsibilities for sub-processes, within an overall process. Each role has a 'swim lane' in the diagram
Swivel chair approach	Manually copying data from one system and then entering the same data into another system
Tooling strategy	A plan for acquiring, deploying, and managing the tools that support an organization's strategic goals, ensuring they effectively facilitate planning, execution, and performance monitoring to achieve long-term objectives. It defines what tools, who will own them and how they will support the flow of data and information between the SIAM layers.
Tower	See Service group
Transformation	The act of transforming or the state of being transformed. It is the, less defined, SIAM program that makes all changes, in all roadmap stages, to realize the full benefits of the SIAM model
Transition	Transition is a defined project to change from one form, state, style or place to another. A SIAM transition takes the ecosystem from the previous state to the start of the Run & Improve stage.
User	The role that uses services
User experience (UX)	The sentiment expressed by the end user when using the service
Value chain	Series of activities performed by an organization's business unit to deliver value to its customers and stakeholders
Vendor	See Service provider
VeriSM™	Service Management for the Digital Age
Waterfall	The method of development and implementation that follows sequential stages and a fixed plan of work, for instance, plan, design, build and deploy
Watermelon effect	The situation when a report or metric is 'green on the outside, red on the inside': the service provider(s) meets individual targets, but the end-to-end service is not meeting the customer's requirements
Win-win	A win-win situation or outcome is one that benefits both or all parties, or that has two distinct benefits
Working group	A structural element, typically formed on a reactive ad hoc or fixed term basis from different organizations and different specialist areas to address specific issues or projects

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List of acronyms

This list expands the acronyms used in this document.

Acronym	Meaning
ADKAR®	Awareness, Desire, Knowledge, Ability, and Reinforcement
AG	Aktiengesellschaft (German: Stock Corporation)
Agile SM	Agile Service Management
AI	Artificial Intelligence
BAU	Business As Usual
BoK	Body of Knowledge
BRM	Business Relationship Management
CFO	Chief Financial Officer
CIO	Chief Information Officer
CISO	Chief Information Security Officer
CMM(I)®	Capability Maturity Model (Integration)
COBIT®	Control Objectives for Information and Related Technologies
CTO	Chief Technology Officer
CX	Customer Experience
DevOps	Development and Operations
ELS	Early Life Support
GDPR	General Data Protection Regulation
HR	Human Resources
IaaS	Infrastructure as a Service
IEC	International Electrotechnical Commission
IFDC	International Federation for Digital Competence
ISO®	International Organization for Standardization
IT	Information Technology
IT4IT™	IT for IT
ITO	IT outsourcer (or organization)
ITSM	IT Service Management
KPI	Key Performance Indicator
LAN	Local Area Network
Ltd	Limited
MSI	Multi-Sourcing integration
OCM	Organizational Change Management
OLA	Operational Level Agreement
OSI	Open Systems Interconnect
PaaS	Platform as a Service
PRINCE2®	PRojects IN Controlled Environments
RACI	Responsible, Accountable, Consulted, Informed
SaaS	Software as a Service
SFIA®	Skills Framework for the Information Age
SI	Service Integration (or Integrator)
SIAM	Service Integration and Management
SLA	Service Level Agreement

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Acronym	Meaning
SMAI	Service Management and Integration
SME	Subject Matter Expert
SMI	Service Management Integration
SMS	Service Management System (or Short Message Service -telephony-)
TOGAF®	The Open Group Architecture Framework
UK	United Kingdom
UX	User Experience
VeriSM™	Value-driven, Evolving, Responsive, Integrated, Service Management
WAN	Wide Area Network

Appendix B: License information

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Section 1 – Definitions.

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